

One-hour equation audit: base vs. evolved PySR

Nine EmpiricalBench-overlap tasks in the **archived SRBench2 setup**; ten paired seeds (10000–10009). Each search uses one core, a 3600-second search budget, and restarts capped at 10^6 evaluations. Warm-up/scoring are excluded. These are saved final portfolio frontiers; the final restart can overshoot the budget. This is the comparison behind **72/90 vs. 74/90**, not the later native EmpiricalBench or 90-second-restart experiments.

How to read each page. Left: base PySR; right: evolved PySR (run 709715). The *structural candidate* is the equation selected by the archived review/audit, including near matches. The *best numerical fit* is independently selected by maximum saved training R^2 on the final frontier (ties: lower complexity). It need not be the witness supporting recovery, the search’s chosen output, or the minimum-native-loss equation. Native losses differ between methods and are not directly comparable. No constants are refitted here.

Checking correctness. Exact means the accepted variable dependence, with only the reference’s free constants adjustable. Fixed powers/relative coefficients, tiny extra terms, and transformed targets matter. Near/miss do not count as recovery. Bode is special: the archived broad phenomenological criterion allows a zero offset; its 20 positives are not strict recovery of the full nonzero-offset law. Formula displays use eight significant digits for readability; **the exact saved expression is printed beneath each formula**. Use those full-precision strings to inspect coefficients near 1 or 2. These pages preserve archived decisions, not a new exhaustive correctness certificate.

Task	Base / 10	Evolved / 10	Jump to first seed
Hubble	10	10	Hubble
Ideal gas	10	10	Ideal gas
Kepler	10	10	Kepler
Leavitt	10	10	Leavitt
Newton	10	10	Newton
Planck	0	0	Planck
Rydberg	2	4	Rydberg
Schechter	10	10	Schechter
Bode	10	10	Bode
Total (including broad Bode matches)	72	74	

All 180 frontiers, original reviews, selections, and source fingerprints are saved in `audit_data.json`. One page per task–seed pair follows. The unselected Rydberg miss also has a full-frontier appendix.

Hubble | seed 10000

Accepted target: $v = cx_0$

$x_0 = D$

Dataset columns: D, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 317 restarts; actual search 3600.0 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.72478362; native loss = 142.50496.

$$474.87469 \cdot x_0$$

Exact saved expression (use this for coefficient checks):

`x0 * 474.8746948297984`

Review rationale: Proportional to distance: $c \cdot x_0$.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 22; training R2 = 0.88369025; native loss = 86.55602.

$$x_0 \cdot \left(500.35715 + \frac{\frac{0.34713918}{1.9982982 - x_0} + \frac{15.276395}{x_0}}{x_0 - 0.3519965} \right) + \frac{2.7488492}{\ln(x_0)}$$

Exact saved expression (use this for coefficient checks):

`(((((0.3471391824845384 / (1.998298189745659 - x0)) - (-15.276395107980669 / x0)) / (x0 + -0.3519964962309075)) + 500.3571511603751) * x0) + (2.7488492482469926 / log(x0))`

Evolved PySR (709715)

Archived/audited label: exact. 148 restarts; actual search 3600.1 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -1.584504; native loss = 0.52699528.

$$x_0$$

Exact saved expression (use this for coefficient checks):

`x0`

Review rationale: Proportional to distance: $c \cdot x_0$.

Best numerical fit (maximum training R2)

Frontier index 13; complexity 26; training R2 = 0.47456773; native loss = 0.30333283.

$$308.51922 \cdot \sqrt{x_0 + 0.34755999 - \frac{0.032797408}{0.34731588 - x_0} - \frac{9.9956768}{0.92607777 - x_0} + \frac{1.5761851}{0.90955696 - x_0}}$$

Exact saved expression (use this for coefficient checks):

`((((0.005108871628739366 / (0.9095569620961703 - x0)) - (0.03239887895795362 / (0.926077775036416 - x0))) + sqrt((x0 + 0.3475599941737374) - ((0.035339689089004984 / (0.34731588293778426 - x0)) * 0.9280615798693348))) * 308.5192194558132`

Hubble | seed 10001

Accepted target: $v = cx_0$

$x_0 = D$

Dataset columns: D, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 315 restarts; actual search 3600.6 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.72478362; native loss = 142.50496.

$$474.87469 \cdot x_0$$

Exact saved expression (use this for coefficient checks):

`x0 * 474.8746949306655`

Review rationale: Proportional to distance: $c \cdot x_0$.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 23; training R2 = 0.86515313; native loss = 83.390414.

$$x_0 \cdot \left(\frac{12.389466 \cdot \left(4.4034764 - \frac{0.3344152}{x_0 - 1.7221803} \right)}{x_0 - 0.36293547} + 385.62644 + \frac{6.5999519}{\ln(x_0^2)} \right)$$

Exact saved expression (use this for coefficient checks):

`x0 * ((385.62643851662114 + ((((-0.3344152047016276 / (x0 + -1.722180261865646)) -
-4.4034763482682076) / 0.08071372986968048) / (x0 + -0.36293546916578007))) +
(6.599951942383051 / log(square(x0))))`

Evolved PySR (709715)

Archived/audited label: exact. 150 restarts; actual search 3600.1 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -1.584504; native loss = 0.52699528.

$$x_0$$

Exact saved expression (use this for coefficient checks):

`x0`

Review rationale: Proportional to distance: $c \cdot x_0$.

Best numerical fit (maximum training R2)

Frontier index 9; complexity 13; training R2 = -0.16566148; native loss = 0.37562039.

$$\ln \left(\left(-x_0^3 + \frac{0.010022475}{0.75393357 - x_0^3} \right)^2 \right) + 279.84825$$

Exact saved expression (use this for coefficient checks):

`log(square((0.010022475304174843 / (0.7539335657430845 - cube(x0))) - cube(x0))) +
279.84824532547947`

Hubble | seed 10002

Accepted target: $v = cx_0$

$x_0 = D$

Dataset columns: D, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 315 restarts; actual search 3600.5 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.72478362; native loss = 142.50496.

$$474.87469 \cdot x_0$$

Exact saved expression (use this for coefficient checks):

`x0 * 474.8746947891507`

Review rationale: Proportional to distance: $c \cdot x_0$.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 25; training R2 = 0.8638463; native loss = 80.589284.

$$371.48283 \cdot x_0 + \frac{3.1390234 \cdot x_0}{\ln(x_0)} + \frac{3.1390234 \cdot \left(-\frac{x_0}{x_0 - 1.7166046} + 16.473277 \right)}{\ln(x_0) + 1.0765914} + 68.331084$$

Exact saved expression (use this for coefficient checks):

`(((((16.473277337959455 - (x0 / (x0 + -1.716604564526174)))) / (log(x0) -
-1.076591446335422)) + (x0 / 0.00844998227464194)) + 21.768262302862958) + (x0 /
log(x0))) * 3.139023356959021`

Evolved PySR (709715)

Archived/audited label: exact. 150 restarts; actual search 3600.1 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -1.584504; native loss = 0.52699528.

$$x_0$$

Exact saved expression (use this for coefficient checks):

`x0`

Review rationale: Proportional to distance: $c \cdot x_0$.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 30; training R2 = 0.038499278; native loss = 0.33044388.

$$1.6720398 \cdot x_0 + 653.02966 \cdot e^{-\frac{0.9185992}{x_0}} + \frac{0.8751358}{x_0 - \frac{0.65827504}{1.0212851 - \ln(x_0)}} + \frac{1.5066917 \cdot \ln(x_0)}{x_0 - 0.90383828}$$

Exact saved expression (use this for coefficient checks):

`exp((-0.918599198686661 / x0) + 6.481622546266859) - ((((-0.9011099421122617 / ((x0 +
-0.9038382755248312) / log(x0))) - x0) / 0.5980718862147626) + (-0.8751357982105101
/ ((-0.6582750399779413 / (1.0212851076268084 - log(x0))) + x0)))`

Hubble | seed 10003

Accepted target: $v = cx_0$

$x_0 = D$

Dataset columns: D, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 314 restarts; actual search 3600.1 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.72478362; native loss = 142.50496.

$$474.87469 \cdot x_0$$

Exact saved expression (use this for coefficient checks):

$x_0 * 474.8746947893252$

Review rationale: Proportional to distance: $c*x_0$.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 28; training R2 = 0.87828098; native loss = 85.054673.

$$x_0 \cdot \left(522.01951 - \frac{0.18137767}{x_0 - 1.9986754} \right) - \frac{-2.3444937 \cdot x_0 - 0.56768802}{\ln(x_0)} - \frac{5.9855602}{-2.7492797 + \frac{17.993926}{x_0 + 6.1908272}}$$

Exact saved expression (use this for coefficient checks):

$((x_0 * (522.0195110529559 - (0.18137767363179638 / (x_0 + -1.998675396673268)))) - (5.985560169557812 / ((17.99392601669823 / (x_0 - -6.190827184844493)) - 2.7492796745709986))) - ((-0.5676880183576987 - (x_0 / 0.42653133026360307)) / \log(x_0))$

Evolved PySR (709715)

Archived/audited label: exact. 120 restarts; actual search 3600.7 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -1.584504; native loss = 0.52699528.

$$x_0$$

Exact saved expression (use this for coefficient checks):

x_0

Review rationale: Proportional to distance: $c*x_0$.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 22; training R2 = -1.1676851; native loss = 0.32479373.

$$\frac{13.842156 \cdot \left(13.845037 + \frac{1.7491992}{x_0 - 0.40751388} \right)}{x_0 + \frac{3.3377359}{\left(x_0 - \frac{0.010721211}{0.91242799 - x_0} \right)^2}}$$

Exact saved expression (use this for coefficient checks):

$((((1.749199181395731 / (x_0 - 0.40751388314311027)) + 13.845037004065938) * 13.84215611213484) / (x_0 + \text{square}(-1.8269471601925027 / (x_0 - (0.010721211435885636 / (0.9124279902562951 - x_0)))))$

Hubble | seed 10004

Accepted target: $v = cx_0$

$x_0 = D$

Dataset columns: D, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 318 restarts; actual search 3599.8 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.72478362; native loss = 142.50496.

$$474.87469 \cdot x_0$$

Exact saved expression (use this for coefficient checks):

$x_0 * 474.87469477898105$

Review rationale: Proportional to distance: $c \cdot x_0$.

Best numerical fit (maximum training R2)

Frontier index 19; complexity 26; training R2 = 0.86347087; native loss = 83.338031.

$$x_0 \cdot \left(\frac{x_0}{\left(-0.78980067 + \frac{0.43217818}{\ln(x_0)}\right) \cdot \ln(x_0)} + 397.19832 + \frac{3.1428828}{\ln(x_0)} - \frac{58.142329}{0.36365113 - x_0} \right)$$

Exact saved expression (use this for coefficient checks):

$((((397.19831641510837 + (-58.14232856820065 / (0.3636511259707619 - x_0))) + ((x_0 / \log(x_0)) / (-0.7898006735636526 - (-0.4321781780466818 / \log(x_0))))) - (-3.14288279781156 / \log(x_0))) * x_0$

Evolved PySR (709715)

Archived/audited label: exact. 120 restarts; actual search 3600.9 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -1.584504; native loss = 0.52699528.

$$x_0$$

Exact saved expression (use this for coefficient checks):

x_0

Review rationale: Proportional to distance: $c \cdot x_0$.

Best numerical fit (maximum training R2)

Frontier index 14; complexity 21; training R2 = -1.5466395; native loss = 0.35363712.

$$5.2573442 \cdot x_0 + \frac{0.086097955}{x_0 - 0.97718211} - \frac{0.033957688}{(0.2825948 - x_0) \cdot (1.9952547 - x_0)}$$

Exact saved expression (use this for coefficient checks):

$(x_0 / 0.19021010673233163) + ((0.08609795502567708 / (x_0 + -0.9771821074163067)) - ((0.03395768783850729 / (1.0088326538891244 - (x_0 + -0.9864220346719487)))) / (0.28259480031128065 - x_0))$

Hubble | seed 10005

Accepted target: $v = cx_0$

$x_0 = D$

Dataset columns: D, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 316 restarts; actual search 3599.8 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.72478362; native loss = 142.50496.

$$474.87469 \cdot x_0$$

Exact saved expression (use this for coefficient checks):

`x0 * 474.8746948858074`

Review rationale: Proportional to distance: $c \cdot x_0$.

Best numerical fit (maximum training R2)

Frontier index 20; complexity 30; training R2 = 0.88861071; native loss = 79.414105.

$$x_0 \cdot \left(\frac{2.9938709 \cdot x_0^3}{\ln(x_0)} + 438.67711 \right) + x_0 + \frac{44.093301 + \frac{0.41153444}{1.9988418 - x_0}}{\ln(2.7816964 \cdot x_0)} + \frac{1.1932222}{x_0}$$

Exact saved expression (use this for coefficient checks):

`((1.1932221843360789 / x0) + (x0 + (((44.09330109171803 - (-0.4115344427388087 / (1.9988418176325695 - x0))) / log(x0 * 2.7816964511036915)) + (x0 * (((2.9938708541911785 * cube(x0)) / log(x0)) + 438.6771086303138)))))`

Evolved PySR (709715)

Archived/audited label: exact. 120 restarts; actual search 3599.5 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -1.584504; native loss = 0.52699528.

$$x_0$$

Exact saved expression (use this for coefficient checks):

`x0`

Review rationale: Proportional to distance: $c \cdot x_0$.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 29; training R2 = -1.5600711; native loss = 0.31449502.

$$x_0 - \frac{x_0 - 0.20066909}{-\frac{x_0 - 0.99785496}{x_0 - \frac{0.51495548}{x_0 - 0.32112195}} - \frac{0.017324483}{x_0 - 0.90190871}} + 0.71737301$$

Exact saved expression (use this for coefficient checks):

`((x0 - (((x0 + -0.20066908657969235) / ((-0.017324482695884404 / (x0 - 0.9019087055774397)) - ((x0 - 0.9978549599428658) / ((-0.5149554789607103 / (x0 - 0.32112195210237104)) + x0)))) - x0)) - x0) + 0.7173730147022148`

Hubble | seed 10006

Accepted target: $v = cx_0$

$x_0 = D$

Dataset columns: D, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 312 restarts; actual search 3600.6 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.72478362; native loss = 142.50496.

$$474.87469 \cdot x_0$$

Exact saved expression (use this for coefficient checks):

`x0 * 474.87469477941744`

Review rationale: Proportional to distance: $c \cdot x_0$.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 26; training R2 = 0.86754691; native loss = 83.970406.

$$x_0 \cdot \left(-3.7224843 \cdot x_0^3 + 424.35972 + \frac{45.06}{x_0 - 0.35399552} \right) - \frac{e^{x_0}}{x_0 \cdot (x_0 - 1.7255599) \cdot \ln(x_0)}$$

Exact saved expression (use this for coefficient checks):

`(((((45.05999993786244 / (x0 + -0.3539955167216214)) + (cube(x0) * -3.7224842995621144)) + 424.3597235159382) * x0) - ((exp(x0) / log(x0)) / (x0 * (x0 + -1.725559899942646))))`

Evolved PySR (709715)

Archived/audited label: exact. 122 restarts; actual search 3600.1 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -1.584504; native loss = 0.52699528.

$$x_0$$

Exact saved expression (use this for coefficient checks):

`x0`

Review rationale: Proportional to distance: $c \cdot x_0$.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 21; training R2 = -1.5361096; native loss = 0.33084063.

$$8.4533904 \cdot \left(0.66408509 + e^{-x_0} + \frac{0.026492023}{0.89474992 - x_0} \right)^2 + \frac{0.31570871}{0.51954144 - 1.7466269 \cdot x_0}$$

Exact saved expression (use this for coefficient checks):

`square((exp(1.067283794585005 - x0) + 1.9308084515586048) + (0.07702480003640225 / (0.8947499161406764 - x0))) + (0.3157087130785581 / (0.519541441216797 - (x0 / 0.5725321205510425))))`

Hubble | seed 10007

Accepted target: $v = cx_0$

$x_0 = D$

Dataset columns: D, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 314 restarts; actual search 3599.9 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.72478362; native loss = 142.50496.

$$474.87469 \cdot x_0$$

Exact saved expression (use this for coefficient checks):

`x0 * 474.8746946955793`

Review rationale: Proportional to distance: $c \cdot x_0$.

Best numerical fit (maximum training R2)

Frontier index 13; complexity 21; training R2 = 0.85137691; native loss = 87.982143.

$$885.12627 \cdot (0.24004026 \cdot x_0 - 1)^3 + 925.72168 - \frac{43.21235}{\ln\left(\frac{0.32308901}{x_0}\right)} + \frac{9.1152937}{x_0 - 0.97461047}$$

Exact saved expression (use this for coefficient checks):

`(cube(x0 + -4.165967795014631) + ((0.7445830697814654 / (x0 + -0.9746104719383847))) + ((-3.529802263816484 / log(0.3230890135587621 / x0)) + 75.61760649128897))) * 12.242144670272147`

Evolved PySR (709715)

Archived/audited label: exact. 118 restarts; actual search 3600.6 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -1.584504; native loss = 0.52699528.

$$x_0$$

Exact saved expression (use this for coefficient checks):

`x0`

Review rationale: Proportional to distance: $c \cdot x_0$.

Best numerical fit (maximum training R2)

Frontier index 22; complexity 30; training R2 = -1.498186; native loss = 0.31152733.

$$1.9995737 \cdot x_0 + 1.9995737 \cdot e^{x_0 - \frac{0.02297807 + \frac{0.025448977}{0.97550435 - x_0}}{1.9976426 - x_0}} + 0.024926444 + \frac{2 \cdot x_0 - 0.38819483}{x_0 - 0.35502311}$$

Exact saved expression (use this for coefficient checks):

`((x0 + (x0 + -0.38819482869411515)) / (x0 + -0.3550231119165023)) + ((exp(x0 - (((0.025448976742351474 / (0.9755043506826777 - x0)) + 0.022978069529206998) / (1.9976425855663147 - x0)))) + (x0 + 0.012465879047436694)) * 1.9995737439841872)`

Hubble | seed 10008

Accepted target: $v = cx_0$

$x_0 = D$

Dataset columns: D, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 318 restarts; actual search 3600.3 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.72478362; native loss = 142.50496.

$$474.87469 \cdot x_0$$

Exact saved expression (use this for coefficient checks):

`x0 * 474.8746948826243`

Review rationale: Proportional to distance: $c \cdot x_0$.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 26; training R2 = 0.86717339; native loss = 84.485325.

$$\frac{x_0 \cdot \left(373.24735 + \frac{1.3729065 + \frac{2.0323911}{\ln(x_0)} + \frac{0.10662312}{x_0}}{1.7419174 - x_0} \right)}{0.93095555 + \frac{0.12048695}{0.22280082 - x_0}}$$

Exact saved expression (use this for coefficient checks):

`(((((2.03239107105452 / log(x0)) + ((0.10662311540938112 / x0) + 1.3729064785535654)) / (1.7419173628048759 - x0)) + 373.2473536969058) * (x0 / ((0.12048694697401542 / (0.22280081593465195 - x0)) + 0.9309555473828363)))`

Evolved PySR (709715)

Archived/audited label: exact. 121 restarts; actual search 3600.2 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -1.584504; native loss = 0.52699528.

$$x_0$$

Exact saved expression (use this for coefficient checks):

`x0`

Review rationale: Proportional to distance: $c \cdot x_0$.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 29; training R2 = 0.80123644; native loss = 0.33208793.

$$493.56076 \cdot x_0 - 60.352955 \cdot \ln(x_0) - \frac{30.689764}{-\ln(x_0) - 0.93770007} + \frac{3.0454813}{x_0 \cdot (-x_0 \cdot \ln(x_0) + 1.3794625) \cdot \ln(x_0)}$$

Exact saved expression (use this for coefficient checks):

`((((x0 / -0.12228069988418) + (0.5085047437286387 / (-0.9377000660749294 - log(x0)))) + (((-0.050461179819862964 / (x0 * log(x0))) / (1.3794625124992295 - (x0 * log(x0)))) + log(x0))) / -0.016569197019706668`

Hubble | seed 10009

Accepted target: $v = cx_0$

$x_0 = D$

Dataset columns: D, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 305 restarts; actual search 3599.8 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.72478362; native loss = 142.50496.

$$474.87469 \cdot x_0$$

Exact saved expression (use this for coefficient checks):

`x0 * 474.8746948963056`

Review rationale: Proportional to distance: $c \cdot x_0$.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 22; training R2 = 0.8650732; native loss = 85.245279.

$$422.30762 \cdot x_0 + \frac{18.721275 + \frac{2.091539}{\ln(x_0)}}{x_0 - 0.36096641} + 24.467761 - \frac{8.509983}{x_0 - 1.7351772}$$

Exact saved expression (use this for coefficient checks):

`((((x0 * 422.3076168573912) + (-8.509982979917302 / (x0 + -1.7351771584487419)))) + 24.467761027454653) + (((2.0915390417979274 / log(x0)) + 18.72127513337357) / (x0 + -0.36096640657135637))`

Evolved PySR (709715)

Archived/audited label: exact. 121 restarts; actual search 3600.8 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -1.584504; native loss = 0.52699528.

$$x_0$$

Exact saved expression (use this for coefficient checks):

`x0`

Review rationale: Proportional to distance: $c \cdot x_0$.

Best numerical fit (maximum training R2)

Frontier index 16; complexity 19; training R2 = -1.5718945; native loss = 0.35343727.

$$2 \cdot x_0 - \frac{0.04532851}{0.98313671 - x_0} - \frac{0.11685702}{(0.34139691 - x_0) \cdot (1.7689333 - x_0)}$$

Exact saved expression (use this for coefficient checks):

`(x0 - (((0.11685701851556915 / (0.34139690644360093 - x0)) / (1.7689332757791534 - x0)) - x0)) - (0.04532850954058434 / (0.9831367118973613 - x0))`

Ideal gas | seed 10000

Accepted target: $\log P = c_0 + \log x_0 + \log x_1 - \log x_2$

$(x_0, x_1, x_2) = (n, T, V)$

Dataset columns: n, T, V, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 277 restarts; actual search 3600.4 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 8; training R2 = 0.99576942; native loss = 0.066443048.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2}\right) + 2.054419$$

Exact saved expression (use this for coefficient checks):

$2.0544190302536536 + \log((x_0 / x_2) * x_1)$

Review rationale: Equals $\log(x_0 \cdot x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 22; training R2 = 0.99787431; native loss = 0.046884453.

$$\ln\left(\frac{8.092041 \cdot x_0 \cdot (x_1 + 0.55987112)}{x_2 + 0.47718224}\right) - \frac{0.50736001}{x_2 \cdot \left(6.871252 - \frac{x_1}{x_0}\right)}$$

Exact saved expression (use this for coefficient checks):

$\log(((x_0 * 8.092040972538877) / (x_2 + 0.47718223969779683)) * (x_1 - -0.5598711194966941) - (0.5073600141635786 / ((6.871252023409586 - (x_1 / x_0)) * x_2)))$

Evolved PySR (709715)

Archived/audited label: exact. 113 restarts; actual search 3600.0 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 6; training R2 = -1.4029727; native loss = 0.059460259.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2}\right)$$

Exact saved expression (use this for coefficient checks):

$\log((x_0 * x_1) / x_2)$

Review rationale: Equals $\log(x_0 \cdot x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 14; complexity 18; training R2 = 0.99787151; native loss = 0.045162047.

$$\ln\left(\frac{8.3316574 \cdot x_0 \cdot x_1}{x_2 + 0.62488373}\right) + \frac{4.134628}{x_1 - 8.1934166 \cdot x_2}$$

Exact saved expression (use this for coefficient checks):

$\log((x_0 * (x_1 / 0.12002413764747767)) / (x_2 + 0.6248837296425431)) - (-4.134628028213558 / (x_1 - (x_2 / 0.1220492064942589)))$

Ideal gas | seed 10001

Accepted target: $\log P = c_0 + \log x_0 + \log x_1 - \log x_2$

$$(x_0, x_1, x_2) = (n, T, V)$$

Dataset columns: n, T, V, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 279 restarts; actual search 3600.5 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 8; training R2 = 0.99597135; native loss = 0.066443048.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2}\right) + 2.0630418$$

Exact saved expression (use this for coefficient checks):

$$2.0630418172317264 + \log((x_0 * x_1) / x_2)$$

Review rationale: Equals $\log(x_0 * x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 13; complexity 27; training R2 = 0.99815213; native loss = 0.043893489.

$$\frac{1.1199146 \cdot x_2}{-0.24587907 \cdot x_2 - 5.610936} + \ln(x_0 \cdot x_1) + 1.2519774 + \frac{2.2764181}{-x_1 + 7.8959085 \cdot (0.3558762 \cdot x_0 - 0.3558762 \cdot x_2 + 1)}^2$$

Exact saved expression (use this for coefficient checks):

$$(((x_2 / ((x_2 * -0.2458790702380066) + -5.610935998916273)) / 0.892925203804584) + (2.276418130332265 / (\text{square}((x_0 - x_2) + 2.809965918830331) - x_1))) + (\log(x_1 * x_0) + 1.2519773831289216)$$

Evolved PySR (709715)

Archived/audited label: exact. 112 restarts; actual search 3599.6 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 6; training R2 = -1.4029727; native loss = 0.059460259.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2}\right)$$

Exact saved expression (use this for coefficient checks):

$$\log((x_0 * x_1) / x_2)$$

Review rationale: Equals $\log(x_0 * x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 14; complexity 20; training R2 = 0.99779486; native loss = 0.045425962.

$$\ln\left(\frac{7.768453 \cdot x_0 \cdot x_1}{x_2 + 0.34567213} + x_0\right) - \frac{0.097211513}{x_0 - 1.4522379 \cdot x_2}$$

Exact saved expression (use this for coefficient checks):

$$\log(x_0 + (((x_0 / 0.12872575848708745) * x_1) / (x_2 + 0.345672129723848))) - (0.09721151303738257 / (x_0 - (x_2 * 1.4522378746193578)))$$

Ideal gas | seed 10002

Accepted target: $\log P = c_0 + \log x_0 + \log x_1 - \log x_2$

$(x_0, x_1, x_2) = (n, T, V)$

Dataset columns: n, T, V, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 277 restarts; actual search 3599.8 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 8; training R2 = 0.995776; native loss = 0.066443048.

$$2.0546534 - \ln\left(\frac{x_2}{x_0 \cdot x_1}\right)$$

Exact saved expression (use this for coefficient checks):

2.0546533657838446 - log(x2 / (x1 * x0))

Review rationale: Equals $\log(x_0 \cdot x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 12; complexity 18; training R2 = 0.99757522; native loss = 0.046734595.

$$\ln\left(\frac{8.1119273 \cdot x_0 \cdot x_1}{x_2 + 0.64254027}\right) + \frac{0.098108592}{-x_0 + x_2 + 5.014406}$$

Exact saved expression (use this for coefficient checks):

log(((x0 / (x2 + 0.6425402655080223)) / 0.12327526701170496) * x1) + (0.09810859148113132 / ((x2 - 5.014406007278853) - x0))

Evolved PySR (709715)

Archived/audited label: exact. 114 restarts; actual search 3601.1 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 6; training R2 = -1.4029727; native loss = 0.059460259.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2}\right)$$

Exact saved expression (use this for coefficient checks):

log((x0 * x1) / x2)

Review rationale: Equals $\log(x_0 \cdot x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 27; training R2 = 0.99810753; native loss = 0.043489711.

$$1.8967466 \cdot \ln\left(\sqrt{\frac{x_1}{\frac{1}{x_0} \cdot x_2}} - 0.23633081\right) + 2.4060024 - \frac{0.017270797 \cdot \ln(x_1)}{-x_0 + 2.7993212 \cdot \sqrt{x_1} + 0.44646259}$$

Exact saved expression (use this for coefficient checks):

((log(sqrt(x1 / (x2 / x0))) + -0.23633081347602355) * 1.8967465764670501) + 2.4060024251782566 + ((log(x1) / ((sqrt(x1) / 0.3572294610050653) + (0.4464625889112891 - x0))) * -0.017270796628683246)

Ideal gas | seed 10003

Accepted target: $\log P = c_0 + \log x_0 + \log x_1 - \log x_2$

$$(x_0, x_1, x_2) = (n, T, V)$$

Dataset columns: n, T, V, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 277 restarts; actual search 3599.6 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 8; training R2 = 0.99594052; native loss = 0.066443048.

$$2.0614617 - \ln\left(\frac{x_2}{x_0 \cdot x_1}\right)$$

Exact saved expression (use this for coefficient checks):

$$2.061461673069181 - \log((x_2 / x_0) / x_1)$$

Review rationale: Equals $\log(x_0 \cdot x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 28; training R2 = 0.9979054; native loss = 0.04508827.

$$\ln\left(\frac{8.0757373 \cdot x_0 \cdot x_1}{x_2 + 0.6322394}\right) + \frac{7.299802}{x_0 \cdot (x_0 - 1.586217 \cdot x_2 + 1.586217 \cdot \ln(x_2)) \cdot \ln(x_1)}$$

Exact saved expression (use this for coefficient checks):

$$\log((x_0 / (x_2 + 0.6322394040551043)) * (x_1 * 8.075737295090097)) + (0.9011053041911985 / (((x_0 * ((\log(x_2) - x_2) / 0.6304307479085768) + x_0)) / 8.10094218704312) * \log(x_1)))$$

Evolved PySR (709715)

Archived/audited label: exact. 135 restarts; actual search 3600.1 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 6; training R2 = -1.4029727; native loss = 0.059460259.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2}\right)$$

Exact saved expression (use this for coefficient checks):

$$\log((x_0 * x_1) / x_2)$$

Review rationale: Equals $\log(x_0 \cdot x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 9; complexity 12; training R2 = 0.99701104; native loss = 0.054211048.

$$\ln\left(\frac{8.1079214 \cdot x_0 \cdot x_1}{x_2 + \frac{1.873466}{x_2}}\right)$$

Exact saved expression (use this for coefficient checks):

$$\log(((x_1 / ((1.8734660302128128 / x_2) + x_2)) / 0.12333617405111251) * x_0)$$

Ideal gas | seed 10004

Accepted target: $\log P = c_0 + \log x_0 + \log x_1 - \log x_2$

$(x_0, x_1, x_2) = (n, T, V)$

Dataset columns: n, T, V, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 279 restarts; actual search 3600.0 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 8; training R2 = 0.99585214; native loss = 0.066443048.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2}\right) + 2.0575445$$

Exact saved expression (use this for coefficient checks):

`2.0575444966579615 + log((x0 * x1) / x2)`

Review rationale: Equals $\log(x_0 \cdot x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 13; complexity 20; training R2 = 0.99777259; native loss = 0.047780126.

$$\ln\left(\frac{x_0 \cdot (x_1 + 1.3459355)}{0.12239466 \cdot x_2 + 0.077210992}\right) - \frac{0.39936493}{-0.12320049 \cdot x_1 + x_2}$$

Exact saved expression (use this for coefficient checks):

`log((x0 / ((x2 - -0.6308362587385731) * 0.12239466362236347)) * (x1 + 1.3459354920809319)) + (-0.3993649289559462 / (x2 - (x1 * 0.12320049335936865)))`

Evolved PySR (709715)

Archived/audited label: exact. 134 restarts; actual search 3600.8 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 6; training R2 = -1.4029727; native loss = 0.059460259.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2}\right)$$

Exact saved expression (use this for coefficient checks):

`log((x0 * x1) / x2)`

Review rationale: Equals $\log(x_0 \cdot x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 24; training R2 = 0.99758774; native loss = 0.043286782.

$$\ln\left(x_0 \cdot \left(x_1 + \frac{0.29234358}{-0.59848396 + \frac{23.50678}{x_2}}\right)\right) - 3.3473962 + \frac{105.91474}{x_2 + 23.906359} + \frac{0.29631923}{x_2}$$

Exact saved expression (use this for coefficient checks):

`log(((0.29234358235613145 / ((23.506779988992626 / x2) + -0.5984839634326817)) + x1) * x0) + (((0.29631922869012306 / x2) + -3.347396234657772) - (-105.91474168389178 / (x2 + 23.906359069522427)))`

Ideal gas | seed 10005

Accepted target: $\log P = c_0 + \log x_0 + \log x_1 - \log x_2$

$(x_0, x_1, x_2) = (n, T, V)$

Dataset columns: n, T, V, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 276 restarts; actual search 3599.2 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 8; training R2 = 0.9958975; native loss = 0.066443048.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2}\right) + 2.0594607$$

Exact saved expression (use this for coefficient checks):

`2.0594607489650163 + log((x0 * x1) / x2)`

Review rationale: Equals $\log(x_0 \cdot x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 16; complexity 26; training R2 = 0.99786554; native loss = 0.046955.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2}\right) + 2.10501 - \frac{0.22820818}{\frac{x_0 \cdot x_2}{x_0} - 0.12690393 \cdot x_1} - \frac{0.264774}{x_2 - 2.1054243}$$

Exact saved expression (use this for coefficient checks):

`((-0.22820818475650273 / ((-0.1269039271461884 * x1) + ((x2 / x0) * x0))) + 2.1050100032527204) + (log((x0 / x2) * x1) - (0.2647739950805346 / (x2 - 2.1054243415590164)))`

Evolved PySR (709715)

Archived/audited label: exact. 134 restarts; actual search 3600.1 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 6; training R2 = -1.4029727; native loss = 0.059460259.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2}\right)$$

Exact saved expression (use this for coefficient checks):

`log((x0 * x1) / x2)`

Review rationale: Equals $\log(x_0 \cdot x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 26; training R2 = 0.99717202; native loss = 0.043687361.

$$\ln(x_1) - \ln\left(\frac{0.11947619 \cdot x_2}{x_0}\right) - \frac{0.040038463}{\left(-\frac{0.12153618 \cdot x_1}{x_2} + 0.98798833\right) \cdot \ln(x_0)} - \frac{0.35288974}{x_2}$$

Exact saved expression (use this for coefficient checks):

`(log(x1) - ((0.3528897399193215 / x2) - (-0.040038462973233865 / (log(x0) * (0.9879883319482564 - ((x1 / x2) * 0.12153617956871549)))))) - log((x2 / x0) * 0.11947619346327111)`

Ideal gas | seed 10006

Accepted target: $\log P = c_0 + \log x_0 + \log x_1 - \log x_2$

$$(x_0, x_1, x_2) = (n, T, V)$$

Dataset columns: n, T, V, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 275 restarts; actual search 3600.2 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 8; training R2 = 0.99598282; native loss = 0.066443048.

$$2.0636689 - \ln\left(\frac{x_2}{x_0 \cdot x_1}\right)$$

Exact saved expression (use this for coefficient checks):

$$2.0636688881161622 - \log((x_2 / x_1) / x_0)$$

Review rationale: Equals $\log(x_0 \cdot x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 13; complexity 20; training R2 = 0.99780554; native loss = 0.047228469.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2 + 0.5806093}\right) + 2.1146863 + \frac{0.075071049}{\frac{x_1}{x_2 - 0.43637114} - 8.1839896}$$

Exact saved expression (use this for coefficient checks):

$$(\log((x_0 * x_1) / (x_2 - -0.5806092981604545))) - 2.1146863447546433) + (0.07507104905582071 / ((x_1 / (x_2 - 0.43637113944784056)) - 8.183989634129926))$$

Evolved PySR (709715)

Archived/audited label: exact. 133 restarts; actual search 3601.0 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 6; training R2 = -1.4029727; native loss = 0.059460259.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2}\right)$$

Exact saved expression (use this for coefficient checks):

$$\log((x_0 * x_1) / x_2)$$

Review rationale: Equals $\log(x_0 \cdot x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 16; complexity 25; training R2 = 0.99814716; native loss = 0.042999507.

$$-\ln\left(0.67451788 \cdot x_2 \cdot \frac{1}{x_0 \cdot x_1}\right) + 1.6998519 + \frac{0.97420923}{-0.05521608 \cdot x_0 \cdot x_2 + 127.08137} - \frac{9.5608715}{x_2^3}$$

Exact saved expression (use this for coefficient checks):

$$((((-9.560871516044013 / \text{cube}(x_2)) + (0.9742092259691698 / (127.08137182144134 - ((x_0 * x_2) * 0.05521607992142216)))) - \log((x_2 * 0.6745178839571563) / (x_0 * x_1))) - -1.6998519125778149$$

Ideal gas | seed 10007

Accepted target: $\log P = c_0 + \log x_0 + \log x_1 - \log x_2$

$(x_0, x_1, x_2) = (n, T, V)$

Dataset columns: n, T, V, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 278 restarts; actual search 3600.2 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 8; training R2 = 0.99594666; native loss = 0.066443048.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2}\right) + 2.0617655$$

Exact saved expression (use this for coefficient checks):

`2.0617655008861164 + log((x0 / x2) * x1)`

Review rationale: Equals $\log(x_0 \cdot x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 16; complexity 22; training R2 = 0.99753331; native loss = 0.046710505.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2 + 0.63010069}\right) + 2.090001 + \frac{0.097130957}{x_2 - (x_0 - \sqrt{\sqrt{x_1}}) + 0.65444281}$$

Exact saved expression (use this for coefficient checks):

`(0.09713095671540735 / ((x2 - -0.6544428058653493) - (x0 - sqrt(sqrt(x1))))) + (log((x0 * x1) / (x2 - -0.6301006863136727))) + 2.090000998043819`

Evolved PySR (709715)

Archived/audited label: exact. 135 restarts; actual search 3600.9 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 6; training R2 = -1.4029727; native loss = 0.059460259.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2}\right)$$

Exact saved expression (use this for coefficient checks):

`log((x0 * x1) / x2)`

Review rationale: Equals $\log(x_0 \cdot x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 27; training R2 = 0.99799698; native loss = 0.043639836.

$$\ln\left(\frac{x_1}{0.006132642 + \frac{x_2}{x_0}}\right) + 2.0928356 + \frac{0.09831491}{x_0 - \left(\ln(x_2) + \frac{x_1}{\sqrt{x_0}}\right)}$$

Exact saved expression (use this for coefficient checks):

`((log(x1 / ((x2 / x0) + 0.006132642006360095)) * cube(x1 / x1)) - (-0.09831491037969789 / (x0 - (log(x2) + (x1 / sqrt(x0))))) - -2.0928355519715223`

Ideal gas | seed 10008

Accepted target: $\log P = c_0 + \log x_0 + \log x_1 - \log x_2$

$(x_0, x_1, x_2) = (n, T, V)$

Dataset columns: n, T, V, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 278 restarts; actual search 3599.7 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 8; training R2 = 0.99592612; native loss = 0.066443048.

$$\ln\left(\frac{x_0}{\frac{1}{x_1} \cdot x_2}\right) + 2.0607687$$

Exact saved expression (use this for coefficient checks):

`log(x0 / (x2 / x1)) + 2.060768707760328`

Review rationale: Equals $\log(x_0 \cdot x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 13; complexity 30; training R2 = 0.99814456; native loss = 0.042844992.

$$0.003217211 \cdot x_1 + \ln(x_0) + \frac{4.43307}{2.0477376 \cdot \sqrt[3]{0.056872565 \cdot x_2 + 1} - 1.3169419} - \frac{2.1361028}{\left(x_0 + \frac{x_2}{x_0}\right) \cdot (x_0 - x_2 - 17.381959)}$$

Exact saved expression (use this for coefficient checks):

`((log(x0) + (x1 * 0.0032172109655069833)) + (4.433069960837305 / (sqrt(sqrt(x2 + 17.58317023595703)) + -1.3169419195802052))) - (2.1361027758835336 / ((x0 + (x2 / x0)) * (x0 - (x2 + 17.381959178193533)))))`

Evolved PySR (709715)

Archived/audited label: exact. 134 restarts; actual search 3600.6 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 6; training R2 = -1.4029727; native loss = 0.059460259.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2}\right)$$

Exact saved expression (use this for coefficient checks):

`log((x0 * x1) / x2)`

Review rationale: Equals $\log(x_0 \cdot x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 14; complexity 20; training R2 = 0.99788995; native loss = 0.045127788.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2 + 0.62693205}\right) + 2.1203403 - \frac{0.46082812}{-0.12294855 \cdot x_1 + x_2 + 0.038230772}$$

Exact saved expression (use this for coefficient checks):

`(log(x0 * (x1 / (x2 + 0.6269320500872523)))) - (0.46082812137211204 / (x2 + ((x1 + -0.3109493595849469) * -0.12294854859966388)))) + 2.120340308764076`

Ideal gas | seed 10009

Accepted target: $\log P = c_0 + \log x_0 + \log x_1 - \log x_2$

$$(x_0, x_1, x_2) = (n, T, V)$$

Dataset columns: n, T, V, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 279 restarts; actual search 3599.7 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 8; training R2 = 0.99577435; native loss = 0.066443048.

$$2.0545942 - \ln\left(\frac{x_2}{x_0 \cdot x_1}\right)$$

Exact saved expression (use this for coefficient checks):

$$2.054594231225709 - \log(x_2 / (x_0 * x_1))$$

Review rationale: Equals $\log(x_0 \cdot x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 22; training R2 = 0.99782721; native loss = 0.047026531.

$$\ln\left(\frac{8.3002644 \cdot x_0 \cdot x_1}{x_2 + 0.6304592}\right) + \frac{0.60018817}{\frac{8.058772 \cdot x_1}{x_2} - 65.3414}$$

Exact saved expression (use this for coefficient checks):

$$\log(((x_0 * x_1) / (x_2 + 0.6304592022980495)) * 8.300264454816713) + (0.6001881746991273 / (((x_1 / x_2) * 8.05877198215021) + 0.6307138206342439) - 65.97211341318469))$$

Evolved PySR (709715)

Archived/audited label: exact. 135 restarts; actual search 3600.4 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 6; training R2 = -1.4029727; native loss = 0.059460259.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2}\right)$$

Exact saved expression (use this for coefficient checks):

$$\log((x_0 * x_1) / x_2)$$

Review rationale: Equals $\log(x_0 \cdot x_1 / x_2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 12; complexity 18; training R2 = 0.99750393; native loss = 0.045217945.

$$\ln\left(\frac{x_0 \cdot x_1}{x_2 + 0.63731919}\right) + 2.0956361 + \frac{0.43800957}{0.1241102 \cdot x_1 - x_2}$$

Exact saved expression (use this for coefficient checks):

$$\log((x_1 / (0.6373191861082149 + x_2)) * x_0) + (2.0956360979884754 - (-0.43800957334291907 / ((0.12411020124781307 * x_1) - x_2)))$$

Kepler | seed 10000

Accepted target: $P = cx_0^{3/2}$

$x_0 = a$

Dataset columns: a, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 343 restarts; actual search 3600.4 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 5; training R2 = 0.99999637; native loss = 4.1765893.

$$366.86747 \cdot x_0^{\frac{3}{2}}$$

Exact saved expression (use this for coefficient checks):

`cube(sqrt(x0)) * 366.8674702618343`

Review rationale: Proportional to $x_0^{3/2}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 19; complexity 28; training R2 = 1; native loss = 0.0066531371.

$$x_0^{\frac{3}{2}} \cdot \left(\frac{\sqrt[4]{x_0}}{0.11837407 \cdot x_0^3 + 0.33256294 + \frac{0.020915285}{x_0^3}} + 0.35304849 \cdot x_0 + 363.51775 \right) - 0.73715061$$

Exact saved expression (use this for coefficient checks):

`(((((x0 * 0.3530484898570999) + 363.5177487331747) + (sqrt(sqrt(x0)) / (((cube(x0) * 0.11837407067011722) - (-0.020915284959690354 / cube(x0))) + 0.33256293955171573)))) * x0) * sqrt(x0)) + -0.7371506084428843`

Evolved PySR (709715)

Archived/audited label: exact. 155 restarts; actual search 3600.3 s. Archived review / positive-equation audit

Recovery witness

Frontier index 2; complexity 3; training R2 = -0.49300262; native loss = 0.0037158111.

$$\sqrt{x_0^3}$$

Exact saved expression (use this for coefficient checks):

`sqrt(cube(x0))`

Review rationale: Proportional to $x_0^{3/2}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 26; training R2 = 1; native loss = 1.1849504e-07.

$$0.69068993 \cdot \sqrt{2} \cdot x_0 \cdot (0.45345889 \cdot x_0 + 370.61333) \cdot \sqrt{x_0 + 0.041926686} - 4.7209253 + \frac{0.0011185775}{0.71708219 - x_0}$$

Exact saved expression (use this for coefficient checks):

`((((370.61333398151 + (x0 * 0.45345888559088765)) * (sqrt((0.08385337135148377 + x0) + x0) * (x0 / 1.4478276783927244)))) + (-4.7002587906681095 + (0.0011185774945241004 / (0.7170821919494348 - x0)))) + -0.020666541254374324`

Kepler | seed 10001

Accepted target: $P = cx_0^{3/2}$

$x_0 = a$

Dataset columns: a, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 342 restarts; actual search 3599.7 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 5; training R2 = 0.99999637; native loss = 4.17659.

$$366.86747 \cdot x_0^{\frac{3}{2}}$$

Exact saved expression (use this for coefficient checks):

`cube(sqrt(x0 * 51.2475140422566))`

Review rationale: Proportional to $x_0^{3/2}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 20; complexity 30; training R2 = 1; native loss = 0.001451077.

$$\sqrt{x_0} \cdot \left(365.71032 \cdot x_0 - \frac{0.025524856}{x_0^2 - x_0 - 0.096195265} \right) - x_0 + 0.0031624283 \cdot e^{x_0} + 0.83036742 - \frac{0.56467233}{x_0}$$

Exact saved expression (use this for coefficient checks):

`(sqrt(x0) * ((x0 * 365.7103171242234) + (-0.02552485560220043 / ((square(x0) + -0.09619526457305054) - x0)))) + (((exp(x0) * 0.003162428333422299) - (x0 + -1.709675987125125)) + -0.8793085695125208) + (-0.5646723270961623 / x0))`

Evolved PySR (709715)

Archived/audited label: exact. 155 restarts; actual search 3600.9 s. Archived review / positive-equation audit

Recovery witness

Frontier index 2; complexity 3; training R2 = -0.49300262; native loss = 0.0037158111.

$$\sqrt{x_0^3}$$

Exact saved expression (use this for coefficient checks):

`sqrt(cube(x0))`

Review rationale: Proportional to $x_0^{3/2}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 16; complexity 20; training R2 = 1; native loss = 4.0422486e-07.

$$0.10017433 \cdot x_0^3 + 363.51622 \cdot (x_0 + 0.0093716667)^{\frac{3}{2}} - 3.634894 + \frac{0.001319306}{x_0 - 0.99093038}$$

Exact saved expression (use this for coefficient checks):

`((cube(x0) * 0.100174325179032) + (((3.6292906548117065e-6 / (x0 + -0.9909303799314183)) + cube(sqrt(x0 + 0.009371666745819093))) / 0.0027509088611274305)) + -3.6348939671798637`

Kepler | seed 10002

Accepted target: $P = cx_0^{3/2}$

$x_0 = a$

Dataset columns: a, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 342 restarts; actual search 3600.1 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 5; training R2 = 0.99999637; native loss = 4.1765893.

$$366.86747 \cdot \sqrt{x_0^3}$$

Exact saved expression (use this for coefficient checks):

`sqrt(cube(x0)) * 366.8674702693471`

Review rationale: Proportional to $x_0^{(3/2)}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 20; training R2 = 1; native loss = 0.0019334529.

$$-x_0^{\frac{3}{2}} \cdot \left(\frac{2.0128663 \cdot x_0^3}{e^{x_0} + 1.1776786} - 367.03897 \right) - 1.2674094$$

Exact saved expression (use this for coefficient checks):

`-0.11105293555253244 - (((sqrt(x0) * ((cube(x0 * 1.2626170394612686) / (exp(x0) + 1.1776786230408334)) + -367.03897040960635)) * x0) + 1.1563564725715694)`

Evolved PySR (709715)

Archived/audited label: exact. 155 restarts; actual search 3600.9 s. Archived review / positive-equation audit

Recovery witness

Frontier index 2; complexity 3; training R2 = -0.49300262; native loss = 0.0037158111.

$$\sqrt{x_0^3}$$

Exact saved expression (use this for coefficient checks):

`sqrt(cube(x0))`

Review rationale: Proportional to $x_0^{(3/2)}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 14; complexity 20; training R2 = 1; native loss = 3.3490161e-07.

$$9.6152356 \cdot x_0 \cdot e^{-0.19806989 \cdot x_0^2} + 371.06025 \cdot (\sqrt{x_0} - 0.011565022)^3 - 0.98558931$$

Exact saved expression (use this for coefficient checks):

`(exp((log(x0) - square(x0 / 2.246936331394049)) + 2.263348876503449) + cube(sqrt(x0) + -0.011565021716440246) * 7.185905099155586)) - 0.9855893106175418`

Kepler | seed 10003

Accepted target: $P = cx_0^{3/2}$

$x_0 = a$

Dataset columns: a, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 342 restarts; actual search 3600.3 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 5; training R2 = 0.99999637; native loss = 4.17659.

$$366.86747 \cdot \sqrt{x_0^3}$$

Exact saved expression (use this for coefficient checks):

`sqrt(cube(x0)) * 366.8674699291666`

Review rationale: Proportional to $x_0^{3/2}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 30; training R2 = 1; native loss = 0.0012821522.

$$x_0 \cdot \left(367.93175 \cdot \sqrt{x_0} - 1.6443384 \cdot (x_0 - 1.4932724) \cdot e^{(-x_0 + \ln(x_0^2 - 0.13644231) + 0.9476381)^3} - 4.2575122 \right) + x_0 - 0.22899521$$

Exact saved expression (use this for coefficient checks):

`(x0 + (((sqrt(x0) * 367.9317512019038) - (4.257512204071257 - (((x0 + -1.4932724100031833) * exp(cube(log(square(x0) + -0.1364423078042478) + (0.947638098630608 - x0)))) * -1.6443384265947172))) * x0)) + -0.22899521206646078`

Evolved PySR (709715)

Archived/audited label: exact. 158 restarts; actual search 3600.3 s. Archived review / positive-equation audit

Recovery witness

Frontier index 2; complexity 3; training R2 = -0.49300262; native loss = 0.0037158111.

$$\sqrt{x_0^3}$$

Exact saved expression (use this for coefficient checks):

`sqrt(cube(x0))`

Review rationale: Proportional to $x_0^{3/2}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 21; complexity 28; training R2 = 1; native loss = 3.1594154e-08.

$$0.10014621 \cdot x_0^3 + 363.5184 \cdot \sqrt{(x_0 + 0.0093607815)^3} - 3.6283966 + \frac{0.082044548}{-(0.009350602 - 33.789447 \cdot x_0) \cdot \ln(x_0^3) + 0.55188439}$$

Exact saved expression (use this for coefficient checks):

`((sqrt(cube((x0 - -0.009360781522434653) / 0.019632807281780976)) + (0.08204454777902448 / (0.5518843873811378 - (log(cube(x0)) * ((x0 * -33.78944656667474) - -0.009350601986899297)))))) + -3.628396600466599) + (cube(x0) * 0.10014620590685666)`

Kepler | seed 10004

Accepted target: $P = cx_0^{3/2}$

$x_0 = a$

Dataset columns: a, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 344 restarts; actual search 3600.1 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 5; training R2 = 0.99999637; native loss = 4.176593.

$$366.86747 \cdot x_0^{\frac{3}{2}}$$

Exact saved expression (use this for coefficient checks):

`cube(sqrt(x0)) * 366.86746864857804`

Review rationale: Proportional to $x_0^{3/2}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 16; complexity 22; training R2 = 1; native loss = 0.0015505047.

$$x_0^{\frac{3}{2}} \cdot \left(-\frac{x_0^3}{0.035270797 \cdot x_0 + 0.4959834 \cdot e^{x_0} + 0.53310561} + 367.03916 \right) - 1.267113$$

Exact saved expression (use this for coefficient checks):

`((367.03916167327765 - (cube(x0) / (exp(x0 + -0.7012128210685734) + (0.5331056069085585 + (x0 * 0.035270796579794116)))))) * (sqrt(x0) * x0)) + -1.2671129687521936`

Evolved PySR (709715)

Archived/audited label: exact. 158 restarts; actual search 3600.2 s. Archived review / positive-equation audit

Recovery witness

Frontier index 2; complexity 3; training R2 = -0.49300262; native loss = 0.0037158111.

$$\sqrt{x_0^3}$$

Exact saved expression (use this for coefficient checks):

`sqrt(cube(x0))`

Review rationale: Proportional to $x_0^{3/2}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 27; training R2 = 1; native loss = 8.8283645e-08.

$$371.62895 \cdot x_0 \cdot \left(\sqrt{x_0} - 0.039430208 + \frac{0.041067434 + \frac{0.021545806}{x_0}}{-0.03784736 + \frac{e^{x_0}}{x_0}} \right) - 0.40318473$$

Exact saved expression (use this for coefficient checks):

`(x0 + ((((((0.02154580582027296 / x0) + 0.04106743385655004) / ((exp(x0) / x0) + -0.03784735976369429)) + (-0.039430208427600665 + sqrt(x0))) / (0.00269085604258507 / x0))) - (x0 + 0.40318472734188493)`

Kepler | seed 10005

Accepted target: $P = cx_0^{3/2}$

$x_0 = a$

Dataset columns: a, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 343 restarts; actual search 3600.1 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 5; training R2 = 0.99999637; native loss = 4.1765901.

$$366.86747 \cdot x_0^{\frac{3}{2}}$$

Exact saved expression (use this for coefficient checks):

`cube(sqrt(x0)) * 366.8674698836498`

Review rationale: Proportional to $x_0^{3/2}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 27; training R2 = 1; native loss = 0.0038191095.

$$366.91647 \cdot x_0^{\frac{3}{2}} - \frac{1.0015171 \cdot (x_0^3 + x_0)^3}{x_0 \cdot ((x_0 + e^{x_0})^2 + 0.094829018) - 0.94432174} - 1.0496921$$

Exact saved expression (use this for coefficient checks):

`(cube(sqrt(x0)) * 7.1590556165126635) - (1.0015170999788907 / (((square(exp(x0) + x0) + 0.09482901826213616) * x0) - 0.9443217425644599) / cube(x0 + cube(x0))))) + -1.049692090722572`

Evolved PySR (709715)

Archived/audited label: exact. 151 restarts; actual search 3600.7 s. Archived review / positive-equation audit

Recovery witness

Frontier index 2; complexity 3; training R2 = -0.49300262; native loss = 0.0037158111.

$$\sqrt{x_0^3}$$

Exact saved expression (use this for coefficient checks):

`sqrt(cube(x0))`

Review rationale: Proportional to $x_0^{3/2}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 19; training R2 = 1; native loss = 4.6424228e-07.

$$371.62967 \cdot x_0 \cdot (\sqrt{x_0} + 0.041523855 \cdot (x_0 + 0.53885314) \cdot e^{-x_0} - 0.039415117) - 0.47029565$$

Exact saved expression (use this for coefficient checks):

`((sqrt(x0) - (0.03941511718663358 - ((x0 + 0.5388531429222948) * 0.04152385503237377) / exp(x0)))) / 0.0026908508189881124) * x0 + -0.4702956461090487`

Kepler | seed 10006

Accepted target: $P = cx_0^{3/2}$

$x_0 = a$

Dataset columns: a, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 339 restarts; actual search 3599.7 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 5; training R2 = 0.99999637; native loss = 4.1765907.

$$366.86747 \cdot \sqrt{x_0^3}$$

Exact saved expression (use this for coefficient checks):

`sqrt(cube(x0)) * 366.86746965485463`

Review rationale: Proportional to $x_0^{3/2}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 20; complexity 29; training R2 = 1; native loss = 0.001315255.

$$x_0^{\frac{3}{2}} \cdot (0.025221928 \cdot x_0^2 + 364.46436) + 1.3269083 \cdot \ln\left(x_0 + \frac{0.0027437539 \cdot x_0}{1.0200488 - x_0^2}\right) + 0.59022913$$

Exact saved expression (use this for coefficient checks):

`((log(((x0 / (1.0200487929030835 - square(x0)))) / 364.46417418800075) + x0) / 0.7536315888953226) + 0.5902291262752878) + (cube(sqrt(x0)) * (((x0 * 9.272849684735906) / (367.6503085569789 / x0)) + 364.4643624276532))`

Evolved PySR (709715)

Archived/audited label: exact. 157 restarts; actual search 3601.3 s. Archived review / positive-equation audit

Recovery witness

Frontier index 2; complexity 3; training R2 = -0.49300262; native loss = 0.0037158111.

$$\sqrt{x_0^3}$$

Exact saved expression (use this for coefficient checks):

`sqrt(cube(x0))`

Review rationale: Proportional to $x_0^{3/2}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 21; complexity 30; training R2 = 1; native loss = 1.2619342e-07.

$$365.36086 \cdot x_0^{\frac{3}{2}} - \frac{\sqrt{x_0} \cdot (0.14664744 - x_0)}{(x_0^2)^2 + \frac{0.20834028}{x_0^4}} - \frac{x_0}{x_0 - 9.7208157} - 0.92923326$$

Exact saved expression (use this for coefficient checks):

`cube(sqrt(x0) / 0.13988119258327478) + (((-0.9292332626276243 - ((0.14664743592749746 - x0) * (sqrt(x0) / (square(square(x0)) - (-0.20834027818275722 / square(square(x0)))))) - (x0 / (x0 + -9.720815727012281)))`

Kepler | seed 10007

Accepted target: $P = cx_0^{3/2}$

$x_0 = a$

Dataset columns: a, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 313 restarts; actual search 3600.2 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 5; training R2 = 0.99999637; native loss = 4.17659.

$$366.86747 \cdot x_0^{\frac{3}{2}}$$

Exact saved expression (use this for coefficient checks):

`cube(sqrt(x0)) * 366.86746993828194`

Review rationale: Proportional to $x_0^{3/2}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 14; complexity 20; training R2 = 1; native loss = 0.0013773897.

$$x_0^{\frac{3}{2}} \cdot \left(-\frac{2.0129716 \cdot x_0^3}{e^{x_0} + 1.1827375} + 367.03906 \right) - 1.2696519$$

Exact saved expression (use this for coefficient checks):

`(((((square(x0) * ((x0 / (exp(x0) + 1.1827375377178455)) / -0.49677799326390826)) + 367.03905513716865) * x0) * sqrt(x0)) - 1.2696518685369946`

Evolved PySR (709715)

Archived/audited label: exact. 158 restarts; actual search 3600.9 s. Archived review / positive-equation audit

Recovery witness

Frontier index 2; complexity 3; training R2 = -0.49300262; native loss = 0.0037158111.

$$\sqrt{x_0^3}$$

Exact saved expression (use this for coefficient checks):

`sqrt(cube(x0))`

Review rationale: Proportional to $x_0^{3/2}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 21; complexity 26; training R2 = 1; native loss = 6.8172048e-08.

$$0.10015138 \cdot x_0^3 + 363.51474 \cdot \sqrt{(x_0 + 0.0093638266)^3} - 3.6097127 - \frac{0.0015325626}{-x_0 \cdot x_0 + 0.98969355}$$

Exact saved expression (use this for coefficient checks):

`(sqrt(cube(x0 + 0.00936382655742195)) / 0.002750920101246877) + ((((-0.04898964243605843 / (0.9995968259003173 - ((x0 * x0) + 0.009903272860286083))) + cube(x0 * 1.4738309090635482)) * 0.03128340056174997) + -3.6097126742759764)`

Kepler | seed 10008

Accepted target: $P = cx_0^{3/2}$

$x_0 = a$

Dataset columns: a, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 310 restarts; actual search 3600.6 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 5; training R2 = 0.99999637; native loss = 4.176589.

$$366.86747 \cdot \sqrt{x_0^3}$$

Exact saved expression (use this for coefficient checks):

`sqrt(cube(x0)) * 366.8674703621283`

Review rationale: Proportional to $x_0^{3/2}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 24; training R2 = 1; native loss = 0.0046449745.

$$365.33525 \cdot x_0^{\frac{3}{2}} + 0.0033338044 \cdot e^{x_0} + \frac{0.42817729}{-x_0 + \frac{0.80607055}{x_0^2 - \frac{1.2267867}{x_0^2}}}$$

Exact saved expression (use this for coefficient checks):

`((exp(x0) / 299.95760699047196) - cube(sqrt(x0) * -7.148756864453367)) + (0.42817729073491656 / ((0.8060705520546538 / (square(x0) - square(1.1076040587316704 / x0)))) - x0))`

Evolved PySR (709715)

Archived/audited label: exact. 158 restarts; actual search 3600.5 s. Archived review / positive-equation audit

Recovery witness

Frontier index 2; complexity 3; training R2 = -0.49300262; native loss = 0.0037158111.

$$\sqrt{x_0^3}$$

Exact saved expression (use this for coefficient checks):

`sqrt(cube(x0))`

Review rationale: Proportional to $x_0^{3/2}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 23; complexity 29; training R2 = 1; native loss = 4.9616071e-08.

$$\sqrt{x_0} \cdot (0.024974179 \cdot x_0^3 + 364.51162 \cdot x_0) + \ln(2 \cdot x_0 - 0.26650141) + \frac{0.018398352}{\ln(x_0 \cdot x_0 + 0.13125497)}$$

Exact saved expression (use this for coefficient checks):

`(sqrt(x0) * ((x0 * 364.51161839597916) + cube(x0 * 0.2923010696724383))) + (log((x0 + x0) + -0.26650141202633165) - (-0.018398352307147037 / log(((x0 * x0) + -0.016254792864871817) - -0.14750976229665758))))`

Kepler | seed 10009

Accepted target: $P = cx_0^{3/2}$

$x_0 = a$

Dataset columns: a, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 310 restarts; actual search 3600.0 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 5; training R2 = 0.99999637; native loss = 4.1765891.

$$366.86747 \cdot x_0^{\frac{3}{2}}$$

Exact saved expression (use this for coefficient checks):

`cube(sqrt(x0 * 51.247514078332024))`

Review rationale: Proportional to $x_0^{3/2}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 19; complexity 27; training R2 = 1; native loss = 0.0032399362.

$$x_0^{\frac{3}{2}} \cdot \left(2.3338812 \cdot x_0^2 \cdot \left(-x_0 + \frac{x_0}{1.0854901 \cdot x_0 + 1.5734318} \right) \cdot e^{-x_0} + 367.04699 \right) - 1.2684264$$

Exact saved expression (use this for coefficient checks):

`(sqrt(x0) * (x0 * (367.04698786433335 - (x0 / (exp(x0) * (-0.4284708293057617 / (((x0 / ((x0 - -1.4495127843730917) * 1.0854901169587168)) - x0) * x0)))))) + -1.2684263509150486`

Evolved PySR (709715)

Archived/audited label: exact. 165 restarts; actual search 3600.7 s. Archived review / positive-equation audit

Recovery witness

Frontier index 2; complexity 3; training R2 = -0.49300262; native loss = 0.0037158111.

$$\sqrt{x_0^3}$$

Exact saved expression (use this for coefficient checks):

`sqrt(cube(x0))`

Review rationale: Proportional to $x_0^{3/2}$ on the physical positive domain.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 22; training R2 = 1; native loss = 5.8222241e-07.

$$-\frac{2.0129984 \cdot x_0^{\frac{9}{2}}}{e^{x_0} + 1.1793077} + 367.03914 \cdot x_0^{\frac{3}{2}} - 1.2745791$$

Exact saved expression (use this for coefficient checks):

`(cube(sqrt(x0 / 0.01950705662609561)) + -1.2745790962716785) - ((cube(sqrt(x0) * (x0 * 0.4047218130567444)) / 0.0329326296431404) / (exp(x0) + 1.1793077159491991))`

Leavitt | seed 10000

Accepted target: $M = c_0 + c_1 x_0$

$$x_0 = \log_{10} P$$

Dataset columns: logP, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 270 restarts; actual search 3600.0 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.66823696; native loss = 0.35834894.

$$14.902677 - x_0$$

Exact saved expression (use this for coefficient checks):

14.90267740229158 - x0

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 16; complexity 29; training R2 = 0.98487879; native loss = 0.07593059.

$$x_0 \cdot (0.62048532 \cdot x_0 - 1.6895856) \cdot \left(x_0 \cdot x_0 + \frac{0.001405051 \cdot (1.6704152 - \sqrt[3]{x_0})}{x_0 \cdot x_0 - 0.64048874} + 0.030224472 \right) + 14.822698$$

Exact saved expression (use this for coefficient checks):

((((x0 * 0.6204853197553085) - 1.6895856135776224) * (x0 * (((1.6704151763701582 - sqrt(sqrt(x0))) * (0.0014050510047298212 / ((x0 * x0) + -0.640488741989544)))) + ((x0 * x0) + 0.030224472273476324)))) + 14.822698289140915

Evolved PySR (709715)

Archived/audited label: exact. 148 restarts; actual search 3600.5 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -176.33882; native loss = 0.25949975.

$$x_0$$

Exact saved expression (use this for coefficient checks):

x0

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 8; complexity 9; training R2 = 0.97789822; native loss = 0.14297627.

$$14.962836 - 3.6525779 \cdot e^{-4.226581 \cdot (1 - 0.48641326 \cdot x_0)^2}$$

Exact saved expression (use this for coefficient checks):

14.962835581051285 - (3.652577876818282 / exp(square(2.0558650280790602 - x0)))

Leavitt | seed 10001

Accepted target: $M = c_0 + c_1 x_0$

$$x_0 = \log_{10} P$$

Dataset columns: logP, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 269 restarts; actual search 3600.2 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.66893186; native loss = 0.35834894.

$$14.900747 - x_0$$

Exact saved expression (use this for coefficient checks):

$$14.900746910985486 - x_0$$

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 14; complexity 27; training R2 = 0.98417145; native loss = 0.076023174.

$$\frac{x_0^3 \cdot (x_0 - 2.7129991)}{1.5840809 - \frac{0.18011271 \cdot e^{-x_0}}{(x_0 + x_0)^2 - \left((x_0^2)^3\right)^3}} + 14.800203$$

Exact saved expression (use this for coefficient checks):

$$\begin{aligned} & (\text{cube}(x_0) * ((x_0 - 2.7129990731270475) / (1.5840809113588499 + ((\exp(0.2835492170193592 \\ & - x_0) / (\text{square}(x_0 + x_0) - \text{cube}(\text{cube}(\text{square}(x_0))))) / -7.372240717029471)))) + \\ & 14.800202511400853 \end{aligned}$$

Evolved PySR (709715)

Archived/audited label: exact. 118 restarts; actual search 3600.6 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -176.33882; native loss = 0.25949975.

$$x_0$$

Exact saved expression (use this for coefficient checks):

$$x_0$$

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 8; complexity 9; training R2 = 0.97134505; native loss = 0.14127212.

$$4.0507895 \cdot (0.49685557 \cdot x_0 - 1)^2 \cdot e^{x_0} + 10.913712$$

Exact saved expression (use this for coefficient checks):

$$(\text{square}(x_0 - 2.0126573309998665) * \exp(x_0)) + 10.913712361450305$$

Leavitt | seed 10002

Accepted target: $M = c_0 + c_1 x_0$

$$x_0 = \log_{10} P$$

Dataset columns: logP, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 275 restarts; actual search 3600.4 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.66830983; native loss = 0.35834894.

$$14.902476 - x_0$$

Exact saved expression (use this for coefficient checks):

$$14.902475953448175 - x_0$$

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 16; complexity 28; training R2 = 0.98493422; native loss = 0.076328792.

$$0.62838937 \cdot x_0^3 \cdot \left(x_0 + \frac{0.012992416 \cdot x_0}{-x_0 - x_0 \cdot x_0 + 1.4363718} - 2.7127342 \right) + 14.814943$$

Exact saved expression (use this for coefficient checks):

$$\begin{aligned} & (((((x_0 + -2.712734219426612) * x_0) / x_0) + (x_0 * ((-0.01085581157933845 / \\ & ((1.4363718257250693 - (x_0 * x_0)) - x_0)) / -0.8355499033292579))) * (cube(x_0) * \\ & 0.6283893657524448)) + 14.814942790072822 \end{aligned}$$

Evolved PySR (709715)

Archived/audited label: exact. 118 restarts; actual search 3601.4 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -176.33882; native loss = 0.25949975.

$$x_0$$

Exact saved expression (use this for coefficient checks):

$$x_0$$

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 9; complexity 11; training R2 = 0.97573787; native loss = 0.1406634.

$$15.651167 \cdot e^{\frac{7.4658102}{x_0^2 - 4.914238}} + 11.425192$$

Exact saved expression (use this for coefficient checks):

$$\exp((7.46581015522699 / (\text{square}(x_0) + -4.914237972899847))) + 2.750545464981574) - -11.425192211192607$$

Leavitt | seed 10003

Accepted target: $M = c_0 + c_1 x_0$

$$x_0 = \log_{10} P$$

Dataset columns: logP, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 273 restarts; actual search 3600.2 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.66896034; native loss = 0.35834894.

$$14.900667 - x_0$$

Exact saved expression (use this for coefficient checks):

14.900667340044148 - x0

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 17; complexity 27; training R2 = 0.98495386; native loss = 0.076847939.

$$1.5367503 \cdot (x_0 - 0.17941908)^2 \cdot \left(0.022011118 \cdot x_0^4 - 1.061239 - \frac{0.0030621922}{9.8297606 \cdot (0.93364623 \cdot x_0 - 1)^3 + 0.16101365} \right) + 14.808822$$

Exact saved expression (use this for coefficient checks):

14.808822039500072 - ((((((-0.13912024954117858 / (cube((x0 + x0) + -2.1421389947297502) + 0.16101365494943445)) + square(square(x0))) * 0.02201111765972864) + -1.06123898401504) * -1.53675034522516) * square(x0 + -0.17941908225646547))

Evolved PySR (709715)

Archived/audited label: exact. 115 restarts; actual search 3601.1 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -176.33882; native loss = 0.25949975.

$$x_0$$

Exact saved expression (use this for coefficient checks):

x0

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 11; complexity 12; training R2 = 0.97943704; native loss = 0.14111668.

$$12.095418 \cdot \left(0.43456922 \cdot e^{0.20505188 \cdot x_0^2} - 1 \right)^2 + 11.07174$$

Exact saved expression (use this for coefficient checks):

(square(exp(square(x0 / -2.2083511289395816)) - 2.301129383115155) + 4.847045240138608) * 2.284224693570454

Leavitt | seed 10004

Accepted target: $M = c_0 + c_1 x_0$

$$x_0 = \log_{10} P$$

Dataset columns: logP, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 273 restarts; actual search 3599.5 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.66905602; native loss = 0.35834894.

$$14.9004 - x_0$$

Exact saved expression (use this for coefficient checks):

14.900399730232301 - x0

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 14; complexity 20; training R2 = 0.98508823; native loss = 0.08143704.

$$14.925948 - 3.7396688 \cdot e^{-4.3013564 \cdot (1 - 0.48216678 \cdot x_0)^2} + \frac{0.011501573}{-x_0 \cdot e^{x_0} + 1.7584643}$$

Exact saved expression (use this for coefficient checks):

14.92594819643575 - (((14.951319865981583 / exp(square(2.0739711577769033 - x0) -
-1.3858025218571122)) - (0.011501573444984476 / (1.758464328291466 - (exp(x0) * x0)
)))

Evolved PySR (709715)

Archived/audited label: exact. 118 restarts; actual search 3600.6 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -176.33882; native loss = 0.25949975.

$$x_0$$

Exact saved expression (use this for coefficient checks):

x0

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 9; complexity 11; training R2 = 0.83544936; native loss = 0.14160677.

$$-x_0 \cdot x_0 + 13.98663 + \frac{1.2280589}{2.5623749 - x_0}$$

Exact saved expression (use this for coefficient checks):

(13.986629637445652 - (x0 * x0)) + (1.228058901107058 / (2.562374921146075 - x0))

Leavitt | seed 10005

Accepted target: $M = c_0 + c_1 x_0$

$$x_0 = \log_{10} P$$

Dataset columns: logP, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 273 restarts; actual search 3600.5 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.66882696; native loss = 0.35834894.

$$14.90104 - x_0$$

Exact saved expression (use this for coefficient checks):

14.90103971029106 - x0

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 11; complexity 21; training R2 = 0.98378315; native loss = 0.079236351.

$$0.0077477887 \cdot x_0^4 + 10.860367 + 3.9454639 \cdot e^{-0.32896414 \cdot x_0^3} + \frac{0.0015330219}{x_0 - 1.08786}$$

Exact saved expression (use this for coefficient checks):

(exp((cube(x0) + -4.172389589597803) / -3.0398450179271954) + 10.8603670435535) + ((square(square(x0)) + (0.19786573272840566 / (x0 - 1.087860017745279))) * 0.007747788682746546)

Evolved PySR (709715)

Archived/audited label: exact. 118 restarts; actual search 3600.3 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -176.33882; native loss = 0.25949975.

$$x_0$$

Exact saved expression (use this for coefficient checks):

x0

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 8; complexity 10; training R2 = 0.78507637; native loss = 0.1418949.

$$5.9821388 \cdot \left(0.4088573 \cdot e^{0.22217888 \cdot x_0^2} - 1\right)^2 + 12.336091$$

Exact saved expression (use this for coefficient checks):

square(exp(square(x0 * 0.47135854870925425)) + -2.445841130525825) + 12.336091009769746

Leavitt | seed 10006

Accepted target: $M = c_0 + c_1 x_0$

$x_0 = \log_{10} P$

Dataset columns: logP, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 313 restarts; actual search 3600.5 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.66854751; native loss = 0.35834894.

$$14.901817 - x_0$$

Exact saved expression (use this for coefficient checks):

14.90181730337909 - x0

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 14; complexity 26; training R2 = 0.98389978; native loss = 0.077158043.

$$-0.61305093 \cdot x_0^3 \cdot \left(-x_0 + \frac{0.10663322}{x_0^3 + \sqrt{x_0^3 - 0.17958843}} \right) - 1.6696738 \cdot x_0^3 + 14.801574$$

Exact saved expression (use this for coefficient checks):

(((((0.1066332166459657 / (sqrt(cube(x0)) + (cube(x0) - 0.17958842820082635)))) - x0) / -1.6311858425655053) * (x0 * square(x0))) + 14.80157415046847) + (cube(x0) * -1.6696737742177175)

Evolved PySR (709715)

Archived/audited label: exact. 118 restarts; actual search 3601.0 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -176.33882; native loss = 0.25949975.

$$x_0$$

Exact saved expression (use this for coefficient checks):

x0

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 8; complexity 9; training R2 = 0.97134505; native loss = 0.14127212.

$$4.0507896 \cdot (1 - 0.49685557 \cdot x_0)^2 \cdot e^{x_0} + 10.913712$$

Exact saved expression (use this for coefficient checks):

(exp(x0) * square(2.0126573361453812 - x0)) + 10.913712306342847

Leavitt | seed 10007

Accepted target: $M = c_0 + c_1 x_0$

$$x_0 = \log_{10} P$$

Dataset columns: logP, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 315 restarts; actual search 3599.9 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.66874437; native loss = 0.35834894.

$$14.90127 - x_0$$

Exact saved expression (use this for coefficient checks):

14.901269884573786 - x0

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 17; complexity 26; training R2 = 0.98500822; native loss = 0.076383003.

$$x_0^4 \cdot \left(0.63116196 - \frac{1.7134549}{x_0} \right) + 14.822427 - \frac{0.015719219}{1.6376453 - \frac{0.83748647}{x_0^6 \cdot \ln(8.9535322 \cdot x_0)}}$$

Exact saved expression (use this for coefficient checks):

((square(square(x0)) * ((-1.7134549487240165 / x0) + 0.6311619565902711)) - (0.01571921940556852 / (1.6376453429792412 - (0.8374864655638684 / (log(x0 * 8.953532180581242) * square(cube(x0))))))) + 14.822427340400841

Evolved PySR (709715)

Archived/audited label: exact. 120 restarts; actual search 3601.1 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -176.33882; native loss = 0.25949975.

$$x_0$$

Exact saved expression (use this for coefficient checks):

x0

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 8; complexity 10; training R2 = 0.95289584; native loss = 0.14140156.

$$0.0016770026 \cdot x_0^9 - x_0^2 + 14.741368$$

Exact saved expression (use this for coefficient checks):

14.74136846068554 - (square(x0) - cube(cube(x0) * 0.1188076983159697))

Leavitt | seed 10008

Accepted target: $M = c_0 + c_1 x_0$

$$x_0 = \log_{10} P$$

Dataset columns: logP, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 316 restarts; actual search 3600.2 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.66828482; native loss = 0.35834894.

$$14.902545 - x_0$$

Exact saved expression (use this for coefficient checks):

14.902545132284331 - x0

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 16; complexity 24; training R2 = 0.98443595; native loss = 0.072398803.

$$0.63288193 \cdot x_0^3 \cdot (x_0 - 2.7125144) - \frac{0.001809889 \cdot x_0}{1.7495897 - 1.6072638 \cdot x_0} + 14.801816$$

Exact saved expression (use this for coefficient checks):

(((((x0 * -0.0018098890061232777) / (((x0 * -0.6072638304113224) - x0) + 1.7495896729529818)) - 0.03963443981456202) + (cube(x0) * ((x0 + -2.7125144222961937) * 0.6328819257545442)))) + 14.841450688874415

Evolved PySR (709715)

Archived/audited label: exact. 119 restarts; actual search 3600.2 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -176.33882; native loss = 0.25949975.

$$x_0$$

Exact saved expression (use this for coefficient checks):

x0

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 9; complexity 10; training R2 = 0.94222634; native loss = 0.1418711.

$$-x_0^2 + 0.017643795 \cdot e^{x_0^2} + 14.69784$$

Exact saved expression (use this for coefficient checks):

((exp(square(x0)) * 0.01764379490843616) + 14.697840430628334) - square(x0)

Leavitt | seed 10009

Accepted target: $M = c_0 + c_1 x_0$

$$x_0 = \log_{10} P$$

Dataset columns: logP, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 316 restarts; actual search 3599.7 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 3; training R2 = 0.66884738; native loss = 0.35834894.

$$14.900983 - x_0$$

Exact saved expression (use this for coefficient checks):

$$14.900982738893074 - x_0$$

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 13; complexity 18; training R2 = 0.98506998; native loss = 0.079680627.

$$14.93258 - 3.7125892 \cdot e^{-4.2564608 \cdot (0.48470297 \cdot x_0 - 1)^2} + \frac{0.00046732652}{-0.19574974 \cdot \ln(x_0) - 0.044525744}$$

Exact saved expression (use this for coefficient checks):

$$((0.0004673265242999728 / ((\log(x_0) - -0.22746259400563842) * -0.19574974367339693)) + 14.932579967318617) - \exp(1.3117295412342156 - \text{square}(x_0 + -2.0631191836953446))$$

Evolved PySR (709715)

Archived/audited label: exact. 119 restarts; actual search 3600.4 s. Archived review / positive-equation audit

Recovery witness

Frontier index 0; complexity 1; training R2 = -176.33882; native loss = 0.25949975.

$$x_0$$

Exact saved expression (use this for coefficient checks):

$$x_0$$

Review rationale: Affine in period's logarithm (SRBench2 input is already log period).

Best numerical fit (maximum training R2)

Frontier index 10; complexity 11; training R2 = 0.79622526; native loss = 0.1419061.

$$6.0773482 \cdot \left(1 - 0.40564202 \cdot e^{0.22438124 \cdot x_0^2}\right)^2 + 12.274026$$

Exact saved expression (use this for coefficient checks):

$$\text{square}(2.4652278289038145 - \exp((x_0 * x_0) * 0.22438123563203788)) + 12.274026231374759$$

Newton | seed 10000

Accepted target: $\log F = c_0 + \log x_1 + \log x_2 - 2 \log x_0$

$(x_0, x_1, x_2) = (r, m_1, m_2)$

Dataset columns: r, m1, m2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 132 restarts; actual search 3600.4 s. Archived review / positive-equation audit

Recovery witness

Frontier index 6; complexity 9; training R2 = 0.99989722; native loss = 0.079480472.

$$\ln\left(\frac{x_1 \cdot x_2}{x_0^2}\right) - 23.420864$$

Exact saved expression (use this for coefficient checks):

`log((x2 * x1) / square(x0)) + -23.420863922221596`

Review rationale: Equals $\log(x_1 \cdot x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 23; training R2 = 0.99993225; native loss = 0.0638535.

$$\ln\left(\frac{x_1}{x_0 + 8178638.2 + \frac{36208.943 \cdot x_2}{x_1}}\right) + \ln\left(-0.00024999538 \cdot x_0 + \frac{x_2}{x_0}\right) - 23.404653$$

Exact saved expression (use this for coefficient checks):

`(log((x0 / -4000.073964586456) + (x2 / x0)) + -23.40465266081022) + log(x1 / (((x2 / x1) + 225.87343170249514) * 36208.94280333897) + x0))`

Evolved PySR (709715)

Archived/audited label: exact. 69 restarts; actual search 3601.0 s. Archived review / positive-equation audit

Recovery witness

Frontier index 8; complexity 9; training R2 = 0.99990415; native loss = 0.0098100805.

$$-\ln\left(\frac{x_0^2}{x_1 \cdot x_2}\right) - 23.449487$$

Exact saved expression (use this for coefficient checks):

`-23.449487145618345 - log((square(x0) / x1) / x2)`

Review rationale: Equals $\log(x_1 \cdot x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 20; complexity 25; training R2 = 0.99993737; native loss = 0.0073580698.

$$-\frac{x_0^2 \cdot \left(0.00030399422 + \frac{0.018320151}{-\frac{x_0^2}{x_2} + 1.0203829}\right)}{x_2} + \ln\left(\frac{6.5335169 \cdot 10^{-11} \cdot x_1 \cdot x_2}{x_0^2}\right)$$

Exact saved expression (use this for coefficient checks):

`log((square(8.083017285053575e-6 / x0) * x2) * x1) - ((square(x0) / x2) * (0.0003039942249663163 - (-0.018320150965464924 / (1.0203829312004946 - (square(x0) / x2))))))`

Newton | seed 10001

Accepted target: $\log F = c_0 + \log x_1 + \log x_2 - 2 \log x_0$

$$(x_0, x_1, x_2) = (r, m_1, m_2)$$

Dataset columns: r, m1, m2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 96 restarts; actual search 3600.9 s. Archived review / positive-equation audit

Recovery witness

Frontier index 7; complexity 10; training R2 = 0.9998975; native loss = 0.079480472.

$$\ln(x_1) + \ln\left(\frac{x_2}{x_0^2}\right) - 23.42144$$

Exact saved expression (use this for coefficient checks):

`(log(x2 / square(x0)) + -23.421440208686985) + log(x1)`

Review rationale: Equals $\log(x_1 \cdot x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 16; complexity 20; training R2 = 0.99992497; native loss = 0.063675924.

$$-0.99097119 \cdot \ln(x_0) + \ln\left(-7.2289938 \cdot 10^{-5} \cdot x_0^3 + \frac{x_2}{x_0 \cdot \frac{1}{x_1}}\right) - 23.646966$$

Exact saved expression (use this for coefficient checks):

`(-24.005313525676755 - ((log(x0) / 1.0091110768558313) + -0.3583474601715867)) + log((x2 / (x0 / x1)) + cube(x0 / -24.005313525676755))`

Evolved PySR (709715)

Archived/audited label: exact. 75 restarts; actual search 3601.5 s. Archived review / positive-equation audit

Recovery witness

Frontier index 8; complexity 9; training R2 = 0.99990415; native loss = 0.0098100805.

$$\ln\left(6.54662 \cdot 10^{-11} \cdot x_1 \cdot \frac{x_2}{x_0^2}\right)$$

Exact saved expression (use this for coefficient checks):

`log((x1 / (square(x0) / x2)) * 6.546620020671808e-11)`

Review rationale: Equals $\log(x_1 \cdot x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 19; complexity 27; training R2 = 0.99993957; native loss = 0.0077797722.

$$\ln\left(-x_0 + \frac{2.6587676 \cdot 10^{-9} \cdot x_1 \cdot (x_2 + 0.11781369)}{x_0^2}\right) - 3.6797431 - \frac{19393.661 \cdot \left(1643.2254 + \frac{1.3424193 \cdot x_2}{x_1}\right)}{x_0}$$

Exact saved expression (use this for coefficient checks):

`(log(((x2 + 0.11781369421763238) / (square(x0 * 19393.661342376166) / x1)) - x0) + -3.6797430672884452) - (((x2 / (x1 / 1.3424192961054984)) + 1643.2254370009844) * (19393.661342990137 / x0))`

Newton | seed 10002

Accepted target: $\log F = c_0 + \log x_1 + \log x_2 - 2 \log x_0$

$(x_0, x_1, x_2) = (r, m_1, m_2)$

Dataset columns: r, m1, m2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 96 restarts; actual search 3600.8 s. Archived review / positive-equation audit

Recovery witness

Frontier index 6; complexity 9; training R2 = 0.99989682; native loss = 0.079480472.

$$\ln\left(\frac{x_1 \cdot x_2}{x_0^2}\right) - 23.42004$$

Exact saved expression (use this for coefficient checks):

`log((x1 * x2) / square(x0)) + -23.42003997285071`

Review rationale: Equals $\log(x_1 x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 28; training R2 = 0.99994039; native loss = 0.060620852.

$$-27304.519 \cdot x_2 \cdot \frac{1}{x_0 \cdot x_1} + \ln(x_1) + \ln\left(\frac{x_2}{(x_0 + 0.91761064)^2}\right) - 23.416378 + \frac{0.010499241}{\ln\left(\frac{x_0^2}{x_2}\right)}$$

Exact saved expression (use this for coefficient checks):

`(log(x2 / square(x0 - -0.9176106346689079)) + (log(x1) - 23.416377994303446)) + ((x2 / (x0 * x1)) * -27304.518657136374) - (-0.010499241150818638 / log(square(x0) / x2)))`

Evolved PySR (709715)

Archived/audited label: exact. 80 restarts; actual search 3601.6 s. Archived review / positive-equation audit

Recovery witness

Frontier index 8; complexity 9; training R2 = 0.99990415; native loss = 0.0098100805.

$$-\ln\left(\frac{x_0^2}{x_1 \cdot x_2}\right) - 23.449487$$

Exact saved expression (use this for coefficient checks):

`-23.449487161241304 - log((square(x0) / x1) / x2)`

Review rationale: Equals $\log(x_1 x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 22; complexity 30; training R2 = 0.99994646; native loss = 0.0062303938.

$$-\frac{x_0^2}{(x_1 - 10.0282 \cdot x_2) \cdot \sqrt{\ln(x_0^3)}} + \ln\left(\frac{0.9999782 \cdot x_1 \cdot x_2}{\left(0.9999891 \cdot x_0 - \left(1 + \frac{0.99999455 \cdot x_2}{x_1}\right)^2\right)^2}\right) - 23.438312$$

Exact saved expression (use this for coefficient checks):

`(log(x2 * (x1 / square(x0 - square((x2 / x1) + 1.0000054501181819)))) + -23.4383118146467) - ((square(x0) / (x1 + (x2 / -0.09971879520934773))) / sqrt(log(cube(x0))))`

Newton | seed 10003

Accepted target: $\log F = c_0 + \log x_1 + \log x_2 - 2 \log x_0$

$$(x_0, x_1, x_2) = (r, m_1, m_2)$$

Dataset columns: r, m1, m2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 242 restarts; actual search 3600.4 s. Archived review / positive-equation audit

Recovery witness

Frontier index 6; complexity 9; training R2 = 0.99989592; native loss = 0.079480472.

$$\ln\left(\frac{x_1 \cdot x_2}{x_0^2}\right) - 23.418284$$

Exact saved expression (use this for coefficient checks):

`log((x2 * x1) / square(x0)) + -23.418284177206814`

Review rationale: Equals $\log(x_1 x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 30; training R2 = 0.99992359; native loss = 0.05976168.

$$-0.48318306 \cdot \ln(x_0 - 89033872.0) + \ln\left(-201.99936 \cdot x_0^2 + \frac{x_1 \cdot x_2}{\sqrt{x_0 \cdot x_0}}\right) - 23.840316$$

Exact saved expression (use this for coefficient checks):

`((log(x0 - cube(exp(square((x0 / x0) + 1.4701233035449093)))) * -0.4831830555594736) + log(((x2 / x0) * (x1 / sqrt(x0))) - square(x0 / -0.07035986528813246))) + -23.840316346328922`

Evolved PySR (709715)

Archived/audited label: exact. 74 restarts; actual search 3601.5 s. Archived review / positive-equation audit

Recovery witness

Frontier index 8; complexity 9; training R2 = 0.99990415; native loss = 0.0098100805.

$$\ln\left(\frac{x_2}{x_0^2 \cdot \frac{1}{x_1}}\right) - 23.449487$$

Exact saved expression (use this for coefficient checks):

`-23.44948717776843 + log(x2 / (square(x0) / x1))`

Review rationale: Equals $\log(x_1 x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 16; complexity 21; training R2 = 0.99993934; native loss = 0.0073952915.

$$\ln(x_1) + \ln\left(-0.00025940896 + \frac{x_2}{x_0^2}\right) - 23.425768 + \frac{0.011311254}{\ln\left(\frac{x_0 \cdot x_0}{x_2}\right)}$$

Exact saved expression (use this for coefficient checks):

`((((0.011311254475630606 / log((x0 / x2) * x0)) + log(x1)) + log((x2 / square(x0)) + -0.00025940896297013655)) - 23.425768348022697`

Newton | seed 10004

Accepted target: $\log F = c_0 + \log x_1 + \log x_2 - 2 \log x_0$

$(x_0, x_1, x_2) = (r, m_1, m_2)$

Dataset columns: r, m1, m2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 203 restarts; actual search 3599.9 s. Archived review / positive-equation audit

Recovery witness

Frontier index 6; complexity 9; training R2 = 0.99989826; native loss = 0.079480472.

$$-\ln\left(\frac{x_0^2}{x_1 \cdot x_2}\right) - 23.423094$$

Exact saved expression (use this for coefficient checks):

`-23.42309423677474 - log((square(x0) / x1) / x2)`

Review rationale: Equals $\log(x_1 \cdot x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 23; training R2 = 0.99992684; native loss = 0.063069615.

$$1.0031287 \cdot \ln(x_1) + 1.9891301 \cdot \ln\left(-\frac{11.598583 \cdot x_0^{\frac{3}{2}}}{x_2} + \frac{\sqrt{x_2}}{x_0}\right) - 23.564046$$

Exact saved expression (use this for coefficient checks):

`((log((sqrt(x2) / x0) + ((x0 / -0.08621742651041686) * (sqrt(x0) / x2)))) * 1.982926233734612) + log(x1)) + -23.49055243811257) * 1.00312865646415`

Evolved PySR (709715)

Archived/audited label: exact. 85 restarts; actual search 3601.3 s. Archived review / positive-equation audit

Recovery witness

Frontier index 8; complexity 9; training R2 = 0.99990415; native loss = 0.0098100805.

$$\ln\left(\frac{6.5466193 \cdot 10^{-11} \cdot x_1 \cdot x_2}{x_0^2}\right)$$

Exact saved expression (use this for coefficient checks):

`log(((x2 * x1) / 1.5275059604322647e10) / square(x0))`

Review rationale: Equals $\log(x_1 \cdot x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 21; complexity 27; training R2 = 0.99993438; native loss = 0.0074238754.

$$\ln\left(\frac{1.57626 \cdot x_2 \cdot (-0.081452369 \cdot x_0^2 + x_1)}{x_0^2}\right) - 23.894625 + \frac{0.011757294}{-\frac{x_2}{x_0^2} + \frac{x_0}{x_0}}$$

Exact saved expression (use this for coefficient checks):

`(0.011757293832734942 / ((x0 / x0) - (x2 / square(x0)))) + (log(((x1 - (square(x0) * 0.08145236881839751)) * square(1.2554919467939558 / x0)) * x2) + -23.894625430363156)`

Newton | seed 10005

Accepted target: $\log F = c_0 + \log x_1 + \log x_2 - 2 \log x_0$

$(x_0, x_1, x_2) = (r, m_1, m_2)$

Dataset columns: r, m1, m2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 238 restarts; actual search 3599.9 s. Archived review / positive-equation audit

Recovery witness

Frontier index 6; complexity 9; training R2 = 0.9998955; native loss = 0.079480472.

$$\ln\left(\frac{x_1 \cdot x_2}{x_0^2}\right) - 23.417497$$

Exact saved expression (use this for coefficient checks):

`log((x1 * x2) / square(x0)) + -23.417496882923803`

Review rationale: Equals $\log(x_1 x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 16; complexity 26; training R2 = 0.99992947; native loss = 0.057216728.

$$-\sqrt{\frac{\sqrt{x_0} \cdot x_2}{x_1^{\frac{3}{2}}}} + \ln(x_2) + 0.99526307 \cdot \ln\left(-0.090781523 + \frac{x_1}{x_0 \cdot x_0}\right) - 23.341901$$

Exact saved expression (use this for coefficient checks):

`log(x2) + (-23.34190080773282 - ((log(((x1 / x0) / x0) - 0.09078152288309964) / -1.004759476001604) + sqrt((x2 / sqrt(x1)) / (x1 / sqrt(x0)))))`

Evolved PySR (709715)

Archived/audited label: exact. 89 restarts; actual search 3600.7 s. Archived review / positive-equation audit

Recovery witness

Frontier index 8; complexity 9; training R2 = 0.99990415; native loss = 0.0098100805.

$$-\ln\left(\frac{x_0^2}{x_1 \cdot x_2}\right) - 23.449487$$

Exact saved expression (use this for coefficient checks):

`-23.44948715911408 - log((square(x0) / x1) / x2)`

Review rationale: Equals $\log(x_1 x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 18; training R2 = 0.99993412; native loss = 0.0080164845.

$$0.99984812 \cdot \ln\left(x_2 \cdot \left(\frac{x_1}{\left(x_0 - \left(\frac{x_2}{x_1}\right)^2\right)^2} - 0.06890505\right)\right) - 23.41984$$

Exact saved expression (use this for coefficient checks):

`(log(x2 * ((x1 / square(x0 - square(x2 / x1))) - 0.06890505008030262)) + -23.423397444460264) * 0.9998481212048367`

Newton | seed 10006

Accepted target: $\log F = c_0 + \log x_1 + \log x_2 - 2 \log x_0$

$(x_0, x_1, x_2) = (r, m_1, m_2)$

Dataset columns: r, m1, m2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 241 restarts; actual search 3599.9 s. Archived review / positive-equation audit

Recovery witness

Frontier index 6; complexity 9; training R2 = 0.99989549; native loss = 0.079480472.

$$\ln\left(\frac{x_1 \cdot x_2}{x_0^2}\right) - 23.417482$$

Exact saved expression (use this for coefficient checks):

`log((x1 * x2) / square(x0)) + -23.41748218403159`

Review rationale: Equals $\log(x_1 \cdot x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 24; training R2 = 0.99992703; native loss = 0.061568295.

$$-\frac{3.0825631 \cdot 10^{-10} \cdot x_0^6}{x_2^3} - 0.66331181 \cdot \ln(x_0^3) + 1.0012213 \cdot \ln(x_1) + \ln(x_2) - 23.74503$$

Exact saved expression (use this for coefficient checks):

`((log(x1) * 1.0012212623043941) + log(x2)) + ((cube((x0 * (x0 / x2)) *
-0.0006755186207279046) + -23.74502984956183) + (log(cube(x0)) *
-0.6633118106677491))`

Evolved PySR (709715)

Archived/audited label: exact. 85 restarts; actual search 3601.3 s. Archived review / positive-equation audit

Recovery witness

Frontier index 8; complexity 9; training R2 = 0.99990415; native loss = 0.0098100805.

$$\ln\left(\frac{6.5466198 \cdot 10^{-11} \cdot x_1 \cdot x_2}{x_0^2}\right)$$

Exact saved expression (use this for coefficient checks):

`log(((x1 / 1.5275058548467606e10) / square(x0)) * x2)`

Review rationale: Equals $\log(x_1 \cdot x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 19; complexity 25; training R2 = 0.99993911; native loss = 0.0073568823.

$$\frac{6973.7648 \cdot \sqrt{x_0}}{x_0 - \frac{x_2}{x_0}} + \ln\left(x_1 \cdot \left(-0.00026021786 + \frac{x_2}{x_0 \cdot (x_0 + 1.7366532)}\right)\right) - 23.437452$$

Exact saved expression (use this for coefficient checks):

`(-23.437452091098724 + ((sqrt(x0) / 0.00014339456888946004) / (x0 - (x2 / x0)))) + log(
x1 * (((x2 / x0) / (x0 - -1.7366532169466709)) - 0.00026021785842921175))`

Newton | seed 10007

Accepted target: $\log F = c_0 + \log x_1 + \log x_2 - 2 \log x_0$

$(x_0, x_1, x_2) = (r, m_1, m_2)$

Dataset columns: r, m1, m2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 238 restarts; actual search 3600.0 s. Archived review / positive-equation audit

Recovery witness

Frontier index 6; complexity 9; training R2 = 0.99989788; native loss = 0.079480472.

$$\ln\left(\frac{x_1 \cdot x_2}{x_0^2}\right) - 23.422247$$

Exact saved expression (use this for coefficient checks):

`log((x1 * x2) / square(x0)) + -23.422247329568386`

Review rationale: Equals $\log(x_1 x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 20; complexity 30; training R2 = 0.99993999; native loss = 0.055823763.

$$\ln\left(-0.00025686096 + \frac{x_2}{x_0 \cdot x_0}\right) + \ln\left(-45925.7 \cdot x_0 + 2 \cdot x_1 - \frac{45925.7 \cdot x_2}{x_0}\right) - 23.873615 - \frac{5.5067003}{\ln(0.40119217 \cdot x_0)}$$

Exact saved expression (use this for coefficient checks):

`(-5.506700330333452 / log(x0 * 0.40119216799300855)) + (log((((x2 / x0) + x0) / -2.177430071250093e-5) + (x1 + x1))) + (log(((x2 / x0) / x0) + -0.0002568609621558116) + -23.873614929324408))`

Evolved PySR (709715)

Archived/audited label: exact. 82 restarts; actual search 3601.4 s. Archived review / positive-equation audit

Recovery witness

Frontier index 8; complexity 9; training R2 = 0.99990415; native loss = 0.0098100805.

$$\ln\left(\frac{6.5466201 \cdot 10^{-11} \cdot x_1 \cdot x_2}{x_0^2}\right)$$

Exact saved expression (use this for coefficient checks):

`log(((x2 * x1) * 6.546620055421425e-11) / square(x0))`

Review rationale: Equals $\log(x_1 x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 20; complexity 27; training R2 = 0.99993788; native loss = 0.0074839879.

$$\ln\left(\frac{x_2 \cdot \left(-0.070023796 \cdot x_0 + \frac{x_1}{x_0}\right)}{x_0}\right) - 23.425753 - \frac{0.0087512861}{-\frac{x_0}{\sqrt{x_2}} + 1.0112349}$$

Exact saved expression (use this for coefficient checks):

`((log((((x0 * 1.6423654709496684) / -23.45439077958023) + (x1 / x0)) * (x2 / x0)) + -23.467226518325386) - -0.041473534837925395) - (0.008751286081779105 / (1.011234878435164 - (x0 / sqrt(x2)))))`

Newton | seed 10008

Accepted target: $\log F = c_0 + \log x_1 + \log x_2 - 2 \log x_0$

$(x_0, x_1, x_2) = (r, m_1, m_2)$

Dataset columns: r, m1, m2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 236 restarts; actual search 3600.1 s. Archived review / positive-equation audit

Recovery witness

Frontier index 6; complexity 9; training R2 = 0.9998963; native loss = 0.079480472.

$$\ln\left(\frac{x_1 \cdot x_2}{x_0^2}\right) - 23.419016$$

Exact saved expression (use this for coefficient checks):

`log((x1 * x2) / square(x0)) + -23.41901563406889`

Review rationale: Equals $\log(x_1 \cdot x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 21; complexity 30; training R2 = 0.99993949; native loss = 0.059551425.

$$\ln\left(-176.57782 \cdot x_0^{\frac{3}{2}} + \frac{x_2}{x_0^2 \cdot \frac{1}{x_1}}\right) - 23.39832 - \frac{\ln(x_2 - 23.395796)^3}{\sqrt[4]{7.434318 \cdot 10^{-8} \cdot x_0^3 + x_1}}$$

Exact saved expression (use this for coefficient checks):

`(log((x2 / (square(x0) / x1)) - cube(sqrt(x0 / 0.03177185458042025)))) +
-23.398319576409257) - (cube(log(x2 + -23.395795986280486)) / sqrt(sqrt(cube(x0 *
0.0042048164580558095) + x1)))`

Evolved PySR (709715)

Archived/audited label: exact. 88 restarts; actual search 3601.2 s. Archived review / positive-equation audit

Recovery witness

Frontier index 8; complexity 9; training R2 = 0.99990415; native loss = 0.0098100805.

$$\ln\left(\frac{6.5466201 \cdot 10^{-11} \cdot x_1 \cdot x_2}{x_0^2}\right)$$

Exact saved expression (use this for coefficient checks):

`log((x2 * (6.546620076955092e-11 * x1)) / square(x0))`

Review rationale: Equals $\log(x_1 \cdot x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 20; training R2 = 0.9999423; native loss = 0.0076144101.

$$\ln\left(\frac{6.6511044 \cdot 10^{-11} \cdot x_1 \cdot x_2}{x_0^2}\right) + \frac{8.2393206 \cdot 10^{20}}{x_0^2 - 0.094885327 \cdot x_1}$$

Exact saved expression (use this for coefficient checks):

`log(((x2 * x0) * x1) * cube(0.00040516437759076216 / x0)) + (8.239320569263625e20 / (
square(x0) + (x1 * -0.09488532733156005)))`

Newton | seed 10009

Accepted target: $\log F = c_0 + \log x_1 + \log x_2 - 2 \log x_0$

$(x_0, x_1, x_2) = (r, m_1, m_2)$

Dataset columns: r, m1, m2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 240 restarts; actual search 3600.0 s. Archived review / positive-equation audit

Recovery witness

Frontier index 7; complexity 10; training R2 = 0.99989768; native loss = 0.079480472.

$$\ln \left(\frac{x_1 \cdot x_2}{x_0 \cdot x_0} \right) - 23.421831$$

Exact saved expression (use this for coefficient checks):

`log((x1 * (x2 / x0)) / x0) + -23.421831400490113`

Review rationale: Equals $\log(x_1 x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 21; complexity 30; training R2 = 0.99993422; native loss = 0.060507671.

$$-\left(17.93185 \cdot \sqrt{\frac{x_0 \cdot x_2}{x_1 \cdot (-x_0 + x_1)}} - 0.012667156\right) \cdot \ln(x_0) + \ln\left(x_2 \cdot \left(\frac{x_1}{x_0 \cdot x_0} - 0.086136616\right)\right) - 23.727123$$

Exact saved expression (use this for coefficient checks):

`(-23.727123381785034 - (log(x0) * (sqrt(x2 / ((x1 - x0) * ((0.003109924116081225 / x0) * x1)))) - 0.01266715617456986))) + log(((x1 / (x0 * x0)) - 0.08613661608680599) * x2)`

Evolved PySR (709715)

Archived/audited label: exact. 85 restarts; actual search 3601.4 s. Archived review / positive-equation audit

Recovery witness

Frontier index 8; complexity 9; training R2 = 0.99990415; native loss = 0.0098100805.

$$-\ln \left(\frac{x_0^2}{x_1 \cdot x_2} \right) - 23.449487$$

Exact saved expression (use this for coefficient checks):

`-23.44948717856577 - log((square(x0) / x1) / x2)`

Review rationale: Equals $\log(x_1 x_2 / x_0^2)$ plus a fitted constant; x_0 is distance.

Best numerical fit (maximum training R2)

Frontier index 22; complexity 26; training R2 = 0.99994184; native loss = 0.0073340793.

$$\ln \left(1.0022839 \cdot x_1 \cdot \left(-0.00026314733 + \frac{x_2}{x_0 \cdot (x_0 + 0.09935789)} \right) \right) - 23.435935 + \frac{0.017364872}{0.98059218 - \frac{x_2}{x_0 \cdot x_0}}$$

Exact saved expression (use this for coefficient checks):

`(log((((x2 / x0) / (x0 + 0.09935789009271205)) - 0.0002631473308913217) * (x1 / 0.9977212982266469)) + -23.43593456432834) + (0.017364871762731156 / (0.9805921783505026 - ((x2 / x0) / x0)))`

Planck | seed 10000

Accepted target: $\log B = \log\left(\frac{c_0 x_0^3}{\exp(c_1 x_0/x_1) - 1}\right)$

$$(x_0, x_1) = (\nu, T)$$

Dataset columns: nu, T, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: near. 214 restarts; actual search 3600.0 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 14; complexity 23; training R2 = 0.99999963; native loss = 0.09600608.

$$-\frac{4.7993622 \cdot 10^{-11} \cdot x_0}{x_1} + \ln\left(-7.1354665 \cdot 10^{-33} \cdot x_0^3 \cdot \left(-20.028624 - \frac{4.2083643 \cdot 10^{11} \cdot x_1}{x_0}\right)\right) - 43.768107$$

Exact saved expression (use this for coefficient checks):

$$(-43.76810687700659 - (-20.197464366610422 / (x_1 * (-4.208364293629795e11 / x_0)))) + \log(((x_1 / x_0) * -4.208364293629795e11) + -20.02862356717487) / \text{cube}(-5.194286216062697e10 / x_0))$$

Review rationale: Candidate 14 contains a Wien-like term proportional to $-x_0/x_1$, but its logarithmic factor is log of a linear x_1/x_0 expression rather than $-\log(\exp(c_1 x_0/x_1) - 1)$. Thus it is only an asymptotic/phenomenological approximation, not exact Planck log-radiance.

Best numerical fit

Same equation as above (maximum saved training R^2).

Evolved PySR (709715)

Archived/audited label: near. 52 restarts; actual search 3602.9 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 12; complexity 16; training R2 = 0.79148266; native loss = 0.0018143549.

$$-\frac{4.8236685 \cdot 10^{-11} \cdot x_0}{x_1} + \ln(x_0^2 \cdot (4.8236685 \cdot 10^{-11} \cdot x_0 + x_1))$$

Exact saved expression (use this for coefficient checks):

$$\log(x_0 * ((x_1 + (x_0 * 4.82366846672854e-11)) * x_0)) - ((4.82366846672854e-11 / x_1) * x_0)$$

Review rationale: This has the Wien core $3 \cdot \log(\text{nu}) - c \cdot \text{nu} / T$ after rewriting, with a rational-style correction through $\log(T + c \cdot \text{nu})$. It approximates the Planck denominator but is not algebraically $\log(\exp(c \cdot \text{nu} / T) - 1)$.

Best numerical fit (maximum training R2)

Frontier index 14; complexity 20; training R2 = 0.99997238; native loss = 0.00061021744.

$$-\frac{4.8236685 \cdot 10^{-11} \cdot x_0}{x_1} + \ln(5.4138947 \cdot 10^{-42} \cdot x_0^2 \cdot (4.8236685 \cdot 10^{-11} \cdot x_0 + x_1)) + 4.2566478$$

Exact saved expression (use this for coefficient checks):

$$(\log(((x_0 * x_0) * 5.4138946871505926e-42) * (x_1 + (x_0 * 4.82366846672854e-11)))) - ((x_0 / x_1) * 4.82366846672854e-11)) + 4.256647811184551$$

Planck | seed 10001

Accepted target: $\log B = \log\left(\frac{c_0 x_0^3}{\exp(c_1 x_0/x_1) - 1}\right)$

$(x_0, x_1) = (\nu, T)$

Dataset columns: nu, T, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: near. 215 restarts; actual search 3599.8 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 7; complexity 12; training R2 = 0.99995847; native loss = 1.0297583.

$$-\frac{4.7836989 \cdot 10^{-11} \cdot x_0}{x_1} + 2.3104929 \cdot \ln(x_0) - 92.220941$$

Exact saved expression (use this for coefficient checks):

$(\log(x_0) * 2.3104929200654545) + ((x_0 * (-4.7836988873129984e-11 / x_1)) + -92.2209411377521)$

Review rationale: This has the recognizable Wien-limit structure: a log-frequency term plus a term proportional to -nu/T. It is not exact Planck log-radiance because $\log(\exp(c_1 \nu/T)-1)$ is replaced by its asymptotic linear form, and the log-frequency power is not 3.

Best numerical fit (maximum training R2)

Frontier index 16; complexity 28; training R2 = 0.99998396; native loss = 0.61560765.

$$-\frac{4.7556358 \cdot 10^{-11} \cdot x_0}{x_1 - 1.0138904} - \frac{0.54231313 \cdot \sqrt{x_1}}{-\ln(x_1) - 0.067265322} - 2.2183905 \cdot \ln\left(\frac{5.4302163 \cdot 10^{11}}{x_0}\right) - 32.02575$$

Exact saved expression (use this for coefficient checks):

$((-32.02574950621395 - ((25.82413122920502 / (5.430216297722761e11 / x_0)) / (x_1 - 1.0138904353072893))) - ((\sqrt{x_1} * 0.5423131313000066) / (-0.06726532238389259 - \log(x_1)))) - (\log(5.430216297722761e11 / x_0) / 0.45077725557424436)$

Evolved PySR (709715)

Archived/audited label: near. 54 restarts; actual search 3602.6 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 12; complexity 17; training R2 = 0.94194677; native loss = 0.001855065.

$$-\frac{4.9046389 \cdot 10^{-11} \cdot x_0}{x_1} + \ln\left(5.0030172 \cdot 10^{-19} \cdot x_0^3 \cdot \left(4.9046389 \cdot 10^{-11} + \frac{x_1}{x_0}\right)\right)$$

Exact saved expression (use this for coefficient checks):

$\log(\text{cube}(7.938601419244054e-7 * x_0) * ((x_1 / x_0) + 4.9046388698749956e-11)) - ((x_0 / x_1) * 4.9046388698749956e-11)$

Review rationale: This contains the Wien-limit core $3*\log(\nu)-c_1*\nu/T$. Its additional $\log(T/\nu+c_1)$ factor approaches a constant at high ν/T , but it is not the exact Planck correction $-\log(\exp(c_1*\nu/T)-1)$.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 25; training R2 = 0.99992608; native loss = 0.00086760207.

$$-\frac{4.7590374 \cdot 10^{-11} \cdot x_0}{x_1} + 4.7590374 \cdot 10^{-11} \cdot (7.0042175 \cdot 10^9 - x_1) \cdot \left(\ln\left(9.2777494 \cdot 10^{30} \cdot x_0^6 \cdot (4.7590374 \cdot 10^{-11} \cdot x_0 + x_1)^3\right) - 345.49725\right) - 0.11083177$$

Exact saved expression (use this for coefficient checks):

$(((((7.00421750254904e9 - x_1) * (\log(\text{cube}(((x_1 * 2.1012652508010555e10) + x_0) * x_0) * x_0))) + -345.4972463755991)) - (x_0 / x_1)) / 2.101265250801055e10) - 0.1108317686846414$

Planck | seed 10002

Accepted target: $\log B = \log\left(\frac{c_0 x_0^3}{\exp(c_1 x_0/x_1) - 1}\right)$

$$(x_0, x_1) = (\nu, T)$$

Dataset columns: nu, T, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: near. 214 restarts; actual search 3601.1 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 8; complexity 14; training R2 = 0.9999383; native loss = 0.92683941.

$$-\frac{4.7644182 \cdot 10^{-11} \cdot x_0}{x_1} + \ln(x_0^2) + \ln(x_1) - 90.874364$$

Exact saved expression (use this for coefficient checks):

$$((x_0 * (-4.7644182055013605e-11 / x_1)) + (\log(\text{square}(x_0)) + \log(x_1))) + -90.87436404287921$$

Review rationale: This is the Wien-limit form: $2 \log(\nu) + \log(T) - c_1 \nu/T + \text{constant}$. It lacks the exact nonconstant Planck factor $-\log(\exp(c_1 \nu/T) - 1)$, so it is only an asymptotic approximation.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 27; training R2 = 0.99999346; native loss = 0.21697906.

$$-\frac{4.7643742 \cdot 10^{-11} \cdot x_0}{x_1 - 1.3849219} + \ln(x_0^2) + \ln(4.7643742 \cdot 10^{-11} \cdot x_0 + x_1) - 91.108544$$

Exact saved expression (use this for coefficient checks):

$$((-90.8713543559987 + ((\log(x_1 - (x_0 * -4.7643742447139955e-11)) + (-4.7643742447139955e-11 * (x_0 / (-1.3849218728624313 + x_1)))))) - (0.10312130799148918 - 0.645203318005024))) + \log(1.6671817383279148)) + \log(\text{square}(x_0))$$

Evolved PySR (709715)

Archived/audited label: near. 51 restarts; actual search 3601.7 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 9; complexity 11; training R2 = 0.64018543; native loss = 0.0055198613.

$$-\frac{6.3455402 \cdot 10^{-11} \cdot x_0}{x_1} + \ln(x_0^3 \cdot x_1)$$

Exact saved expression (use this for coefficient checks):

$$\log(x_1 * \text{cube}(x_0)) + ((x_0 * -6.345540253085546e-11) / x_1)$$

Review rationale: This has the Planck prefactor's $3 \log(\nu)$ and a Wien-like $-c \nu/T$ term, but includes an extra $\log(T)$ and replaces $-\log(\exp(c \nu/T) - 1)$ with a linear asymptotic term.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 22; training R2 = 0.99936191; native loss = 0.0010554992.

$$\ln(4.7884196 \cdot 10^{-42} \cdot x_0^2 \cdot (2.0810106 \cdot 10^{-11} \cdot x_0 + x_1)) + 2.7800745 - \frac{4.6778689 \cdot 10^{-11} \cdot (x_0 - 2.7800745)}{x_1}$$

Exact saved expression (use this for coefficient checks):

$$((((x_0 - 2.7800745131142715) * -4.6778689408041506e-11) / x_1) + \log((x_0 * 4.7884196056531995e-42) * ((x_1 - (x_0 * -2.081010596626242e-11)) * x_0))) + 2.7800745131142715$$

Planck | seed 10003

Accepted target: $\log B = \log\left(\frac{c_0 x_0^3}{\exp(c_1 x_0/x_1) - 1}\right)$

$(x_0, x_1) = (\nu, T)$

Dataset columns: nu, T, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: near. 210 restarts; actual search 3599.8 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 9; complexity 14; training R2 = 0.99995856; native loss = 1.0248091.

$$-\frac{4.7868142 \cdot 10^{-11} \cdot x_0}{x_1} + 2.3280024 \cdot \ln(x_0) - 92.675279$$

Exact saved expression (use this for coefficient checks):

$(-92.67527934687016 - ((x_0 / (x_1 / 0.013668196837628013)) / 2.855384862534358e8)) + (\log(x_0) * 2.328002422744021)$

Review rationale: This has the Wien-limit structure constant + k*log(nu) - k'*nu/T. It lacks the essential exact Planck term -log(exp(c1*nu/T)-1), so it is only an asymptotic approximation.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 30; training R2 = 0.99998861; native loss = 0.35661241.

$$2 \cdot \ln(x_0) + \ln(4.7647262 \cdot 10^{-11} \cdot x_0 + x_1) - 90.808712 - \frac{4.7647262 \cdot 10^{-11} \cdot x_0 + \sqrt{x_1} + 4.7647262 \cdot 10^{-11} \cdot x_1^3 + 0.76566939}{x_1 - 1.0726704}$$

Exact saved expression (use this for coefficient checks):

$((\log((4.764726156183372e-11 * x_0) + x_1) + -90.80871180284285) - ((\sqrt{x_1} + ((4.764726156183372e-11 * (x_0 + \text{cube}(x_1))) - 0.7656693896471966)) / (x_1 + -1.072670354657737))) + \log(x_0)) + \log(x_0)$

Evolved PySR (709715)

Archived/audited label: near. 76 restarts; actual search 3601.7 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 12; complexity 15; training R2 = 0.64868314; native loss = 0.0041615776.

$$-\frac{4.5762926 \cdot 10^{-11} \cdot x_0}{x_1} + \ln(x_0^2 \cdot (x_0 + 2.1930121 \cdot 10^{10} \cdot x_1))$$

Exact saved expression (use this for coefficient checks):

$\log(\text{square}(x_0) * (x_0 - (x_1 / -4.5599384355882984e-11))) + ((x_0 / x_1) * -4.5762925696361896e-11)$

Review rationale: This has the Wien-like core 3*log(x0) - c*x0/x1 after factoring its logarithm. It lacks the exact Planck correction -log(exp(c*x0/x1)-1) and adds a nonconstant log(1+c*x1/x0) factor.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 24; training R2 = 0.99925217; native loss = 0.0010340993.

$$\ln(4.7644014 \cdot 10^{-42} \cdot x_0^2) + \ln(4.5657999 \cdot 10^{-11} \cdot x_0 + 2 \cdot x_1) + 0.90535958 + \frac{4.671992 \cdot 10^{-11} \cdot (14.884497 - x_0)}{x_1}$$

Exact saved expression (use this for coefficient checks):

$(\log(x_1 + (x_1 + (x_0 * 4.565799889668444e-11))) - (((-4.671991956658082e-11 / x_1) * (14.884496670364557 - x_0)) - \log(\text{square}(x_0 * 2.182750884307781e-21)))) + 0.9053595783071047$

Planck | seed 10004

Accepted target: $\log B = \log\left(\frac{c_0 x_0^3}{\exp(c_1 x_0/x_1) - 1}\right)$

$$(x_0, x_1) = (\nu, T)$$

Dataset columns: nu, T, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: near. 216 restarts; actual search 3600.3 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 7; complexity 12; training R2 = 0.99995808; native loss = 1.0427269.

$$-\frac{4.7833808 \cdot 10^{-11} \cdot x_0}{x_1} + 2.2626373 \cdot \ln(x_0) - 90.956847$$

Exact saved expression (use this for coefficient checks):

$$(\log(x_0) * 2.2626372793018703) + ((x_0 / x_1) * -4.783380843032948e-11) + -90.95684687563889)$$

Review rationale: This is a Wien-like log-radiance approximation: a logarithmic frequency term plus a term proportional to -nu/T. It lacks the exact Planck correction -log(exp(c1*nu/T)-1) and its frequency-log coefficient is not the required 3.

Best numerical fit (maximum training R2)

Frontier index 16; complexity 29; training R2 = 0.99998957; native loss = 0.4981635.

$$-\frac{4.787106 \cdot 10^{-11} \cdot x_0}{x_1} + \frac{11.90705}{\left(\frac{x_0}{x_1}\right)^{0.1400669}} + 2.5043118 \cdot \ln(5.3362157 \cdot 10^{-13} \cdot x_0) - 28.547644$$

Exact saved expression (use this for coefficient checks):

$$((\log(x_0 / 1.873987200766064e12) - (-3.589599532252597 / \exp(\log(x_0 / \text{square}(2.727440336778211 * x_1)) * 0.14006689578768164)))) * 2.5043117736252842) + ((-0.3182223000299829 + ((x_0 / -1.873987200767226e12) / x_1)) * 89.70975423892834)$$

Evolved PySR (709715)

Archived/audited label: near. 77 restarts; actual search 3602.0 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 12; complexity 15; training R2 = 0.79361488; native loss = 0.0018829032.

$$-\frac{4.8786469 \cdot 10^{-11} \cdot x_0}{x_1} + \ln(x_0^2 \cdot (6.5920046 \cdot 10^{-11} \cdot x_0 + x_1))$$

Exact saved expression (use this for coefficient checks):

$$((x_0 / x_1) * -4.878646896807244e-11) + \log(\text{square}(x_0) * ((x_0 * 6.592004573125127e-11) + x_1))$$

Review rationale: This has a Wien-like high-frequency form: as x0/x1 grows, its logarithm approaches 3 log(x0) minus a constant times x0/x1. It lacks the exact nonconstant -log(exp(c1*x0/x1)-1) Planck denominator.

Best numerical fit (maximum training R2)

Frontier index 20; complexity 30; training R2 = 0.99971941; native loss = 0.00072701531.

$$-\frac{4.8786469 \cdot 10^{-11} \cdot x_0}{x_1} + 1.5724806 \cdot 10^{-11} \cdot x_1 + \ln(x_0^2 \cdot (6.5920046 \cdot 10^{-11} \cdot x_0 + x_1)) - 90.083795$$

Exact saved expression (use this for coefficient checks):

$$\log(((x_0 * 6.592004573125127e-11) + x_1) * (x_0 * x_0)) + ((((((x_0 / x_1) * -4.878646896807244e-11) - 82.07113870340538) - (1.4040017726784801 / 0.2002096832964193)) - 1.0) - (x_1 * (-4.878646896807244e-11 / 3.102516348407367))))$$

Planck | seed 10005

Accepted target: $\log B = \log\left(\frac{c_0 x_0^3}{\exp(c_1 x_0/x_1) - 1}\right)$

$$(x_0, x_1) = (\nu, T)$$

Dataset columns: nu, T, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: near. 215 restarts; actual search 3599.7 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 7; complexity 12; training R2 = 0.99995948; native loss = 1.0223832.

$$-\frac{4.7857798 \cdot 10^{-11} \cdot x_0}{x_1} + 2.2980349 \cdot \ln(x_0) - 91.831999$$

Exact saved expression (use this for coefficient checks):

$$((x_0 * -4.785779751176414e-11) / x_1) + ((\log(x_0) * 2.2980349244243286) + -91.83199866282513)$$

Review rationale: This has the recognizable Wien-limit structure: a negative nu/T term plus log(nu). It is not exact Planck log-radiance: the log(nu) coefficient is 2.298 rather than 3, and it replaces -log(exp(c1*nu/T)-1) with a linear asymptotic term.

Best numerical fit (maximum training R2)

Frontier index 12; complexity 23; training R2 = 0.99998265; native loss = 0.62949636.

$$4.4653385 \cdot \ln(\sqrt{x_0} + 5.694897 \cdot x_1) - 89.241432 - \frac{4.7771839 \cdot 10^{-11} \cdot (x_0 + 7.963673 \cdot 10^{12})}{x_1}$$

Exact saved expression (use this for coefficient checks):

$$((\log((x_1 / 0.1755958013734038) + \sqrt{x_0})) * 4.46533846000568) - 28.422178219802298) - (((((x_0 - 7.963673021724376e12) / x_1) / -7.963673021724376e12) * -380.4393057605288) + 60.81925357624339)$$

Evolved PySR (709715)

Archived/audited label: near. 75 restarts; actual search 3600.9 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 13; complexity 15; training R2 = 0.79367504; native loss = 0.0018869408.

$$-\frac{4.8804399 \cdot 10^{-11} \cdot x_0}{x_1} + \ln(x_0^2 \cdot (6.6755309 \cdot 10^{-11} \cdot x_0 + x_1))$$

Exact saved expression (use this for coefficient checks):

$$\log(((6.675530899751416e-11 * x_0) + x_1) * \text{square}(x_0)) + ((x_0 * -4.880439879826403e-11) / x_1)$$

Review rationale: This is a Wien/interpolating approximation: its exponential-tail term is proportional to -x0/x1 and it approaches the required 3 log(x0) dependence. However, log((a*x0+x1)*x0^2) is not the exact -log(exp(c*x0/x1)-1) Planck factor.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 18; training R2 = 0.99987409; native loss = 0.00069406939.

$$-\frac{4.8526842 \cdot 10^{-11} \cdot x_0}{x_1} + \ln(x_0^2) + \ln(x_0 + 1.5631907 \cdot 10^{10} \cdot x_1) - 114.00263$$

Exact saved expression (use this for coefficient checks):

$$(\log(x_0 - (x_1 * -1.5631906884574743e10)) + \log(\text{square}(x_0))) + (-114.0026308978055 + (x_0 / (x_1 * -2.0607151990727787e10)))$$

Planck | seed 10006

Accepted target: $\log B = \log\left(\frac{c_0 x_0^3}{\exp(c_1 x_0/x_1) - 1}\right)$

$$(x_0, x_1) = (\nu, T)$$

Dataset columns: nu, T, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: near. 221 restarts; actual search 3599.7 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 8; complexity 14; training R2 = 0.99993844; native loss = 0.9287433.

$$-\frac{4.7633445 \cdot 10^{-11} \cdot x_0}{x_1} + \ln(x_0 \cdot x_0 \cdot x_1) - 90.874954$$

Exact saved expression (use this for coefficient checks):

$$(-90.87495434387183 - ((4.763344521031002e-11 * x_0) / x_1)) + \log(x_1 * (x_0 * x_0))$$

Review rationale: This has a Wien-like exponential-tail term, $\log(x_0)$ -c x_0/x_1 , but it lacks the exact $-\log(\exp(c_1 x_0/x_1)-1)$ Planck structure and adds a nonconstant $\log(x_1)$ factor.

Best numerical fit (maximum training R2)

Frontier index 11; complexity 19; training R2 = 0.99998178; native loss = 0.71167151.

$$-\frac{4.7822463 \cdot 10^{-11} \cdot x_0}{x_1} + 2.2658739 \cdot \ln(x_0) - 89.318943 - \frac{37.18719}{\sqrt{x_1}}$$

Exact saved expression (use this for coefficient checks):

$$(\log(x_0) + ((-39.41920217368402 + (\sqrt{x_1}) / (x_1 * -0.0609315714419632))) - (x_0 / (x_1 * 4.73809544952129e10)))) / 0.44133081716104533$$

Evolved PySR (709715)

Archived/audited label: near. 71 restarts; actual search 3601.7 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 9; complexity 11; training R2 = 0.63943638; native loss = 0.0055206464.

$$-\frac{6.3547779 \cdot 10^{-11} \cdot x_0}{x_1} + \ln(x_0^3 \cdot x_1)$$

Exact saved expression (use this for coefficient checks):

$$\log(\text{cube}(x_0) * x_1) - (x_0 * (6.354777932362058e-11 / x_1))$$

Review rationale: This has the Wien-like core $3 \cdot \log(x_0)$ -c x_0/x_1 , but it adds the nonconstant $\log(x_1)$ factor and lacks the exact $-\log(\exp(c x_0/x_1)-1)$ Planck term.

Best numerical fit (maximum training R2)

Frontier index 23; complexity 30; training R2 = 0.99996863; native loss = 0.00058005345.

$$1.9787754 \cdot \ln(x_0) - \ln\left(\frac{44.092163}{4.7705953 \cdot 10^{-11} \cdot x_0 + x_1}\right) - 86.857932 - \frac{4.7705953 \cdot 10^{-11} \cdot \left(x_0 + \frac{x_0}{\sqrt{x_1^3}}\right)}{x_1}$$

Exact saved expression (use this for coefficient checks):

$$((-86.85793222330325 - \log(44.09216261988004 / (x_1 + (x_0 / 2.096174450591571e10)))) + \log(x_0)) - (((x_0 + (x_0 / \sqrt{\text{cube}(x_1)})) / 2.0961744505915695e10) / x_1) - (\log(x_0) / 1.0216848248764883))$$

Planck | seed 10007

Accepted target: $\log B = \log\left(\frac{c_0 x_0^3}{\exp(c_1 x_0/x_1) - 1}\right)$

$(x_0, x_1) = (\nu, T)$

Dataset columns: nu, T, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: near. 218 restarts; actual search 3600.3 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 13; complexity 22; training R2 = 0.99992319; native loss = 0.62767506.

$$-\frac{4.8396272 \cdot 10^{-11} \cdot x_0}{x_1} + \ln(9.6201475 \cdot 10^{-32} \cdot x_0^3) + \ln\left(8.6268809 \cdot 10^{-9} + \frac{188.27525 \cdot x_1}{x_0}\right) - 24.594386$$

Exact saved expression (use this for coefficient checks):

```
((x0 / x1) * -4.839627203550224e-11) + (log(cube(x0 * 4.582057944858791e-11))) + (log(((x1 / x0) + 4.582057944858791e-11) * 188.27524656090057) + -24.59438562356642))
```

Review rationale: This contains the Wien-like core $3 \cdot \log(x_0) - c \cdot x_0/x_1$, but adds $\log(x_1/x_0 + c)$, a genuine nonconstant factor. It also lacks the exact $\log(\exp(c \cdot x_0/x_1) - 1)$ Planck denominator.

Best numerical fit (maximum training R2)

Frontier index 12; complexity 20; training R2 = 0.99998322; native loss = 0.65418299.

$$\frac{0.0013173177 \cdot x_0}{5603774.5 - 27580284.0 \cdot x_1} + 2.2366485 \cdot \ln(x_0) + \ln(x_1) - 96.622907$$

Exact saved expression (use this for coefficient checks):

```
((log(x1) - (-2.236648453589069 * log(x0))) + -96.62290652666988) + ((x0 * 0.0013173176761415632) / (2.75802838128958e7 * (sqrt(0.04128229500607973) - x1)))
```

Evolved PySR (709715)

Archived/audited label: near. 73 restarts; actual search 3601.2 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 13; complexity 16; training R2 = 0.792465; native loss = 0.0018766992.

$$-\frac{4.8433038 \cdot 10^{-11} \cdot x_0}{x_1} + \ln(x_0^2 \cdot (4.8433038 \cdot 10^{-11} \cdot x_0 + x_1))$$

Exact saved expression (use this for coefficient checks):

```
log(x0 * (x0 * (x1 + (4.843303826059307e-11 * x0)))) - ((x0 * 4.843303826059307e-11) / x1)
```

Review rationale: With $z = c \cdot x_0/x_1$, this has $3 \log(x_0) - z$ plus $\log((1+z)/z)$, a rational-style approximation to the Planck correction $-\log(1 - \exp(-z))$. It lacks the exact $\exp(z) - 1$ dependence.

Best numerical fit (maximum training R2)

Frontier index 21; complexity 26; training R2 = 0.99959965; native loss = 0.0007111947.

$$\ln(4.8389055 \cdot 10^{-11} \cdot x_0 + x_1) - 2.0460054 + \frac{-4.8389055 \cdot 10^{-11} \cdot x_0 + x_1 \cdot (2 \cdot \ln(x_0) - 84.936113) - 9.6778111 \cdot 10^{-11}}{x_1}$$

Exact saved expression (use this for coefficient checks):

```
-2.0460053750413105 + (((x1 * ((log(x0) - 84.93611288632873) + log(x0))) - ((x0 * 4.838905545932113e-11) + 9.677811091864227e-11)) / x1) + log(x1 + (x0 * 4.838905545932113e-11)))
```

Planck | seed 10008

Accepted target: $\log B = \log\left(\frac{c_0 x_0^3}{\exp(c_1 x_0/x_1) - 1}\right)$

$$(x_0, x_1) = (\nu, T)$$

Dataset columns: nu, T, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: near. 218 restarts; actual search 3599.9 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 17; complexity 28; training R2 = 0.99999661; native loss = 0.27754302.

$$-\frac{4.7912071 \cdot 10^{-11} \cdot x_0}{x_1} + \ln(5.8764947 \cdot 10^{-14} \cdot x_0) + \ln(5.7553119 \cdot 10^{-14} \cdot x_1) + \ln\left(5.7553119 \cdot 10^{-14} \cdot x_0 \cdot \left(\frac{4.6085809 \cdot 10^{-11} \cdot x_0}{x_1} + 1.962515\right)\right)$$

Exact saved expression (use this for coefficient checks):

$$(\log(((x_0 * 4.608580879431399e-11) / x_1) + 1.9625149515273996) * (x_0 * 5.755311906494394e-14)) + (\log(x_1 * 5.755311906494394e-14) - ((x_0 * 4.7912070664111556e-11) / x_1))) + \log(x_0 * 5.876494738176631e-14)$$

Review rationale: Candidate 17 has the Wien-like linear ν/T decay and logarithmic frequency structure, but its $\log(\text{linear } \nu/T \text{ plus constant})$ correction is not exactly $-\log(\exp(c_1 \nu/T) - 1)$. Thus it is an approximation, not the Planck log-radiance form.

Best numerical fit

Same equation as above (maximum saved training R^2).

Evolved PySR (709715)

Archived/audited label: near. 74 restarts; actual search 3601.5 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 9; complexity 11; training R2 = 0.64021038; native loss = 0.0055198599.

$$-\frac{6.3452307 \cdot 10^{-11} \cdot x_0}{x_1} + \ln(x_0^3 \cdot x_1)$$

Exact saved expression (use this for coefficient checks):

$$\log(x_1 * \text{cube}(x_0)) - ((x_0 * 6.345230685048109e-11) / x_1)$$

Review rationale: This is the Wien-limit structure: $\log(T \nu^3)$ minus $c \nu/T$. It omits the nonconstant Planck correction implicit in $-\log(\exp(c_1 \nu/T) - 1)$, so it is not exactly the stored log spectral-radiance law.

Best numerical fit (maximum training R2)

Frontier index 14; complexity 17; training R2 = 0.99943263; native loss = 0.004227099.

$$-\frac{4.8944579 \cdot 10^{-11} \cdot x_0}{x_1} + 0.78265371 \cdot \ln(2.3331831 \cdot 10^{-34} \cdot x_0^3 \cdot x_1) - 36.71999$$

Exact saved expression (use this for coefficient checks):

$$-36.719990359249415 + (((x_0 * 6.253669844541694e-11) / x_1) - \log(x_1 * \text{cube}(x_0 * 6.156250382500707e-12))) * -0.7826537090003268)$$

Planck | seed 10009

Accepted target: $\log B = \log\left(\frac{c_0 x_0^3}{\exp(c_1 x_0/x_1) - 1}\right)$

$(x_0, x_1) = (\nu, T)$

Dataset columns: nu, T, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: near. 219 restarts; actual search 3599.8 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 9; complexity 15; training R2 = 0.999981; native loss = 0.6565118.

$$-\frac{4.7739873 \cdot 10^{-11} \cdot x_0}{x_1} + 2.1938733 \cdot \ln(x_0) + \ln(x_1) - 95.569529$$

Exact saved expression (use this for coefficient checks):

$\log(x_1) + (((x_0 * -4.7739873038294416e-11) / x_1) + (\log(x_0) * 2.1938732694657666)) + -95.56952925571339)$

Review rationale: This has logarithmic frequency/temperature terms plus a nu/T exponential-tail-like term, a recognizable approximation to log Planck radiance. It lacks the required $-\log(\exp(c_1 \nu/T)-1)$ structure and has incorrect logarithmic powers, so it is not exact.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 27; training R2 = 0.99998572; native loss = 0.59644054.

$$-\frac{4.7544566 \cdot 10^{-11} \cdot x_0}{x_1 - 1.3842429} + 2.4326479 \cdot \ln\left(1.8732443 \cdot 10^{-11} \cdot x_0 \cdot \left(0.012855304 + \frac{x_1}{\sqrt{x_0}}\right)\right) - 25.487887$$

Exact saved expression (use this for coefficient checks):

$((((x_0 / -2.1032898018639374e10) / (x_1 - 1.3842428910691826)) + (-0.016980696827387037 - 25.470906031730145)) + (\log(x_0 * (((x_1 / \sqrt{x_0})) + 0.012855303945999642) / 6.266404032098117e10) * 1.1738505910992714)) / 0.4110746985629424)$

Evolved PySR (709715)

Archived/audited label: near. 135 restarts; actual search 3601.4 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 14; complexity 18; training R2 = 0.95292768; native loss = 0.0013255762.

$$-\frac{4.8375693 \cdot 10^{-11} \cdot x_0}{x_1} + \ln(x_0^2 \cdot (1.1320917 \cdot 10^{-31} \cdot x_0 + 2.3402077 \cdot 10^{-21} \cdot x_1))$$

Exact saved expression (use this for coefficient checks):

$\log((x_0 / (4.273124995626382e20 / (x_1 + (x_0 / 2.067153839371028e10)))) * x_0) - ((x_0 / x_1) / 2.067153839371028e10)$

Review rationale: This is a structured Wien/Padé-like approximation: it has the exponential argument x_0/x_1 and a rational correction, but it does not contain the required $-\log(\exp(c_1 x_0/x_1)-1)$ dependence. Therefore it cannot be algebraically identical to log Planck radiance.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 24; training R2 = 0.99999256; native loss = 0.00059512735.

$$-\frac{4.8116748 \cdot 10^{-11} \cdot x_0}{x_1} + \ln(4.8375693 \cdot 10^{-11} \cdot x_0^2 \cdot (7.6813677 \cdot 10^{-13} \cdot x_0 + 0.015878569 \cdot x_1)) - 62.977968$$

Exact saved expression (use this for coefficient checks):

$(\log(((x_0 / 2.067153839371028e10) / (62.97796814571842 / ((x_0 / 2.067153839371028e10) + x_1)))) * x_0 - 62.97796814571842) - (((x_0 / x_1) / 1.0053815951227618) / 2.067153839371028e10)$

Rydberg | seed 10000

Accepted target: $\log \lambda = c_0 - \log(x_0^{-2} - x_1^{-2})$

$$(x_0, x_1) = (n_1, n_2), \quad n_2 > n_1 > 0$$

Dataset columns: n_1, n_2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: miss. 216 restarts; actual search 3600.8 s. Report-selected structural proxy (review selected none)

Closest structural candidate

Frontier index 7; complexity 11; training R2 = 0.99971279; native loss = 0.021287117.

$$\frac{1.9921975 \cdot x_0^3}{x_1^3} + 1.9921975 \cdot \ln(x_0) - 16.175523$$

Exact saved expression (use this for coefficient checks):

$$((\text{cube}(x_0 / x_1) + -8.119437542038874) + \log(x_0)) / 0.501958254706728$$

Review rationale: No candidate has the required logarithm of the reciprocal Rydberg difference, i.e. a log dependence on $1/x_0^2 - 1/x_1^2$. The frontier expressions use isolated $\log(x_0)$, polynomial ratio terms, or rational/square-root forms instead. Report proxy: compact $\log(n_1)$ plus a ratio-dependent correction. Its cubic correction is not $-\log(1-(n_1/n_2)^2)$, and its log coefficient is not exactly 2. This is a qualitative choice, not a proven nearest expression; the full frontier is included in the appendix. The archived miss label is retained.

Best numerical fit (maximum training R2)

Frontier index 16; complexity 30; training R2 = 0.99996298; native loss = 0.0073909492.

$$1.960148 \cdot \ln(x_0) - 16.263867 + \frac{x_0 - 0.1713685}{(2.1148526 - (-x_0^3 + x_0)^2 \cdot e^{-2 \cdot x_0^2}) \cdot (-0.84988632 \cdot x_0 + x_1)}$$

Exact saved expression (use this for coefficient checks):

$$((x_0 - 0.1713685019882606) / ((x_1 - (x_0 / 1.1766279564497872)) * ((0.5654397407720653 - \text{square}((x_0 - \text{cube}(x_0)) / \exp(\text{square}(x_0)))) + 1.5494128506082108))) + (-16.263866787587084 + (\log(x_0) * 1.9601479519257683))$$

Evolved PySR (709715)

Archived/audited label: near. 139 restarts; actual search 3600.3 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 10; complexity 14; training R2 = 0.99996539; native loss = 0.0058932978.

$$-\ln\left(-\frac{x_1}{x_1^3} + \frac{1.0004365}{x_0^2}\right) - 16.209212$$

Exact saved expression (use this for coefficient checks):

$$-16.209212162900634 - \log((1.0004364774902117 / (\text{cube}(x_0) / x_0)) - (x_1 / \text{cube}(x_1)))$$

Review rationale: This is constant minus $\log(1.0004364774902117/x_0^2 - 1/x_1^2)$, matching the reciprocal-difference log structure. The non-unit relative coefficient on $1/x_0^2$ cannot be absorbed into an overall constant, so it is not exact.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 26; training R2 = 0.99996585; native loss = 0.0058552243.

$$\frac{1.5815264 \cdot 10^{-6} \cdot x_1^2}{\left(0.99941172 - \frac{x_1}{x_0^2}\right)^2} + \ln\left(-\frac{1.806589 \cdot 10^{-6} \cdot x_1^2}{0.99941172 - \frac{x_1^2}{x_0^2}}\right) - 2.9853198$$

Exact saved expression (use this for coefficient checks):

$$(1.5815264236182803e-6 * \text{square}(x_1 / (0.9994117190979162 - (x_1 / \text{square}(x_0))))) + (-2.985319821041777 + \log((-1.8065889977231895e-6 / (0.9994117190979162 - \text{square}(x_1 / x_0))) * \text{square}(x_1)))$$

Rydberg | seed 10001

Accepted target: $\log \lambda = c_0 - \log(x_0^{-2} - x_1^{-2})$ $(x_0, x_1) = (n_1, n_2), \quad n_2 > n_1 > 0$

Dataset columns: n_1, n_2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: near. 213 restarts; actual search 3599.7 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 13; complexity 21; training R2 = 0.99995743; native loss = 0.0081845459.

$$-\sqrt{\frac{x_0}{x_1} + 0.073310388} + \ln(x_0) + \ln(x_1) - \ln\left(\frac{-x_0 + x_1}{x_0}\right) - 15.861922$$

Exact saved expression (use this for coefficient checks):

$$((-15.861922142327213 - \text{sqrt}((x_0 / x_1) - 0.07331038821915609)) - \log((x_1 - x_0) / x_0)) + (\log(x_1) + \log(x_0))$$

Review rationale: Candidate 13 contains the key logarithmic difference factor, with $\log(x_1 - x_0)$ and log powers of x_0, x_1 . However, it lacks the required $\log(x_1 + x_0)$ dependence and adds a nonconstant square-root term, so it is not exact.

Best numerical fit

Same equation as above (maximum saved training R^2).

Evolved PySR (709715)

Archived/audited label: exact. 137 restarts; actual search 3600.9 s. Archived review / positive-equation audit

Recovery witness

Frontier index 10; complexity 12; training R2 = 0.66562312; native loss = 0.0073031114.

$$-\ln\left(\frac{x_0}{x_0^3} - \frac{x_1}{x_1^3}\right) - 15.265847$$

Exact saved expression (use this for coefficient checks):

$$-15.26584703272647 - \log((x_0 / \text{cube}(x_0)) - (x_1 / \text{cube}(x_1)))$$

Review rationale: Equals $-\log(1/x_0^2 - 1/x_1^2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 27; training R2 = 0.99996583; native loss = 0.0058183475.

$$-\ln\left(\frac{-\frac{x_0}{x_0 \cdot x_1} + x_1}{x_0 \cdot x_0 \cdot x_1}\right) - 16.20966 + \frac{1.0624079 \cdot 10^{-6}}{(x_0 - 0.29342679 \cdot x_1)^3}$$

Exact saved expression (use this for coefficient checks):

$$-16.2096604833451 - (\log((x_1 - (x_0 / (x_1 / x_0))) / ((x_0 * x_1) * x_0)) - (4.90982190609086 \text{e-}7 / \text{cube}(x_0 - ((x_0 + x_1) * 0.22685999407954702))))$$

Rydberg | seed 10002

Accepted target: $\log \lambda = c_0 - \log(x_0^{-2} - x_1^{-2})$

$$(x_0, x_1) = (n_1, n_2), \quad n_2 > n_1 > 0$$

Dataset columns: n_1, n_2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: near. 213 restarts; actual search 3600.3 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 13; complexity 19; training R2 = 0.99992477; native loss = 0.010907533.

$$-\frac{0.66454355 \cdot x_0}{x_1} + \ln(x_1) + \ln\left(\frac{x_0 \cdot x_0}{-x_0 + x_1}\right) - 16.273685$$

Exact saved expression (use this for coefficient checks):

$$((\log(x_1) - (x_0 / (x_1 * 1.5047922652816335))) + -16.27368456605708) + \log((x_0 * x_0) / (x_1 - x_0))$$

Review rationale: It contains the key $\log[x_0^2 x_1 / (x_1 - x_0)]$ Rydberg-like factor, but the exact relationship additionally requires the nonconstant factor $x_1 / (x_1 + x_0)$. The added linear ratio term is not algebraically equivalent to that missing logarithmic factor.

Best numerical fit (maximum training R2)

Frontier index 19; complexity 29; training R2 = 0.99996516; native loss = 0.0074470077.

$$\ln(11.57005 \cdot x_1) + \ln\left(\frac{x_0 \cdot x_0}{-x_0 + x_1}\right) - 19.353574 + \frac{0.50582785}{\frac{x_1}{-x_0 + x_1} - 0.26385766}$$

Exact saved expression (use this for coefficient checks):

$$((\log(x_1 * 11.570049890025869) + 0.5255730240617683) + \log((x_0 / (x_1 - x_0)) * x_0)) + (\exp(-0.6815588907256903 - \log((x_1 / (x_1 - x_0)) - 0.2638576587012943)) + -19.879146763436385)$$

Evolved PySR (709715)

Archived/audited label: near. 141 restarts; actual search 3600.9 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 9; complexity 12; training R2 = 0.99996539; native loss = 0.0058932978.

$$-\ln\left(-\frac{3.6976244}{x_1^2} + \frac{3.6992383}{x_0^2}\right) - 14.901522$$

Exact saved expression (use this for coefficient checks):

$$-14.901521617069308 - \log((3.699238308438355 / \text{square}(x_0)) - (3.6976243819745913 / \text{square}(x_1)))$$

Review rationale: Candidate 9 has the required constant-minus-log reciprocal-square difference structure. However, the coefficients of $1/x_0^2$ and $1/x_1^2$ are unequal, so they cannot be absorbed into one common constant multiplying $(1/x_0^2 - 1/x_1^2)$.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 30; training R2 = 0.99996959; native loss = 0.0054140825.

$$-\ln\left(-\frac{3.7277557}{x_1^2} + \frac{3.729742}{x_0^2}\right) - 14.895165 + \frac{0.018771859 \cdot \ln(\sqrt{x_0})}{(x_1 - 4.5064932) \cdot (x_0 - x_1 + 4.5064932)}$$

Exact saved expression (use this for coefficient checks):

$$-14.89932546683593 + (((\log(\text{sqrt}(x_0)) / (-4.506493223292811 + x_1)) / (x_0 - (x_1 + -4.506493223292811))) * 0.01877185932371742) + (0.004160062215121512 - \log((3.729742010359537 / \text{square}(x_0)) - (3.7277557269457335 / \text{square}(x_1)))))$$

Rydberg | seed 10003

Accepted target: $\log \lambda = c_0 - \log(x_0^{-2} - x_1^{-2})$

$$(x_0, x_1) = (n_1, n_2), \quad n_2 > n_1 > 0$$

Dataset columns: n_1, n_2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: near. 219 restarts; actual search 3600.2 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 11; complexity 18; training R2 = 0.99994817; native loss = 0.0090499224.

$$1.9940146 \cdot \ln(x_0) - 16.270503 - \frac{0.46448538}{-\frac{1.141075 \cdot x_1}{x_0} + \frac{x_0}{x_0}}$$

Exact saved expression (use this for coefficient checks):

$$(-0.4644853852060177 / ((x_0 / x_0) + ((x_1 * -1.141075009304426) / x_0))) - ((\log(x_0) * -1.9940145888455112) - -16.270503170600268)$$

Review rationale: Candidate 11 has the leading $2 \cdot \log(x_0)$ dependence and a correction depending on x_1/x_0 , but its rational reciprocal correction is not algebraically equal to the required $-\log(1-(x_0/x_1)^2)$.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 29; training R2 = 0.99996432; native loss = 0.0074942672.

$$-0.14908208 \cdot \sqrt{\frac{1}{x_1}} + 1.96289 \cdot \ln(x_0) - 16.214113 - \frac{0.56506563}{0.057009014 - \frac{1.2401651 \cdot x_1}{x_0} + \frac{x_0}{x_0}}$$

Exact saved expression (use this for coefficient checks):

$$(-0.5650656289653883 / (((x_0 / x_0) + 0.0570090135861533) + ((x_1 / x_0) * -1.240165128723841))) - ((\log(x_0) * -1.9628900356962964) - (-16.214112619197692 - \sqrt{\log(((x_0 / x_1) * 0.022225466432725535) / x_0)}))$$

Evolved PySR (709715)

Archived/audited label: near. 142 restarts; actual search 3600.3 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 10; complexity 12; training R2 = 0.99996534; native loss = 0.0058933152.

$$\ln \left(\frac{9.1323405 \cdot 10^{-8} \cdot x_0^2}{1.0004279 - \frac{x_0^2}{x_1^2}} \right)$$

Exact saved expression (use this for coefficient checks):

$$\log(\text{square}(x_0 * -0.0003021976261105718) / (1.0004278831361049 - \text{square}(x_0 / x_1)))$$

Review rationale: This has the Rydberg log structure $\log(\text{const} \cdot x_0^2 / (1 - (x_0/x_1)^2))$, but the denominator constant is 1.0004278831361049 rather than 1. That relative coefficient cannot be absorbed into the overall wavelength constant, so it is a numerical near-match.

Best numerical fit (maximum training R2)

Frontier index 22; complexity 29; training R2 = 0.99996825; native loss = 0.0056535768.

$$-\frac{1.0280407 \cdot 10^{-5} \cdot x_0 \cdot x_1}{-\frac{x_0 \cdot x_1}{e^{x_0}} + 0.60393373} + \ln \left(\frac{8.9626318 \cdot 10^{-6} \cdot x_0^2}{1.0008485 - \frac{x_0^2}{x_1^2}} \right) - 4.5850124$$

Exact saved expression (use this for coefficient checks):

$$(-4.585012397070042 - ((x_1 * 1.0280407130342221e-5) / ((0.6039337279793623 - ((x_1 * x_0) / \exp(x_0))) / x_0))) + \log(\text{square}(x_0 * -0.0029937654841948906) / (1.0008484908555362 - \text{square}(x_0 / x_1)))$$

Rydberg | seed 10004

Accepted target: $\log \lambda = c_0 - \log(x_0^{-2} - x_1^{-2})$

$$(x_0, x_1) = (n_1, n_2), \quad n_2 > n_1 > 0$$

Dataset columns: n_1, n_2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: near. 209 restarts; actual search 3600.2 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 13; complexity 22; training R2 = 0.99996273; native loss = 0.0074998404.

$$\ln(x_0 \cdot (x_0 + 0.06163758)) - 16.840512 + \frac{0.51715967 \cdot \left(x_1 - \frac{0.10305819}{x_0}\right)}{-0.85708278 \cdot x_0 + x_1}$$

Exact saved expression (use this for coefficient checks):

$$(((x_1 - (0.10305819453260662 / x_0)) * 0.5171596715121317) / (x_1 - (x_0 * 0.8570827793328586))) - (16.84051195549363 - \log(x_0 * (x_0 - -0.06163757989905879)))$$

Review rationale: This has the leading approximately $2 \log(x_0)$ dependence and a rational singular structure in $x_1 - \text{const} \cdot x_0$, resembling the Rydberg denominator. However, it replaces the required logarithmic dependence on $x_1^2 - x_0^2$ with a rational term and lacks the required $\log(x_1)$ factors.

Best numerical fit

Same equation as above (maximum saved training R^2).

Evolved PySR (709715)

Archived/audited label: near. 109 restarts; actual search 3601.1 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 11; complexity 13; training R2 = 0.99996362; native loss = 0.0058945725.

$$\ln\left(\frac{x_0^2}{1.000131 - \frac{x_0^2}{x_1^2}}\right) - 16.211895$$

Exact saved expression (use this for coefficient checks):

$$\log((x_0 / (1.0001309684893134 - \text{square}(x_0 / x_1))) * x_0) - 16.211895347988587$$

Review rationale: Near: fitted relative coefficient differs from the required 1; it cannot be absorbed into the overall Rydberg constant.

Best numerical fit (maximum training R2)

Frontier index 20; complexity 29; training R2 = 0.99996722; native loss = 0.0056176976.

$$-\ln\left(-\frac{0.078239636}{x_1^2} + \frac{0.078239636 \cdot x_0}{x_0^3}\right) - 18.75762 - \frac{0.02548701}{x_1^3 - e^{x_1} + 18.75762 + \frac{x_1}{x_1}}$$

Exact saved expression (use this for coefficient checks):

$$((-0.025487009869418185 / (((x_1 / x_1) + \text{cube}(x_1)) + (18.757619787233455 - \exp(x_1)))) - 18.757619787233455) - \log(((x_0 / \text{cube}(x_0)) - \text{square}(x_1 / \text{square}(x_1))) * 0.07823963570594022)$$

Original label: exact; changed by the saved audit.

Rydberg | seed 10005

Accepted target: $\log \lambda = c_0 - \log(x_0^{-2} - x_1^{-2})$

$$(x_0, x_1) = (n_1, n_2), \quad n_2 > n_1 > 0$$

Dataset columns: n_1, n_2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 217 restarts; actual search 3600.5 s. Archived review / positive-equation audit

Recovery witness

Frontier index 11; complexity 21; training R2 = 0.99996511; native loss = 0.0074945544.

$$\ln(4.9559088 \cdot x_0^2) + \ln\left(\frac{x_1 \cdot x_1}{(-x_0 + x_1) \cdot (x_0 + x_1)}\right) - 17.810257$$

Exact saved expression (use this for coefficient checks):

$\log(x_0 * (x_0 * 4.955908750499148)) + (\log((x_1 * x_1) / ((x_1 - x_0) * (x_0 + x_1)))) + -17.810257033004675)$

Review rationale: Equals $-\log(1/x_0^2 - 1/x_1^2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 12; complexity 23; training R2 = 0.99996559; native loss = 0.0074526363.

$$\ln(1.2662717 \cdot x_0^2) + \ln\left(\frac{x_1^2}{(-x_0 + x_1) \cdot (x_0 + 0.99077436 \cdot x_1)}\right) - 16.451869$$

Exact saved expression (use this for coefficient checks):

$(\log((x_0 * x_0) * 1.26627171132) + \log((x_1 * x_1) / ((x_1 - x_0) * (x_0 + (x_1 * 0.9907743618184403))))) + -16.451868501444398$

Evolved PySR (709715)

Archived/audited label: exact. 109 restarts; actual search 3600.7 s. Archived review / positive-equation audit

Recovery witness

Frontier index 12; complexity 14; training R2 = 0.99996517; native loss = 0.0058948268.

$$-\ln\left(\frac{-\frac{x_0^2}{x_1} + x_1}{x_0^2 \cdot x_1}\right) - 16.210072$$

Exact saved expression (use this for coefficient checks):

$-16.21007152094707 - \log((x_1 - (\text{square}(x_0) / x_1)) / (\text{square}(x_0) * x_1))$

Review rationale: Equals $-\log(1/x_0^2 - 1/x_1^2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 13; complexity 15; training R2 = 0.99996539; native loss = 0.0058932978.

$$\ln\left(\frac{x_0^2}{-\frac{x_0}{\frac{1}{x_0} \cdot x_1 \cdot x_1} + 1.0004365}\right) - 16.209212$$

Exact saved expression (use this for coefficient checks):

$\log(\text{square}(x_0) / (1.000436477446712 - (x_0 / ((x_1 * x_1) / x_0)))) + -16.20921216649759$

Rydberg | seed 10006

Accepted target: $\log \lambda = c_0 - \log(x_0^{-2} - x_1^{-2})$

$$(x_0, x_1) = (n_1, n_2), \quad n_2 > n_1 > 0$$

Dataset columns: n_1, n_2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: near. 216 restarts; actual search 3599.9 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 10; complexity 16; training R2 = 0.99994522; native loss = 0.0091801933.

$$\frac{x_0}{-2.1044261 \cdot x_0 + 2.4197353 \cdot x_1} + \ln(x_0 \cdot x_0) - 16.277921$$

Exact saved expression (use this for coefficient checks):

$$(\log(x_0 * x_0) + (x_0 / (((x_1 * 1.149831421072968) - x_0) / 0.47518892981666544))) + -16.27792078282496$$

Review rationale: It contains the required $2 \cdot \log(x_0)$ component and a ratio-dependent correction, but the Rydberg law requires a logarithm of the inverse-square difference. The added rational $x_0/(a \cdot x_1 - x_0)$ cannot algebraically equal that logarithmic ratio term.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 26; training R2 = 0.99996531; native loss = 0.007487148.

$$\frac{0.83934269 \cdot x_1}{\left(-0.25037742 + \frac{0.39027828 \cdot x_1^3}{x_0^3}\right) \cdot (x_0 + x_0)} + \ln(x_0 \cdot x_0) - 16.212669$$

Exact saved expression (use this for coefficient checks):

$$(((x_1 / (x_0 + x_0)) * 0.8393426861311797) / (-0.2503774223361434 - (cube(x_1) * cube(2.294688758889032 / (x_0 * -3.140019372124438))))) + \log(x_0 * x_0)) + -16.212668794639526$$

Evolved PySR (709715)

Archived/audited label: near. 108 restarts; actual search 3600.9 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 10; complexity 12; training R2 = 0.99996534; native loss = 0.0058933155.

$$\ln \left(\frac{9.128618 \cdot 10^{-8} \cdot x_1^2}{\left(\frac{x_1}{x_0}\right)^2 - 0.99955738} \right)$$

Exact saved expression (use this for coefficient checks):

$$\log(\text{square}(x_1 * -0.00030213602922582434) / (\text{square}(x_1 / x_0) - 0.9995573803866998))$$

Review rationale: This has the essential log of $x_0^2 x_1^2$ divided by a difference of squared quantum numbers. However, 0.9995573803866998 multiplies the x_0^2 contribution implicitly, yielding $x_1^2 - 0.999557 \dots x_0^2$ rather than $x_1^2 - x_0^2$; this nonconstant structural mismatch prevents exactness.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 23; training R2 = 0.999967; native loss = 0.0056075314.

$$\ln \left(\frac{x_1}{-\frac{2.1258511}{x_1} + \frac{2.1249126 \cdot x_1}{x_0^2}} \right) - 15.458643 + \frac{0.00045167673}{\sqrt{x_1} - 2.1863165}$$

Exact saved expression (use this for coefficient checks):

$$(0.0004516767334298397 / (\text{sqrt}(x_1) - 2.186316452202463)) + (\log(x_1 / (((x_1 * 2.1249125835782507) / x_0) / x_0) - (2.1258510946279503 / x_1)))) + -15.458642652486539$$

Rydberg | seed 10007

Accepted target: $\log \lambda = c_0 - \log(x_0^{-2} - x_1^{-2})$

$$(x_0, x_1) = (n_1, n_2), \quad n_2 > n_1 > 0$$

Dataset columns: n_1, n_2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: near. 219 restarts; actual search 3600.6 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 9; complexity 13; training R2 = 0.99978618; native loss = 0.017919556.

$$\ln \left(x_0^2 \cdot \left(\frac{x_1}{-x_0 + x_1} + 0.8697124 \right) \right) - 16.963538$$

Exact saved expression (use this for coefficient checks):

`log(square(x0) * ((x1 / (x1 - x0)) + 0.8697123997515378)) + -16.963538256278138`

Review rationale: It contains the required $\log(x_0^2)$ factor and a singular dependence on $x_1 - x_0$, resembling the Rydberg denominator. However, the exact factor requires $1/(1/x_0^2 - 1/x_1^2)$, including the $x_1 + x_0$ dependence; the added constant also prevents equivalence.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 28; training R2 = 0.9999643; native loss = 0.007784656.

$$\frac{0.49033505 \cdot x_0}{\sqrt{\frac{x_1}{x_0}} \cdot (-0.82529529 \cdot x_0 + x_1)} + 1.0006902 \cdot \ln(x_0) + \ln(1.1025378 \cdot x_0) - 16.315637$$

Exact saved expression (use this for coefficient checks):

`((log(x0 * 1.1025377983535978) + -16.298598465376575) - 0.0170382301749802) + (((x0 / sqrt(x1 / x0)) * 0.49033505447493714) / (x1 - (x0 * 0.8252952945453359))) + (log(x0) / 0.9993102535463333))`

Evolved PySR (709715)

Archived/audited label: near. 100 restarts; actual search 3601.2 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 11; complexity 13; training R2 = 0.99996539; native loss = 0.0058932978.

$$-\ln \left(\frac{1.0004365 - \frac{x_0^2}{x_1^2}}{x_0^2} \right) - 16.209212$$

Exact saved expression (use this for coefficient checks):

`-16.2092121655495 - log((1.000436477757867 - square(x0 / x1)) / (x0 * x0))`

Review rationale: This has the required constant minus log of the Rydberg difference form, but $1.000436477757867 - (x_0/x_1)^2$ is not algebraically proportional to $1 - (x_0/x_1)^2$; the altered relative coefficient is a genuine nonconstant mismatch.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 19; training R2 = 0.99996581; native loss = 0.0058679076.

$$-0.001240854 \cdot x_1 - \ln \left(\frac{0.99644839 - \frac{(x_0 - 0.013109521)^2}{x_1^2}}{x_0^2} \right) - 16.203464$$

Exact saved expression (use this for coefficient checks):

`((x1 * -0.0012408540235685143) - log((0.9964483873964326 - square((x0 + -0.013109520534901458) / x1)) / (x0 * x0))) + -16.203464116624144`

Rydberg | seed 10008

Accepted target: $\log \lambda = c_0 - \log(x_0^{-2} - x_1^{-2})$

$$(x_0, x_1) = (n_1, n_2), \quad n_2 > n_1 > 0$$

Dataset columns: n_1, n_2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: near. 148 restarts; actual search 3601.0 s. Archived review / positive-equation audit

Closest structural candidate

Frontier index 14; complexity 23; training R2 = 0.99995624; native loss = 0.0082566855.

$$1.9599134 \cdot \ln(x_0) + \ln\left(\frac{-x_0 + 2 \cdot x_1 - 0.52787799}{-x_0 + x_1 - 0.11520257}\right) - 16.950964$$

Exact saved expression (use this for coefficient checks):

$$(\log(((x_1 + (x_1 + -0.5278779914552355)) - x_0) / ((x_1 + -0.11520257340509185) - x_0)) + -16.995723732314605) + ((\log(x_0) - -0.02283771387716927) / 0.5102266344781572)$$

Review rationale: Candidate 14 has logarithmic dependence, an approximately doubled $\log(x_0)$ term, and a log denominator resembling an $n_2 - n_1$ factor. However, it has shifted linear factors and lacks the required $n_2^2 / (n_2 + n_1)$ structure, so it is not algebraically exact.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 24; training R2 = 0.99995664; native loss = 0.008213558.

$$-\frac{x_0 - 0.20317683}{1.6666392 \cdot x_0 - 2 \cdot x_1 + 0.024069661} + \ln(x_0 \cdot (x_0 + 0.15105292) - 0.04070716) - 16.368341$$

Exact saved expression (use this for coefficient checks):

$$(\log(((x_0 + 0.15105292087033112) \cdot x_0) + -0.04070715997676717) + -16.36834071681978) - ((x_0 + -0.20317683185091903) / (((x_0 / 0.600009870961791) + (0.024069661297166933 - x_1)) - x_1))$$

Evolved PySR (709715)

Archived/audited label: exact. 101 restarts; actual search 3600.8 s. Archived review / positive-equation audit

Recovery witness

Frontier index 12; complexity 15; training R2 = 0.99996517; native loss = 0.0058948268.

$$\ln(x_1) + \ln\left(\frac{x_0}{-\frac{x_0}{x_1} + \frac{x_1}{x_0}}\right) - 16.210072$$

Exact saved expression (use this for coefficient checks):

$$(\log(x_1) + \log(x_0 / ((x_1 / x_0) - (x_0 / x_1)))) + -16.210071529041972$$

Review rationale: Equals $-\log(1/x_0^2 - 1/x_1^2)$ plus a fitted constant.

Best numerical fit

Same equation as above (maximum saved training R^2).

Rydberg | seed 10009

Accepted target: $\log \lambda = c_0 - \log(x_0^{-2} - x_1^{-2})$

$$(x_0, x_1) = (n_1, n_2), \quad n_2 > n_1 > 0$$

Dataset columns: n_1, n_2, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 147 restarts; actual search 3600.4 s. Archived review / positive-equation audit

Recovery witness

Frontier index 13; complexity 19; training R2 = 0.99996492; native loss = 0.0074945544.

$$\ln(x_0 \cdot x_0) + \ln\left(-\frac{1.1368007 \cdot x_1}{\frac{x_0 \cdot x_0}{x_1} - x_1}\right) - 16.337469$$

Exact saved expression (use this for coefficient checks):

$$(\log(x_1 * (-1.1368007456302722 / ((x_0 * x_0) / x_1) - x_1))) + \log(x_0 * x_0) + -16.337469421869315$$

Review rationale: Equals $-\log(1/x_0^2 - 1/x_1^2)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 14; complexity 21; training R2 = 0.99996526; native loss = 0.0074759961.

$$\ln\left(-\frac{1.3314128 \cdot x_1}{\frac{x_0 \cdot x_0}{x_1} - x_1}\right) + \ln(x_0 \cdot x_0 + 0.0011083729) - 16.496183$$

Exact saved expression (use this for coefficient checks):

$$\log((-1.3314128115607406 / ((x_0 / x_1) * x_0 - x_1)) * x_1) + (\log((x_0 * x_0) + 0.0011083729018925507) + -16.4961833560951)$$

Evolved PySR (709715)

Archived/audited label: exact. 109 restarts; actual search 3600.4 s. Archived review / positive-equation audit

Recovery witness

Frontier index 10; complexity 12; training R2 = 0.99996517; native loss = 0.0058948268.

$$-\ln\left(\frac{x_0}{x_0^3} - \frac{x_1}{x_1^3}\right) - 16.210072$$

Exact saved expression (use this for coefficient checks):

$$-16.210071558334164 - \log((x_0 / \text{cube}(x_0)) - (x_1 / \text{cube}(x_1)))$$

Review rationale: Equals $-\log(1/x_0^3 - 1/x_1^3)$ plus a fitted constant.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 25; training R2 = 0.99996642; native loss = 0.0056520335.

$$-\ln\left(\frac{x_0}{x_0^3} - \frac{x_1}{x_1^3}\right) - 16.212104 - \frac{0.00014561674 \cdot (x_0^3 - 1.2347656)}{x_1 \cdot (4.0550515 - x_1)}$$

Exact saved expression (use this for coefficient checks):

$$(((\text{cube}(x_0) + -1.234765645396773) * -0.00014561674202849377) / (x_1 * (4.0550515101713795 - x_1))) + (-16.212104076293496 - \log((x_0 / \text{cube}(x_0)) - (x_1 / \text{cube}(x_1))))$$

Schechter | seed 10000

Accepted target: $\log \phi = c_0 + c_1 \log x_0 + c_2 x_0$

$x_0 = L$

Dataset columns: L, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 171 restarts; actual search 3600.8 s. Archived review / positive-equation audit

Recovery witness

Frontier index 6; complexity 10; training R2 = 0.99844684; native loss = 0.14178611.

$$-3.9848026 \cdot 10^{-9} \cdot x_0 - 1.2275075 \cdot \ln(x_0) - 1.8698994$$

Exact saved expression (use this for coefficient checks):

$-1.869899388267386 - (((x_0 / 3.0804724465048e8) + \log(x_0)) * 1.2275074533592532)$

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 28; training R2 = 0.99869159; native loss = 0.12477279.

$$(3.3273271 \cdot 10^{-9} \cdot x_0 + \ln(x_0)) \cdot \left(-1.2123562 + \frac{4.3814138}{\sqrt{x_0} - 19.154335} + \frac{19.157361}{-4.3761304 \cdot \sqrt{x_0} + 0.0014745764 \cdot x_0} \right) - 2.0933624$$

Exact saved expression (use this for coefficient checks):

$((\log(x_0) - (x_0 / -3.005415347277156e8)) * ((19.157360676227693 / ((x_0 * 0.0014745764082591901) - \sqrt{x_0} * 19.15051757140033)))) + ((4.381413840534234 / (\sqrt{x_0} - 19.15433531330791)) - 1.2123562262118464))) + -2.09336235927962$

Evolved PySR (709715)

Archived/audited label: exact. 125 restarts; actual search 3600.2 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 6; training R2 = -0.5233451; native loss = 0.040597138.

$$-3.339361 \cdot 10^{-9} \cdot x_0 - \ln(x_0)$$

Exact saved expression (use this for coefficient checks):

$(x_0 * -3.3393609746391253e-9) - \log(x_0)$

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 6; complexity 10; training R2 = 0.9985209; native loss = 0.038583965.

$$-4.0421974 \cdot 10^{-9} \cdot x_0 - 1.2102645 \cdot \ln(x_0) - 2.1784243$$

Exact saved expression (use this for coefficient checks):

$-2.1784242646601446 - ((\log(x_0) - (x_0 * -3.3399288101802323e-9)) * 1.2102645476718425)$

Schechter | seed 10001

Accepted target: $\log \phi = c_0 + c_1 \log x_0 + c_2 x_0$

$x_0 = L$

Dataset columns: L, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 169 restarts; actual search 3600.7 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 10; training R2 = 0.99844531; native loss = 0.14166538.

$$-3.9790033 \cdot 10^{-9} \cdot x_0 + 1.2299837 \cdot \ln\left(\frac{0.22539395}{x_0}\right)$$

Exact saved expression (use this for coefficient checks):

$((x_0 * -3.2350048927475348e-9) + \log(0.22539394891392897 / x_0)) * 1.2299837088476684$

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 14; complexity 30; training R2 = 0.99867713; native loss = 0.12665935.

$$-\frac{8.8623905 \cdot 10^{-8} \cdot x_0 - 0.36924426 - \frac{2.2569093}{\ln(x_0) - 16.238864}}{\ln(x_0)} - 0.29806004 \cdot \ln(x_0) - 0.29806004 \cdot \ln(x_0^3) - 2.4152407$$

Exact saved expression (use this for coefficient checks):

$((((0.6894815754836773 + \log(\text{cube}(x_0))) + \log(x_0)) * -0.29806003942564246) - (2.2097338142691325 + (((-2.256909307479122 / (-16.23886421732702 + \log(x_0))) + -0.3692442601643493) + \exp(-16.23886421732702 + \log(x_0))) / \log(x_0)))$

Evolved PySR (709715)

Archived/audited label: exact. 120 restarts; actual search 3601.0 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 6; training R2 = -0.52334589; native loss = 0.040597138.

$$-3.3393569 \cdot 10^{-9} \cdot x_0 - \ln(x_0)$$

Exact saved expression (use this for coefficient checks):

$(x_0 * -3.3393569140616715e-9) - \log(x_0)$

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 6; complexity 10; training R2 = 0.99843622; native loss = 0.038607709.

$$-4.0152478 \cdot 10^{-9} \cdot x_0 - 1.2098279 \cdot \ln(x_0) - 2.2437953$$

Exact saved expression (use this for coefficient checks):

$-2.2437953098732457 - ((\log(x_0) - (x_0 * -3.318858817533327e-9)) * 1.2098278530846662)$

Schechter | seed 10002

Accepted target: $\log \phi = c_0 + c_1 \log x_0 + c_2 x_0$

$x_0 = L$

Dataset columns: L, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 172 restarts; actual search 3600.3 s. Archived review / positive-equation audit

Recovery witness

Frontier index 6; complexity 10; training R2 = 0.99844893; native loss = 0.14167639.

$$-3.9787023 \cdot 10^{-9} \cdot x_0 - 1.2297422 \cdot \ln(x_0) - 1.8381234$$

Exact saved expression (use this for coefficient checks):

$(x_0 * -3.978702296222435e-9) + ((\log(x_0) + 1.4947225181106636) * -1.2297422353102272)$

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 14; complexity 26; training R2 = 0.99859213; native loss = 0.13171855.

$$-\frac{1.2307584 \cdot x_0}{35.337209 \cdot \left(\frac{0.55202824 \cdot x_0}{3.0841969 \cdot 10^8 - 1.2307584 \cdot x_0} + 1 \right)^6 + 3.0841969 \cdot 10^8} - 1.2307584 \cdot \ln(x_0) - 1.8222855$$

Exact saved expression (use this for coefficient checks):

$-0.41291416909896855 + ((\log(x_0) + ((x_0 / (\text{cube}(\text{square}(0.9057507680092736 + (0.9057507680092736 + (x_0 / ((x_0 * -1.230758428035062) + 3.0841968970575905e8)))) + 3.0841968970575905e8)) - -1.1451242740877177)) * -1.230758428035062)$

Evolved PySR (709715)

Archived/audited label: exact. 121 restarts; actual search 3600.9 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 6; training R2 = -0.52333589; native loss = 0.040597139.

$$-3.3394084 \cdot 10^{-9} \cdot x_0 - \ln(x_0)$$

Exact saved expression (use this for coefficient checks):

$(x_0 * -3.3394083903682945e-9) - \log(x_0)$

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 7; complexity 10; training R2 = 0.99848745; native loss = 0.038592189.

$$-4.0237258 \cdot 10^{-9} \cdot x_0 - 1.2009849 \cdot \ln(x_0) - 2.3430332$$

Exact saved expression (use this for coefficient checks):

$((x_0 * -3.3503550815749697e-9) - (\log(x_0) + 1.9509264401766597)) * 1.2009848943636905$

Schechter | seed 10003

Accepted target: $\log \phi = c_0 + c_1 \log x_0 + c_2 x_0$

$x_0 = L$

Dataset columns: L, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 179 restarts; actual search 3599.7 s. Archived review / positive-equation audit

Recovery witness

Frontier index 6; complexity 10; training R2 = 0.99843999; native loss = 0.14207262.

$$-3.9929026 \cdot 10^{-9} \cdot x_0 - 1.224606 \cdot \ln(x_0) - 1.9114407$$

Exact saved expression (use this for coefficient checks):

$(x_0 * -3.99290264944989e-9) + ((\log(x_0) + 1.5608617973610959) * -1.2246059619577911)$

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 26; training R2 = 0.99861179; native loss = 0.13081699.

$$-\frac{8.9027759 \cdot 10^{-8} \cdot x_0}{\ln(x_0)} - 1.1889512 \cdot \ln(x_0) - 2.4284982 - \frac{0.0012217469 \cdot \ln(x_0)}{0.84107742 - 8.8504704 \cdot 10^{-8} \cdot x_0}$$

Exact saved expression (use this for coefficient checks):

$((-2.4284982354923863 - (\log(x_0) / 0.841077415487874)) - (((x_0 * -7.286923334872655e-5) / \log(x_0)) - (\log(x_0) / (0.841077415487874 - (-7.250532492585971e-5 * (x_0 * -0.0012206648790374901)))))) * -0.0012217468834623095)$

Evolved PySR (709715)

Archived/audited label: exact. 120 restarts; actual search 3600.9 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 6; training R2 = -0.5233459; native loss = 0.040597138.

$$-3.3393569 \cdot 10^{-9} \cdot x_0 - \ln(x_0)$$

Exact saved expression (use this for coefficient checks):

$(x_0 * -3.3393569035766026e-9) - \log(x_0)$

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 6; complexity 10; training R2 = 0.99842079; native loss = 0.038589476.

$$-4.0745457 \cdot 10^{-9} \cdot x_0 - 1.2187923 \cdot \ln(x_0) - 1.9947256$$

Exact saved expression (use this for coefficient checks):

$-1.2187922947538565 * (1.6366411638923597 + (\log(x_0) + (x_0 * 3.343100948247356e-9)))$

Schechter | seed 10004

Accepted target: $\log \phi = c_0 + c_1 \log x_0 + c_2 x_0$

$x_0 = L$

Dataset columns: L, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 172 restarts; actual search 3600.3 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 10; training R2 = 0.99844562; native loss = 0.14177271.

$$-3.9824185 \cdot 10^{-9} \cdot x_0 - 1.2287697 \cdot \ln(x_0) - 1.850315$$

Exact saved expression (use this for coefficient checks):

`-1.8503150327728444 - (((x0 / 3.085485970133762e8) + log(x0)) / 0.8138221818422653)`

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 28; training R2 = 0.99859672; native loss = 0.13035843.

$$\frac{0.32810173 \cdot x_0}{-82429129.0 - \frac{1.7687221 \cdot 10^{-42}}{\left(-7.6076797 \cdot 10^{-24} \cdot x_0 - \frac{1}{x_0^3}\right)^3}} + 1.2272917 \cdot \ln\left(\frac{0.21499026}{x_0}\right)$$

Exact saved expression (use this for coefficient checks):

`((x0 * (-0.2673380202254098 / ((-1.1760206898807639e23 / cube((x0 * -0.030820334348417945) + ((1.1760206898807639e23 / cube(x0)) * -0.0344484877480591))) + -8.242912882332768e7))) - log(-0.884891429385073 / (x0 / -0.24295664766654423))) * -1.2272916966373326`

Evolved PySR (709715)

Archived/audited label: exact. 122 restarts; actual search 3601.5 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 6; training R2 = -0.52334585; native loss = 0.040597138.

$$-3.3393571 \cdot 10^{-9} \cdot x_0 - \ln(x_0)$$

Exact saved expression (use this for coefficient checks):

`(x0 * -3.339357124662399e-9) - log(x0)`

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 6; complexity 10; training R2 = 0.99852143; native loss = 0.038583869.

$$-4.0361713 \cdot 10^{-9} \cdot x_0 - 1.2089026 \cdot \ln(x_0) - 2.2131458$$

Exact saved expression (use this for coefficient checks):

`(-1.830706494427965 - (log(x0) + (x0 * 3.3387068217693354e-9))) * 1.2089025674979668`

Schechter | seed 10005

Accepted target: $\log \phi = c_0 + c_1 \log x_0 + c_2 x_0$

$x_0 = L$

Dataset columns: L, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 177 restarts; actual search 3600.2 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 10; training R2 = 0.99844395; native loss = 0.14154151.

$$-3.9748931 \cdot 10^{-9} \cdot x_0 - 1.231565 \cdot \ln(x_0) - 1.8094454$$

Exact saved expression (use this for coefficient checks):

`(log(x0) * -1.2315650462062842) + ((x0 / -2.515790903851031e8) + -1.8094453564928101)`

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 16; complexity 28; training R2 = 0.99879811; native loss = 0.1340688.

$$-3.9911957 \cdot 10^{-9} \cdot x_0 - 1.227582 \cdot \ln(x_0) - 1.8642964 - \frac{0.019715366}{3.9911957 \cdot 10^{-9} \cdot x_0 - 0.74801295} - \frac{0.50198652}{\sqrt{x_0}}$$

Exact saved expression (use this for coefficient checks):

`(-1.3692277160070407 - ((x0 * 3.991195684916425e-9) + ((1.2275820271874334 * ((log(x0) - 0.08973380674590425) + 0.49302146747859266)) + (0.5019865235042602 / sqrt(x0)))) + (-0.019715365672202907 / (log(0.47330610180638977) + (x0 * 3.991195684916425e-9)))`

Evolved PySR (709715)

Archived/audited label: exact. 123 restarts; actual search 3600.5 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 6; training R2 = -0.52334045; native loss = 0.040597138.

$$-3.3393849 \cdot 10^{-9} \cdot x_0 - \ln(x_0)$$

Exact saved expression (use this for coefficient checks):

`(x0 * -3.3393849120920165e-9) - log(x0)`

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 6; complexity 10; training R2 = 0.99852235; native loss = 0.038583836.

$$-4.0382048 \cdot 10^{-9} \cdot x_0 - 1.2096582 \cdot \ln(x_0) - 2.1941772$$

Exact saved expression (use this for coefficient checks):

`((x0 * 3.3383023780762358e-9) + (log(x0) - -1.8138820163193097)) * -1.2096581865215135`

Schechter | seed 10006

Accepted target: $\log \phi = c_0 + c_1 \log x_0 + c_2 x_0$

$x_0 = L$

Dataset columns: L, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 176 restarts; actual search 3600.7 s. Archived review / positive-equation audit

Recovery witness

Frontier index 6; complexity 11; training R2 = 0.99845404; native loss = 0.14173856.

$$-3.9770702 \cdot 10^{-9} \cdot x_0 + 0.61505076 \cdot \ln\left(\frac{0.050531286}{x_0^2}\right)$$

Exact saved expression (use this for coefficient checks):

```
(log(square(-0.224791650855556826 / x0)) * 0.6150507593126551) - (x0 * 3.9770702019593144e-9)
```

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 13; complexity 24; training R2 = 0.99852584; native loss = 0.13731593.

$$-4.1026796 \cdot 10^{-9} \cdot x_0 - 1.2180716 \cdot \ln\left(-1.4836985 \cdot \sqrt[4]{x_0} \cdot (1 - 5.669307 \cdot 10^{-26} \cdot x_0^3)^2 + x_0\right) - 2.0296354$$

Exact saved expression (use this for coefficient checks):

```
((1.218071630533234 * (-1.6517194915711229 - log(x0 - (square(1.218071630533234 - cube(x0 * 4.102679595149587e-9)) * sqrt(sqrt(x0)))))) - (x0 * 4.102679595149587e-9)) + -0.017722756321939693
```

Evolved PySR (709715)

Archived/audited label: exact. 121 restarts; actual search 3600.3 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 6; training R2 = -0.52334591; native loss = 0.040597138.

$$-3.3393568 \cdot 10^{-9} \cdot x_0 - \ln(x_0)$$

Exact saved expression (use this for coefficient checks):

```
(x0 * -3.3393568431972155e-9) - log(x0)
```

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 6; complexity 10; training R2 = 0.99850491; native loss = 0.038584724.

$$-4.0560201 \cdot 10^{-9} \cdot x_0 - 1.2143793 \cdot \ln(x_0) - 2.1124307$$

Exact saved expression (use this for coefficient checks):

```
-2.1124306613127604 - ((log(x0) + (x0 * 3.3399944622113666e-9)) * 1.2143792841653438)
```

Schechter | seed 10007

Accepted target: $\log \phi = c_0 + c_1 \log x_0 + c_2 x_0$

$x_0 = L$

Dataset columns: L, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 173 restarts; actual search 3600.4 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 10; training R2 = 0.99844478; native loss = 0.14156032.

$$-3.9754219 \cdot 10^{-9} \cdot x_0 - 1.2312909 \cdot \ln(x_0) - 1.8137761$$

Exact saved expression (use this for coefficient checks):

`((x0 * -3.228661774433606e-9) - log(x0)) + -1.4730686460986848) * 1.2312909274386634`

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 13; complexity 28; training R2 = 0.99899999; native loss = 0.12778628.

$$-4.103998 \cdot 10^{-9} \cdot x_0 - 1.2280132 \cdot \ln(x_0) - 1.8064579 + \frac{0.050086176}{4.103998 \cdot 10^{-9} \cdot x_0 - 0.87282561}$$

Exact saved expression (use this for coefficient checks):

`((-0.3442217253236252 - ((x0 * 4.103998041668074e-9) + sqrt(0.19593713290237744))) +
-6.171982717039366e-9) + (((-0.04078634900989698 / ((x0 * 4.103998041668074e-9) +
-0.872825611180429)) + ((log(x0) + square(-0.911194086544126)) + 4.103998041668074e-9)) / -0.8143234841938951)`

Evolved PySR (709715)

Archived/audited label: exact. 142 restarts; actual search 3600.3 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 6; training R2 = -0.52334584; native loss = 0.040597138.

$$-3.3393572 \cdot 10^{-9} \cdot x_0 - \ln(x_0)$$

Exact saved expression (use this for coefficient checks):

`(x0 * -3.339357211047253e-9) - log(x0)`

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 6; complexity 10; training R2 = 0.99830372; native loss = 0.038593811.

$$-4.0043684 \cdot 10^{-9} \cdot x_0 - 1.1989558 \cdot \ln(x_0) - 2.4527864$$

Exact saved expression (use this for coefficient checks):

`1.198955833134476 * (((-3.3398798593917328e-9 * x0) - log(x0)) - 2.045768766558594)`

Schechter | seed 10008

Accepted target: $\log \phi = c_0 + c_1 \log x_0 + c_2 x_0$

$x_0 = L$

Dataset columns: L, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 177 restarts; actual search 3600.7 s. Archived review / positive-equation audit

Recovery witness

Frontier index 6; complexity 10; training R2 = 0.99844824; native loss = 0.14162531.

$$-3.9747653 \cdot 10^{-9} \cdot x_0 - 1.231553 \cdot \ln(x_0) - 1.8120748$$

Exact saved expression (use this for coefficient checks):

$(\log(x_0) + ((x_0 * 3.2274415110468716e-9) + 1.4713737712443369)) / -0.8119829189642603$

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 13; complexity 25; training R2 = 0.99880179; native loss = 0.12005922.

$$-3.9375574 \cdot 10^{-9} \cdot x_0 - 1.240648 \cdot \ln(3.9802386 \cdot x_0) + \frac{0.77492921}{0.48274818 \cdot \ln(x_0) + \frac{0.24264487 \cdot (-x_0 - 3.1508062 \cdot 10^8)}{x_0}}$$

Exact saved expression (use this for coefficient checks):

$((((x_0 / -3.1508061923108405e8) - (-0.6246164825556171 / ((\log(x_0) * 0.48274817914348683) + ((-3.150806192310841e8 - x_0) / (x_0 * 4.121249334563026)))))) - \log(3.980238593940419 * x_0)) * 1.2406480304277496$

Evolved PySR (709715)

Archived/audited label: exact. 144 restarts; actual search 3600.5 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 6; training R2 = -0.52340079; native loss = 0.040597142.

$$-3.3390745 \cdot 10^{-9} \cdot x_0 - \ln(x_0)$$

Exact saved expression (use this for coefficient checks):

$(x_0 * -3.339074483921332e-9) - \log(x_0)$

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 6; complexity 10; training R2 = 0.99852204; native loss = 0.038588542.

$$-4.0330574 \cdot 10^{-9} \cdot x_0 - 1.2114697 \cdot \ln(x_0) - 2.1656751$$

Exact saved expression (use this for coefficient checks):

$((\log(x_0) - -1.787642810510416) + (x_0 / 3.0038494376613945e8)) * -1.2114697087138422$

Schechter | seed 10009

Accepted target: $\log \phi = c_0 + c_1 \log x_0 + c_2 x_0$

$x_0 = L$

Dataset columns: L, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: exact. 171 restarts; actual search 3600.2 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 8; training R2 = 0.99620161; native loss = 0.26220745.

$$-3.8114189 \cdot 10^{-9} \cdot x_0 - 1.3346373 \cdot \ln(x_0)$$

Exact saved expression (use this for coefficient checks):

`((x0 * 2.855771247232405e-9) + log(x0)) * -1.33463732513035`

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 16; complexity 26; training R2 = 0.99850339; native loss = 0.13488456.

$$-3.9759866 \cdot 10^{-9} \cdot x_0 - 1.2303443 \cdot \ln(x_0) - 1.8334874 + \frac{0.5009677}{0.059473102 \cdot \sqrt{x_0} \cdot (1.6303641 \cdot 10^{-8} \cdot x_0 - 1)^2 - 0.24387108}$$

Exact saved expression (use this for coefficient checks):

`((log(x0) + 1.286240548735773) * -1.23034430373494) - (((x0 * 3.9759865978657625e-9) - (0.5009677028595978 / ((square((x0 * 3.9759865978657625e-9) + -0.24387107625885265) * sqrt(x0)) + -0.24387107625885265)))) + 0.2509686393084223)`

Evolved PySR (709715)

Archived/audited label: exact. 145 restarts; actual search 3600.3 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 6; training R2 = -0.5233459; native loss = 0.040597138.

$$-3.3393569 \cdot 10^{-9} \cdot x_0 - \ln(x_0)$$

Exact saved expression (use this for coefficient checks):

`(x0 * -3.339356902313154e-9) - log(x0)`

Review rationale: $c_0 + c_1 \log(x_0) + c_2 x_0$, with both nonconstant terms present.

Best numerical fit (maximum training R2)

Frontier index 16; complexity 28; training R2 = 0.99912273; native loss = 0.029367417.

$$-4.1017127 \cdot 10^{-9} \cdot x_0 - 1.1933685 \cdot \ln(x_0) - 2.4256386 + \frac{0.012372488}{(1 - 3.392166 \cdot 10^{-17} \cdot x_0^2)^3}$$

Exact saved expression (use this for coefficient checks):

`(0.024921370121969308 / cube(1.2629077484725748 - square(x0 / -1.527832119987085e8))) + (((((x0 - (x0 / 1.3393855976910628e8)) + -2.7916700046461567) - x0) - ((log(x0) + 0.7447500616311524) * 2.167701266796345)) * 0.5505225637910837)`

Bode | seed 10000

Accepted target: $a = c_0 + c_1 \exp(c_2 x_0)$ (archived broad criterion allows $c_0 = 0$)

$x_0 = n$

Dataset columns: n, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: phenomenological match. 349 restarts; actual search 3600.0 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 8; training R2 = 0.99975413; native loss = 0.053387336.

$$0.14855406 \cdot e^{0.6916177 \cdot x_0} + 0.40757019$$

Exact saved expression (use this for coefficient checks):

`exp((x0 + -2.757023627077765) * 0.69161769600249) + 0.4075701869986178`

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 16; complexity 28; training R2 = 0.99999972; native loss = 0.0017558052.

$$0.14199443 \cdot e^{0.69858835 \cdot x_0} + 0.39823693 + \frac{0.0020552952}{\left(0.36748521 \cdot x_0 - (0.71627527 \cdot x_0 - 1)^3 + 0.090075772\right)^3} - \frac{0.12898176}{1.5952283 \cdot x_0 - 8.9477628}$$

Exact saved expression (use this for coefficient checks):

`(exp((x0 - 2.7941597329554093) / 1.4314581720348967) + 0.3982369295612157) + ((-0.1289817554162571 / ((x0 * 1.5952282548961962) - 8.947762766000286)) - (-0.04141470996047404 / cube(0.24511400663879207 - (cube(x0 + -1.3961113122455882) - x0))))`

Evolved PySR (709715)

Archived/audited label: phenomenological match. 62 restarts; actual search 3603.0 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 2; training R2 = -4526.5743; native loss = 0.14746443.

$$e^{x_0}$$

Exact saved expression (use this for coefficient checks):

`exp(x0)`

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 21; complexity 29; training R2 = 0.99999227; native loss = 1.468144e-05.

$$0.14847473 \cdot e^{0.69329599 \cdot x_0 + \frac{0.010009336}{5.6824967 - x_0}} + 0.37215076 - \frac{0.0075424183}{x_0 - 2.8804045} + \frac{0.029249367}{x_0^2}$$

Exact saved expression (use this for coefficient checks):

`((exp((x0 * 0.6932959912130935) + ((0.010009335975703552 / (5.682496713260664 - x0)) - 1.9073405240114323)) + 0.16959665629008716) + (0.02924936748309213 / square(x0))) + (0.2001248588154691 + (-0.007542418347343751 / (x0 + -2.880404525066148)))) + 0.002429244671627672`

Bode | seed 10001

Accepted target: $a = c_0 + c_1 \exp(c_2 x_0)$ (archived broad criterion allows $c_0 = 0$) $x_0 = n$

Dataset columns: n, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: phenomenological match. 336 restarts; actual search 3600.0 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 8; training R2 = 0.99975396; native loss = 0.053381896.

$$0.14857318 \cdot e^{0.69159993 \cdot x_0} + 0.40749503$$

Exact saved expression (use this for coefficient checks):
`exp((x0 + -2.7569083560764756) * 0.6915999282369407) + 0.40749502582657854`

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 11; complexity 23; training R2 = 0.9999995; native loss = 0.0031688058.

$$0.14086954 \cdot e^{0.69973957 \cdot x_0} + 0.39618131 + \frac{0.074722671}{6.2700641 \cdot x_0 - e^{x_0}} + \frac{0.084654943}{5.5705271 - x_0}$$

Exact saved expression (use this for coefficient checks):
`((((0.0747226707002445 / ((x0 * 6.270064101815791) - exp(x0))) + exp((x0 * 0.6997395739186845) + -1.9599210364371686)) + (0.08465494294388307 / (5.5705270774696105 - x0))) + 0.3961813149152678`

Evolved PySR (709715)

Archived/audited label: phenomenological match. 60 restarts; actual search 3601.6 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 2; training R2 = -4526.5743; native loss = 0.14746443.

$$e^{x_0}$$

Exact saved expression (use this for coefficient checks):
`exp(x0)`

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 13; complexity 15; training R2 = 0.99998764; native loss = 0.0035293819.

$$0.13789917 \cdot \sqrt{e^{1.4055034 \cdot x_0}} + 0.39867335 + \frac{0.085290701}{5.4947254 - x_0}$$

Exact saved expression (use this for coefficient checks):
`((sqrt(exp(x0 * 1.40550336249511)) * 0.13789916946086703) + 0.3986733511616438) - (-0.08529070145419113 / (5.494725426209017 - x0))`

Bode | seed 10002

Accepted target: $a = c_0 + c_1 \exp(c_2 x_0)$ (archived broad criterion allows $c_0 = 0$)

$x_0 = n$

Dataset columns: n, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: phenomenological match. 352 restarts; actual search 3599.7 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 8; training R2 = 0.99975399; native loss = 0.053380437.

$$0.14856679 \cdot e^{0.69160571 \cdot x_0} + 0.40754354$$

Exact saved expression (use this for coefficient checks):

$(0.1485667935335544 / \exp(x_0 * -0.6916057127879621)) - -0.40754353786275804$

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 29; training R2 = 0.99999674; native loss = 0.007012197.

$$\left(0.14447223 \cdot e^{0.69669021 \cdot x_0} + 0.39032656 - \frac{0.019481754}{x_0 - 2.6147021}\right) \cdot e^{\frac{0.063920294}{-x_0 \cdot (x_0 - 4.586977) + x_0}}$$

Exact saved expression (use this for coefficient checks):

$\exp(0.0639202939305726 / (x_0 - (x_0 * (x_0 + -4.586977028635639)))) * (((((0.01405604117775936 / (x_0 + -2.614702138315707)) + 0.007985385058218093) / -0.7214977356914992) + \exp((x_0 - 2.7769415102490917) * 0.6966902078172466)) + 0.40139435098164317)$

Evolved PySR (709715)

Archived/audited label: phenomenological match. 99 restarts; actual search 3600.6 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 2; training R2 = -4526.5743; native loss = 0.14746443.

$$e^{x_0}$$

Exact saved expression (use this for coefficient checks):

$\exp(x_0)$

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 11; complexity 15; training R2 = 0.99998581; native loss = 0.0034438012.

$$0.13344321 \cdot e^{0.70743803 \cdot x_0} + 0.13344321 \cdot e^{\frac{0.94253224}{5.998974 - x_0}} + 0.27877055$$

Exact saved expression (use this for coefficient checks):

$(\exp(x_0 * 0.7074380279792544) + (\exp(0.9425322399423339 / (5.998974033081171 - x_0)) - -2.0890575654428893)) * 0.13344321301065865$

Bode | seed 10003

Accepted target: $a = c_0 + c_1 \exp(c_2 x_0)$ (archived broad criterion allows $c_0 = 0$) $x_0 = n$

Dataset columns: n, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: phenomenological match. 338 restarts; actual search 3599.7 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 8; training R2 = 0.99975396; native loss = 0.053382703.

$$0.14856847 \cdot e^{0.69160397 \cdot x_0} + 0.40755627$$

Exact saved expression (use this for coefficient checks):
`exp((x0 + -2.756938150030906) * 0.6916039718270762) + 0.4075562672925307`

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 28; training R2 = 0.99999934; native loss = 0.0032199152.

$$-\frac{0.29569958 - \frac{0.0052551888}{(0.099471525 \cdot x_0^2 - 1)^2}}{0.31992043 \cdot x_0^2 - 10.072875} + 0.14170106 \cdot \sqrt{e^{1.3977842 \cdot x_0}} + 0.39275749$$

Exact saved expression (use this for coefficient checks):
`sqrt(exp((x0 * 1.3977842124491726) - 3.9080712361900414)) + (0.39275749328276754 - ((0.29569958374690886 - square(-0.7287782245420767 / ((cube(x0) / x0) + -10.05312829442839))) / ((square(x0) / 3.12577725650547) + -10.072875183440502)))`

Evolved PySR (709715)

Archived/audited label: phenomenological match. 99 restarts; actual search 3600.8 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 2; training R2 = -4526.5743; native loss = 0.14746443.

$$e^{x_0}$$

Exact saved expression (use this for coefficient checks):
`exp(x0)`

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 22; complexity 30; training R2 = 0.99999386; native loss = 2.7070319e-05.

$$-\frac{0.14262061 \cdot x_0}{0.15424191 - x_0} + 0.14262061 \cdot e^{0.6981661 \cdot x_0 - \frac{0.0014556835}{2 \cdot x_0 - 5.9750564}} + 0.22872787 - \frac{0.081000946}{x_0 - 5.6116845}$$

Exact saved expression (use this for coefficient checks):
`0.22872787245385037 - (((x0 / (5.705962497251242 - (x0 + 5.551720584506562))) - exp((x0 - (0.0020850103400363636 / ((x0 - 5.975056415805104) + x0))) * 0.698166100336025)) + (0.5679469967312045 / (x0 - 5.611684531550322))) * 0.14262060820306866`

Bode | seed 10004

Accepted target: $a = c_0 + c_1 \exp(c_2 x_0)$ (archived broad criterion allows $c_0 = 0$)

$x_0 = n$

Dataset columns: n, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: phenomenological match. 336 restarts; actual search 3599.8 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 8; training R2 = 0.99975402; native loss = 0.053381992.

$$0.14856468 \cdot e^{0.6916077 \cdot x_0} + 0.40754386$$

Exact saved expression (use this for coefficient checks):

`exp((x0 * 0.691607704487866) + -1.90673488268886) + 0.40754385655475467`

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 15; complexity 27; training R2 = 0.99999992; native loss = 0.0013686871.

$$0.14403424 \cdot e^{0.69660986 \cdot x_0} + 0.38955055 - \frac{0.077982421}{x_0 - 5.6508754} + \frac{0.04622285 \cdot x_0}{x_0^3 \cdot (2.9200099 - x_0)}$$

Exact saved expression (use this for coefficient checks):

`(exp((x0 * 0.6966098642472842) + -1.9377042140642646) + (((x0 / cube(x0)) / ((2.84081109922326 - x0) + 0.07919879509944819)) * 0.04622285047394004) + (-0.07798242088008844 / (x0 + -5.650875367520969)))) + 0.3895505475937608`

Evolved PySR (709715)

Archived/audited label: phenomenological match. 97 restarts; actual search 3600.8 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 2; training R2 = -4526.5743; native loss = 0.14746443.

$$e^{x_0}$$

Exact saved expression (use this for coefficient checks):

`exp(x0)`

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 26; training R2 = 0.9999876; native loss = 0.00022075887.

$$0.14390447 \cdot e^{0.69710363 \cdot x_0} + 0.36255152 - \frac{0.080245073}{x_0 - 5.6316228} - \frac{0.090025383}{x_0^2 \cdot (2 \cdot x_0 - 5.8346202)}$$

Exact saved expression (use this for coefficient checks):

`((exp(x0 * 0.6971036264994562) - (0.557627384073785 / (x0 + -5.631622807716615)))) * (0.14390447038987028 - ((0.0900253827356204 / (x0 + (x0 + -5.834620212210738)))) / (x0 * x0))) - -0.36255151534627084`

Bode | seed 10005

Accepted target: $a = c_0 + c_1 \exp(c_2 x_0)$ (archived broad criterion allows $c_0 = 0$)

$x_0 = n$

Dataset columns: n, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: phenomenological match. 339 restarts; actual search 3600.5 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 8; training R2 = 0.99975409; native loss = 0.053385832.

$$0.14855316 \cdot e^{0.69161814 \cdot x_0} + 0.40762366$$

Exact saved expression (use this for coefficient checks):

`exp((x0 + -2.7570306369538735) * 0.6916181409056433) + 0.40762366123224875`

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 30; training R2 = 0.99999976; native loss = 0.0022284924.

$$0.13821197 \cdot e^{0.70244933 \cdot x_0} + 0.38440643 + \frac{0.30228495}{(3.0352851 - 0.55169986 \cdot x_0) \cdot \left(x_0 + e^{-\frac{0.037638278}{x_0 \cdot (0.28718634 \cdot x_0 - 1)^3}}\right)}$$

Exact saved expression (use this for coefficient checks):

`(exp((0.7024493313815355 * x0) - 1.9789667776233812) + 0.3844064337268917) - (-0.3022849480197258 / ((exp(cube((0.3844064337268917 * -3.035676503487829) / (x0 + -3.4820597868243968)) / x0) + x0) * ((-0.5516998639812778 * x0) - -3.035285089749335)))`

Evolved PySR (709715)

Archived/audited label: phenomenological match. 91 restarts; actual search 3600.5 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 2; training R2 = -4526.5743; native loss = 0.14746443.

$$e^{x_0}$$

Exact saved expression (use this for coefficient checks):

`exp(x0)`

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 25; training R2 = 0.99999992; native loss = 0.0001806308.

$$0.14360201 \cdot e^{0.69706173 \cdot x_0} + 0.38770211 - \frac{0.079693968}{x_0 - 5.6320371} + \frac{0.01099933}{x_0^2 \cdot (0.70220211 - 0.24064647 \cdot x_0)}$$

Exact saved expression (use this for coefficient checks):

`((exp(x0 / 1.4345931709498958) - ((0.5549641338623968 / (x0 - 5.632037109619216)) - ((0.07659592920170029 / square(x0)) / (0.7022021142826751 - (x0 / 4.1554733239616555))))) * 0.14360201461869562) - -0.3877021118179293`

Bode | seed 10006

Accepted target: $a = c_0 + c_1 \exp(c_2 x_0)$ (archived broad criterion allows $c_0 = 0$) $x_0 = n$

Dataset columns: n, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: **phenomenological match.** 337 restarts; actual search 3600.4 s. Archived review / positive-equation audit

Recovery witness

Frontier index 4; complexity 8; training R2 = 0.99975403; native loss = 0.05337978.

$$0.14856189 \cdot e^{0.69161018 \cdot x_0} + 0.40757281$$

Exact saved expression (use this for coefficient checks):
`exp((x0 - 2.7569773652696274) * 0.6916101808262582) + 0.40757281360370995`

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 16; complexity 25; training R2 = 0.99999777; native loss = 0.0045893512.

$$-\frac{5.0518532 \cdot 10^{-5} \cdot x_0^2}{(0.35998168 \cdot x_0 - 1)^2} + 0.13991856 \cdot e^{0.70056052 \cdot x_0} + 0.41514252 + \frac{0.0075353308}{0.51711562 - 0.0926191 \cdot x_0}$$

Exact saved expression (use this for coefficient checks):
`((((0.0075353308415657665 / ((x0 * -0.09261910016935047) + 0.5171156157844824)) - square(x0 * (0.01974444616279043 / (x0 - 2.7779191065720603)))) + exp((x0 + -2.807316018718971) * 0.7005605192830989)) + 0.41514252281016073`

Evolved PySR (709715)

Archived/audited label: **phenomenological match.** 94 restarts; actual search 3600.6 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 2; training R2 = -4526.5743; native loss = 0.14746443.

$$e^{x_0}$$

Exact saved expression (use this for coefficient checks):
`exp(x0)`

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 10; complexity 14; training R2 = 0.99998717; native loss = 0.0035292631.

$$0.13763931 \cdot e^{0.70295475 \cdot x_0} + 0.3979985 + \frac{0.085258969}{5.4888733 - x_0}$$

Exact saved expression (use this for coefficient checks):
`((exp(x0 / 1.4225666817422382) * 0.13763930696520388) - (-0.0852589693422757 / (5.488873335802308 - x0))) + 0.39799849483540317`

Bode | seed 10007

Accepted target: $a = c_0 + c_1 \exp(c_2 x_0)$ (archived broad criterion allows $c_0 = 0$)

$x_0 = n$

Dataset columns: n, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: phenomenological match. 337 restarts; actual search 3600.1 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 8; training R2 = 0.99975397; native loss = 0.053382272.

$$0.14856735 \cdot e^{0.691605 \cdot x_0} + 0.40756089$$

Exact saved expression (use this for coefficient checks):

`exp(-1.906716879042398 - (x0 * -0.6916050005994446)) + 0.4075608884614683`

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 26; training R2 = 0.99999879; native loss = 0.0037981175.

$$0.14498557 \cdot e^{0.69564898 \cdot x_0} + 0.39286479 + \frac{0.075211885}{5.6739687 - x_0} + \frac{0.0053009538}{2.9247305 - x_0}$$

Exact saved expression (use this for coefficient checks):

`(0.07521188454347494 / (5.709392383650673 - (x0 + 0.035423723651498935))) + (((0.005300953789260041 / (2.9247305040073974 - x0)) + exp(((1.1721791238747148 - (x0 + -1.8397181880526998)) * -0.6956489815890085) - -0.16410223502778667)) - -0.39286478749596)`

Evolved PySR (709715)

Archived/audited label: phenomenological match. 94 restarts; actual search 3600.5 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 2; training R2 = -4526.5743; native loss = 0.14746443.

$$e^{x_0}$$

Exact saved expression (use this for coefficient checks):

`exp(x0)`

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 20; complexity 30; training R2 = 0.99999743; native loss = 6.7007905e-06.

$$\frac{0.86970554 \cdot \left(-2.3380683 + \frac{e^{x_0}}{x_0}\right)}{x_0 + 0.13295106} + 0.39749376 - \frac{0.029663923}{x_0 - 2.6830467} + \frac{0.090385279}{x_0 \cdot (5.0981272 - x_0)}$$

Exact saved expression (use this for coefficient checks):

`(((((exp(x0) / x0) + -2.338068252561891) / (x0 + 0.1329510634269914)) - (0.03410800737383077 / (x0 + -2.6830466813732214))) / 1.1498144524409168) + (0.09036850407815165 - -0.3071252569709798)) + ((0.09038527897487196 / (5.098127154122821 - x0)) / x0)`

Bode | seed 10008

Accepted target: $a = c_0 + c_1 \exp(c_2 x_0)$ (archived broad criterion allows $c_0 = 0$)

$x_0 = n$

Dataset columns: n, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: phenomenological match. 336 restarts; actual search 3599.8 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 8; training R2 = 0.99975405; native loss = 0.053384823.

$$0.14856359 \cdot e^{0.69160887 \cdot x_0} + 0.40752949$$

Exact saved expression (use this for coefficient checks):

`(exp(x0 * 0.6916088745057437) * 0.1485635922288841) + 0.4075294888810522`

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 17; complexity 25; training R2 = 0.99999909; native loss = 0.0052258878.

$$0.14685558 \cdot e^{0.69452355 \cdot x_0 - \frac{0.011217713}{x_0 - 5.6180206}} + 0.38862675 + \frac{0.05433644}{x_0 + 2.2651723 \cdot (1 - 0.76143515 \cdot x_0)^3 + 0.67174422}$$

Exact saved expression (use this for coefficient checks):

`(0.05433643970419888 / ((x0 + cube(1.313309481927391 - x0)) + 0.6717442231616028)) + (exp(((x0 - (0.01615166747655187 / (x0 + -5.618020578567067)))) * 0.6945235510657856) + -1.9183056289143745) + 0.38862675315957995)`

Evolved PySR (709715)

Archived/audited label: phenomenological match. 156 restarts; actual search 3600.4 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 2; training R2 = -4526.5743; native loss = 0.14746443.

$$e^{x_0}$$

Exact saved expression (use this for coefficient checks):

`exp(x0)`

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 28; training R2 = 0.99999408; native loss = 1.0706221e-05.

$$0.3693133 + \frac{0.87250261 \cdot \left(-2.3681766 - \frac{0.095570128}{x_0 - 2.6856257} - \frac{0.10569978}{x_0 - 5.0972309} + \frac{e^{x_0}}{x_0} \right)}{x_0 + 0.13506643}$$

Exact saved expression (use this for coefficient checks):

`((0.8725026109740102 / (x0 + 0.13506643210256053)) * (((-0.09557012792355937 / (x0 + -2.6856256957437075)) + ((exp(x0) / x0) + (-2.3681766003086673 + (-0.1056997807756291 / (x0 + -5.0972308481462445)))))) - (0.13751480894583232 * -2.6856256957437075)`

Bode | seed 10009

Accepted target: $a = c_0 + c_1 \exp(c_2 x_0)$ (archived broad criterion allows $c_0 = 0$)

$x_0 = n$

Dataset columns: n, target. Natural logs unless specified. Constants c_i denote the accepted free parameters.

Base PySR

Archived/audited label: phenomenological match. 355 restarts; actual search 3599.0 s. Archived review / positive-equation audit

Recovery witness

Frontier index 5; complexity 8; training R2 = 0.99975393; native loss = 0.053384482.

$$0.14858158 \cdot e^{0.69159235 \cdot x_0} + 0.40742865$$

Exact saved expression (use this for coefficient checks):

`exp((x0 + -2.7568568831957143) * 0.6915923474610706) + 0.40742864886084657`

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 16; complexity 27; training R2 = 0.99999842; native loss = 0.0034672673.

$$-\frac{0.0042258702 + \frac{0.086730452}{x_0^2}}{2 \cdot x_0 - 5.7213476} + 0.13922511 \cdot e^{0.7014229 \cdot x_0} + 0.39451317 - \frac{0.085220626}{x_0 - 5.5200035}$$

Exact saved expression (use this for coefficient checks):

`exp((x0 * 0.7014229044051187) + -1.9716631313541992) - (((((0.08673045246893929 / square(x0)) + 0.004225870162127872) / (x0 + (x0 - 5.721347572258147))) + (0.08522062570279723 / (x0 - 5.520003461586825))) + -0.39451317268942054)`

Evolved PySR (709715)

Archived/audited label: phenomenological match. 151 restarts; actual search 3600.4 s. Archived review / positive-equation audit

Recovery witness

Frontier index 1; complexity 2; training R2 = -4526.5743; native loss = 0.14746443.

$$e^{x_0}$$

Exact saved expression (use this for coefficient checks):

`exp(x0)`

Review rationale: Broad P rubric only: exponential family, possibly with zero offset. Not counted as exact recovery; this does not validate the full nonzero-offset Bode law.

Best numerical fit (maximum training R2)

Frontier index 18; complexity 30; training R2 = 0.99999305; native loss = 1.5541477e-05.

$$\left(0.10037176 + \frac{0.0067164993}{x_0}\right) \cdot \left(1.4279032 \cdot e^{\frac{x_0}{1.4343145 + \frac{0.0039966994}{x_0 - 5.6369279}}} + 3.3323085 - \frac{0.035721112}{1.8642473 \cdot x_0 - 5.5358092}\right) + 0.035723709$$

Exact saved expression (use this for coefficient checks):

`((0.10037176056119586 - (-0.006716499317754941 / x0)) * ((-0.03572111195223436 / ((x0 * 1.8642472866483137) - 5.53580922325937)) + ((exp(x0 / (1.4343144856565906 - (-0.003996699207723649 / (x0 - 5.6369278562180485)))) + 2.3337075584667866) * 1.4279032031912304))) - -0.035723708720299306`

Appendix: Rydberg, base PySR, seed 10000

The archived reviewer returned miss without selecting an equation. All saved frontier rows follow; the main page uses index 7 as a compact structural proxy. Target: $\log \lambda = c_0 - \log(x_0^{-2} - x_1^{-2})$.

Candidate 0

Frontier index 0; complexity 1; training R2 = -0.1126028; native loss = 1.3698403.

$$-13.051441$$

Exact saved expression (use this for coefficient checks):
-13.05144117063778

Candidate 1

Frontier index 1; complexity 3; training R2 = 0.96484652; native loss = 0.24376.

$$x_0 - 16.857204$$

Exact saved expression (use this for coefficient checks):
x0 + -16.857203527340214

Candidate 2

Frontier index 2; complexity 5; training R2 = 0.98364763; native loss = 0.14996217.

$$-\frac{193.05422}{x_0 + 11.079999}$$

Exact saved expression (use this for coefficient checks):
-193.05421986658754 / (x0 + 11.079999381102906)

Candidate 3

Frontier index 3; complexity 6; training R2 = 0.98460787; native loss = 0.14240746.

$$3.3232082 \cdot \sqrt{x_0} - 19.443545$$

Exact saved expression (use this for coefficient checks):
(sqrt(x0) + -5.850835594423077) * 3.3232081889867366

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Candidate 4

Frontier index 4; complexity 7; training R2 = 0.98880603; native loss = 0.13151311.

$$\frac{-x_1 - 178.57492}{x_0 + 10.416509}$$

Exact saved expression (use this for coefficient checks):

`(-178.5749180558685 - x1) / (x0 + 10.416508669526925)`

Candidate 5

Frontier index 5; complexity 8; training R2 = 0.99332716; native loss = 0.10495783.

$$\sqrt{x_0 \cdot (19.490679 - x_1)} - 20.029901$$

Exact saved expression (use this for coefficient checks):

`sqrt((19.49067901315621 - x1) * x0) + -20.029900588131188`

Candidate 6

Frontier index 6; complexity 10; training R2 = 0.99508234; native loss = 0.092027962.

$$\sqrt{x_0 \cdot (16.221168 - 0.60199299 \cdot x_1)} - 19.791556$$

Exact saved expression (use this for coefficient checks):

`sqrt(x0 * (16.221168400612935 - (x1 * 0.601992994311058))) + -19.7915557946515`

Candidate 7

Frontier index 7; complexity 11; training R2 = 0.99971279; native loss = 0.021287117.

$$\frac{1.9921975 \cdot x_0^3}{x_1^3} + 1.9921975 \cdot \ln(x_0) - 16.175523$$

Exact saved expression (use this for coefficient checks):

`((cube(x0 / x1) + -8.119437542038874) + log(x0)) / 0.501958254706728`

Appendix: Rydberg, base PySR, seed 10000

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Candidate 8

Frontier index 8; complexity 13; training R2 = 0.99972979; native loss = 0.020851882.

$$\frac{2.0080361 \cdot x_0^3}{x_1^3} + 2.0080361 \cdot \ln(x_0 + 0.020985801) - 16.216507$$

Exact saved expression (use this for coefficient checks):

(log(x0 - -0.02098580147295267) + (cube(x0 / x1) + -8.075804185398546)) / 0.4979990093511404

Candidate 9

Frontier index 9; complexity 15; training R2 = 0.99978663; native loss = 0.019574363.

$$2.0613342 \cdot \ln(x_0 + 0.073350559) - 16.308448 + \frac{2.0613342 \cdot (x_0 - 0.047073556)^3}{x_1^3}$$

Exact saved expression (use this for coefficient checks):

(log(x0 - -0.07335055875765933) + (cube((x0 + -0.047073555833158545) / x1) + -7.911598161737125)) / 0.4851226918515233

Candidate 10

Frontier index 10; complexity 16; training R2 = 0.99979951; native loss = 0.018526348.

$$1.9755629 \cdot \left(8.9410074 \cdot 10^{-5} \cdot x_0^3 + \frac{x_0}{x_1} \right)^3 + 1.9755629 \cdot \ln(x_0) - 16.161706$$

Exact saved expression (use this for coefficient checks):

(cube((x0 / x1) + cube(x0 * 0.04471591807187088)) + (log(x0) + -8.180810554733988)) / 0.5061848451419206

Candidate 11

Frontier index 11; complexity 17; training R2 = 0.99983031; native loss = 0.016251903.

$$\frac{1.9297898 \cdot x_0^3}{x_1^3} + 0.01376734 \cdot \sqrt{x_0^3} + 1.9297898 \cdot \ln(x_0) - 16.182171$$

Exact saved expression (use this for coefficient checks):

(log(x0) + ((sqrt(cube(x0)) * 0.007134114137833358) + (cube(x0 / x1) + -8.38545775268773))) / 0.5181911607893088

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Candidate 12

Frontier index 12; complexity 19; training R2 = 0.99993682; native loss = 0.010384228.

$$\frac{x_0^6}{(x_1 + 0.61233119)^6} + e^{\left(\frac{x_0}{x_1}\right)^2} + \ln(x_0^2) - 17.21475$$

Exact saved expression (use this for coefficient checks):

```
(log(square(x0)) + (exp(square(x0 / x1)) + square(cube(x0 / (x1 + 0.6123311911831475)))))) + -17.2147503012161
```

Candidate 13

Frontier index 13; complexity 20; training R2 = 0.99996266; native loss = 0.0075082368.

$$1.960148 \cdot \ln(x_0) - 16.263867 + \frac{x_0 - 0.1713685}{-1.7973843 \cdot x_0 + 2.1148526 \cdot x_1}$$

Exact saved expression (use this for coefficient checks):

```
((x0 - 0.1713685019882606) / ((x1 - (x0 / 1.1766279564497872)) * (0.5654397407720653 + 1.5494128506082108))) + -16.263866787587084 + (log(x0) * 1.9601479519257683)
```

Candidate 14

Frontier index 14; complexity 22; training R2 = 0.99996234; native loss = 0.0074884885.

$$\frac{x_0 - 0.1713685}{-1.7973843 \cdot x_0 + 2.1148526 \cdot x_1 - 0.0044941058} + 1.960148 \cdot \ln(x_0) - 16.263867$$

Exact saved expression (use this for coefficient checks):

```
-16.263866787587084 + ((1.9601479519257683 * log(x0)) + ((x0 - 0.1713685019882606) / ((x1 - ((x0 - -0.0025003589153673165) / 1.1766279564497872)) * (0.5654397407720653 + 1.5494128506082108))))
```

Candidate 15

Frontier index 15; complexity 27; training R2 = 0.99996175; native loss = 0.0074351314.

$$\frac{x_0 - 0.1713685}{-2.1148526 \cdot x_0^4 \cdot e^{-4 \cdot x_0} - 1.7973843 \cdot x_0 + 2.1148526 \cdot x_1} + 1.960148 \cdot \ln(x_0) - 16.263867$$

Exact saved expression (use this for coefficient checks):

```
((x0 - 0.1713685019882606) / (((x1 - (x0 / 1.1766279564497872)) - square(square(x0 / exp(x0)))) * (0.5654397407720653 + 1.5494128506082108))) + -16.263866787587084 + (log(x0) * 1.9601479519257683)
```

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Candidate 16

Frontier index 16; complexity 30; training R2 = 0.99996298; native loss = 0.0073909492.

$$1.960148 \cdot \ln(x_0) - 16.263867 + \frac{x_0 - 0.1713685}{\left(2.1148526 - (-x_0^3 + x_0)^2 \cdot e^{-2 \cdot x_0^2}\right) \cdot (-0.84988632 \cdot x_0 + x_1)}$$

Exact saved expression (use this for coefficient checks):

```
((x0 - 0.1713685019882606) / ((x1 - (x0 / 1.1766279564497872)) * ((0.5654397407720653 - square((x0 - cube(x0)) / exp(square(x0)))) + 1.5494128506082108))) + (-16.263866787587084 + (log(x0) * 1.9601479519257683))
```

Sources and reproduction

Rebuild from the repository root: `python figures/build_empirical_1m_audit.py`. This reads saved artifacts and runs Tectonic locally; it submits no jobs and makes no LLM requests. **Source files (SHA-256 fingerprints in `audit_data.json`):**

`analysis/benchmark_positive_audit_2026-09-08.json`

`reports/srbench2_portfolio_solve_over_time/first_recovery.json`

`manual_solve_check.py`

`runs/srbench2_9-8_gt_baseline_1core_l1_portfolio_1m_no_warmup/srbench_full_results.json`

`runs/srbench2_9-8_gt_baseline_1core_l1_portfolio_1m_no_warmup/manual_solve_check_results.json`

`runs/srbench2_9-8_gt_baseline_1core_l1_portfolio_1m_no_warmup/manifest.json`

`runs/709715/srbench2_9-8_ground_truth_1core_portfolio_1m_no_warmup/srbench_full_results.json`

`runs/709715/srbench2_9-8_ground_truth_1core_portfolio_1m_no_warmup/manual_solve_check_results.json`

`runs/709715/srbench2_9-8_ground_truth_1core_portfolio_1m_no_warmup/manifest.json`

Audit limits. Selected positive witnesses were previously checked algebraically. Archived near/miss decisions were not exhaustively re-reviewed. The closest structural candidate is qualitative, not a globally optimal symbolic distance. All displayed equations are verified to occur verbatim at their stated zero-based final-frontier indices. The numeric-fit selection uses saved training metrics, not held-out performance.